

Math Intellect

Prototype Validation Evidence

Test-case evidence documentation

For Math Intellect Portfolio Support

Actual screenshots and test-case notes will be shown. Final deck will be exported as PDF for uploading to GitHub.

Evidence Chain

Metric Snapshot

PPT screenshots

GitHub repository

Evaluation summary

0580/11 May/June 2026 and 0580/31 May/June 2026
Past Papers are used as test case

Test Case Identity

Test Case ID Format : MI-TC-xxx

Source [Past Paper Reference]: **0580/11
May/June 2026 and 0580/31 May/June 2026**

Evidence Folder: **WILL UPDATE LATER** [GitHub LINK](#)

Evidence Image

System-Level Technical Evidence

1. Representative successful n8n workflow execution screenshot
2. Representative execution log & output table

Test-Case Educational Evidence

3. Original question image sent through LINE
4. Analyzer / question interpretation output
5. LINE input-output screenshot
6. Official Cambridge mark scheme answer
7. Solution section of the LINE reply

Final Recording Rule

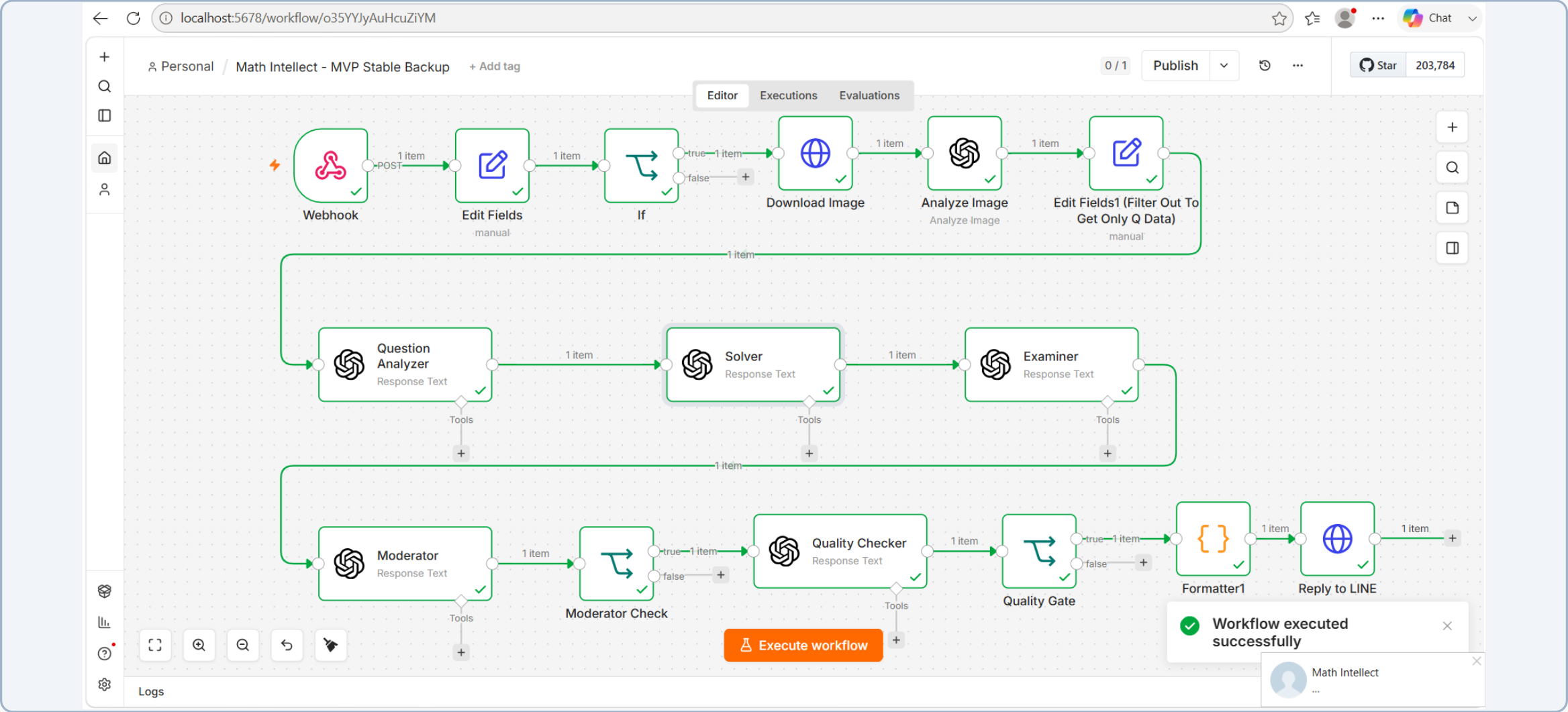
Record the final values in Excel only after reviewing the visual evidence.

PowerPoint/PDF provides the case-level audit trail; Excel is the master validation dataset and statistical summary.

Repository files:

1. Math Intellect Prototype Validation Evidence (.pptx)
2. Reviewer copy (.pdf)
3. Completed Validation Data workbook (.xlsx)

Selected Successful n8n workflow execution example screenshot



Selected Successful Formatter's log & output table

Logs

Clear execution

Formatter1 Success in 55ms

Input

Output

Success in 1m 22s

Overview

Details

Examiner

Moderator

Moderator Check

Quality Checker

Quality Gate

Formatter1

Reply to LINE

INPUT

1 item

output

output[0]

id

msg_0d10aa5cca62a310006aa14024189c87d095b5935d6b8647e2

type

message

status

completed

content

content[0]

type

output_text

OUTPUT

1 item

replyToken	formatted_output
2131f306c073484690a3df7802ef1f10	Topic: Accuracy and approximation Difficulty: Easy Assessment Objective: AO1 Solution Step 1: Lower bound = $12.7 - 0.05 = 12.65$ Step 2: Upper bound = $12.7 + 0.05 = 12.75$ Step 3: $12.65 \leq t < 12.75$ Final Answer: $12.65 \leq t < 12.75$ M1: Steps 1-2 give bounds 12.65 and 12.75 A1: Step 3 gives $12.65 \leq t < 12.75$ Total: 2

Representative system-level evidence. The same validated workflow configuration is used consistently across the complete test set; case-level results are recorded separately in Excel and the corresponding evidence slides.

Original Question Image

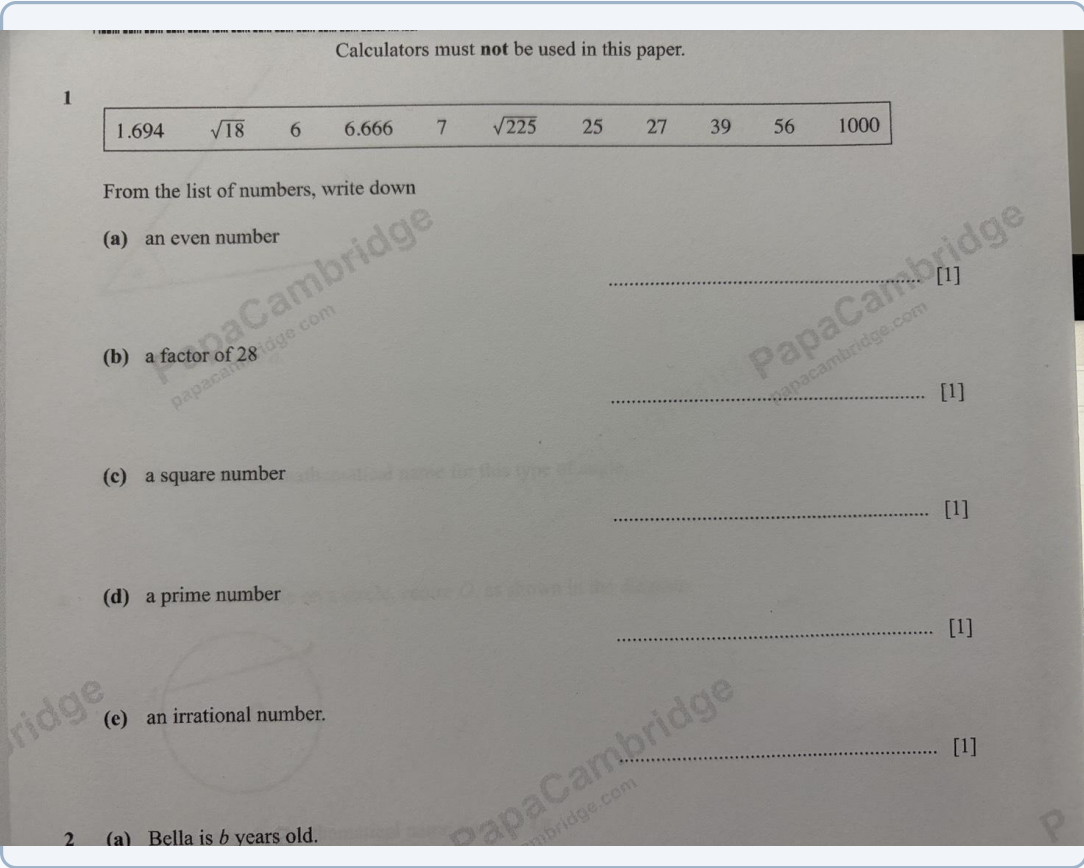


Image Quality	
GOOD	– All texts / symbols clearly readable
MODERATE	– minor blur / skew but all essential information recoverable
POOR	– visible degradation affecting text / symbol yet question remains testable

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR– visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node



1 item

1

conten : 1 \n \n 1.694 $\sqrt{18}$ 6 6.666 7 $\sqrt{225}$ 25

27 39 56 1000

From the list of numbers,

write down (a) an even number

..... [1] \n \n (b) a

factor of 28

[1] \n \n (c) a square number

..... [1] \n \n (d) a

prime number

[1] \n \n (e) an irrational number.

..... [1]

Interpretation

Category	Value
PASS	1
FAIL	1

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_06ed4d69a36c9625006aa38962ea5c87d0a898dc6ab04785fb

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) 6\n\n(b) 7\n\n(c) 25\n\n(d) 7\n\n(e) \n\n√18

role : assistant

Official Mark Scheme Answer

1(a)	6 or 56 or 1000	1
1(b)	7	1
1(c)	25	1
1(d)	7	1
1(e)	√18	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Sub-question Final Answer Accuracy

100 %

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Number

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) 6

(b) 7

(c) 25

(d) 7

(e) √18

B1: (a) 6 is an even number from the list

B1: (b) 7 is a factor of 28

B1: (c) 25 is a square number from the list

B1: (d) 7 is a prime number from the list

B1: (e) √18 is irrational

Total: 5

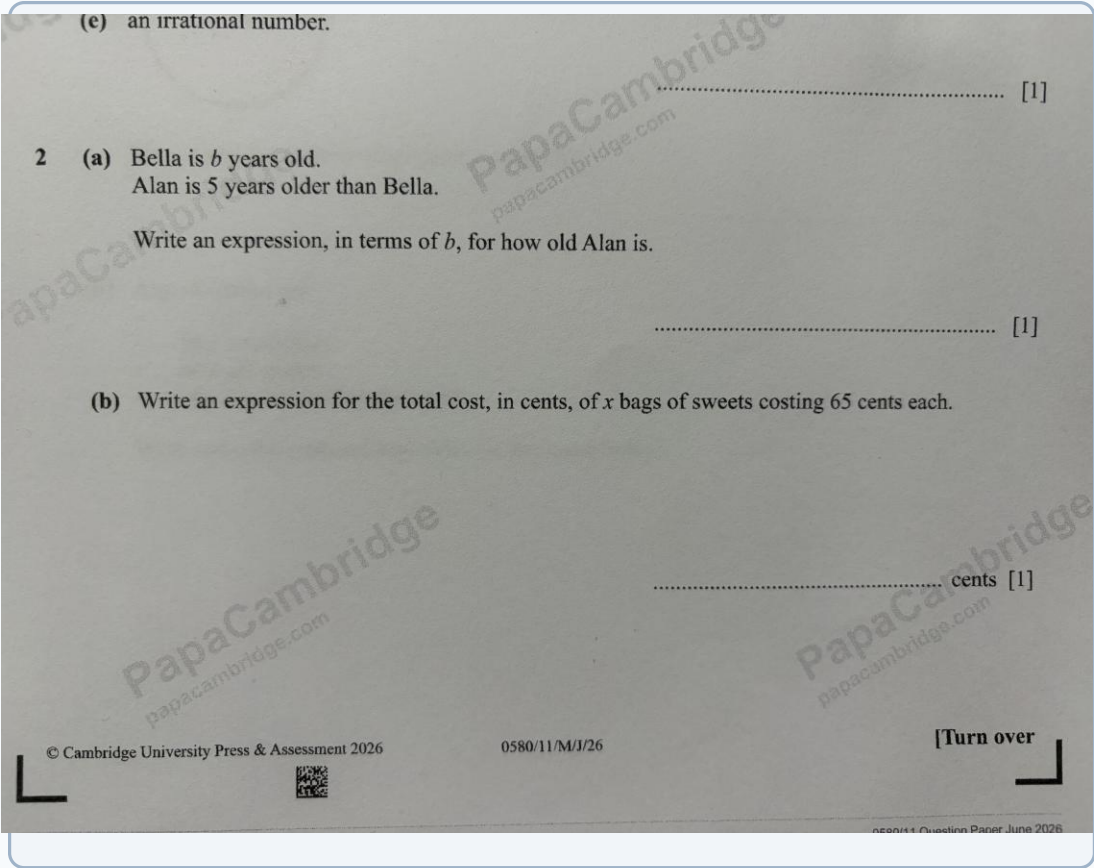
11:54 AM

Aa

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 2 (a) Bella is b years old.
Alan is 5 years older than Bella.
Write an expression, in terms of b , for how old Alan is.
..... [1]
(b) Write an expression for the total cost, in cents, of x bags of sweets costing 65 cents each.
..... cents [1]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0783edd994992797006aa3bf8b709887d0925179ea79ff1a5d

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) b + 5\n\n(b) 65x

role : assistant

Official Mark Scheme Answer

2(a)	$b + 5$ oe final answer	1
2(b)	65x oe final answer	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Sub-question Final Answer Accuracy

100 %

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

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Success Layout – LINE Reply Message

Topic: Algebra

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) $b + 5$

(b) $65x$

B1: (a) $b + 5$

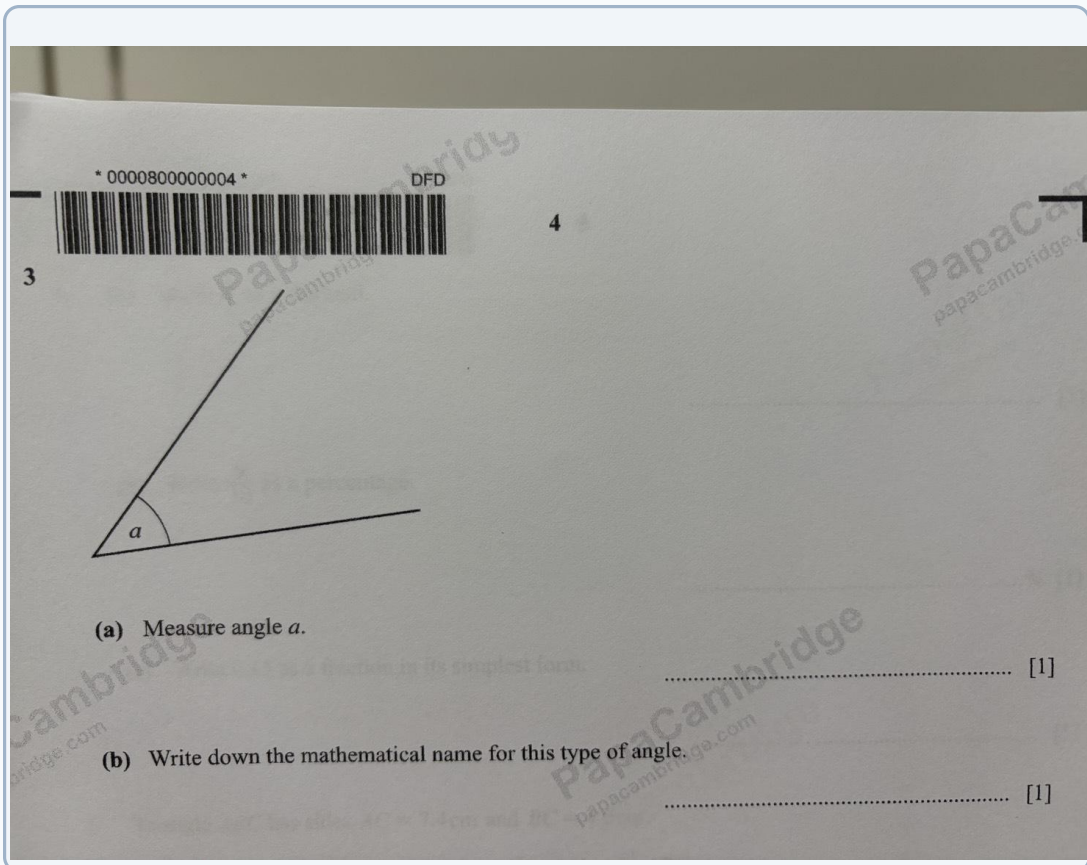
B1: (b) $65x$

Total: 2

3:45 PM

Aa

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : (a) Measure angle a.
..... [1] \n \n (b)
Write down the mathematical name for this type of
angle. [1]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

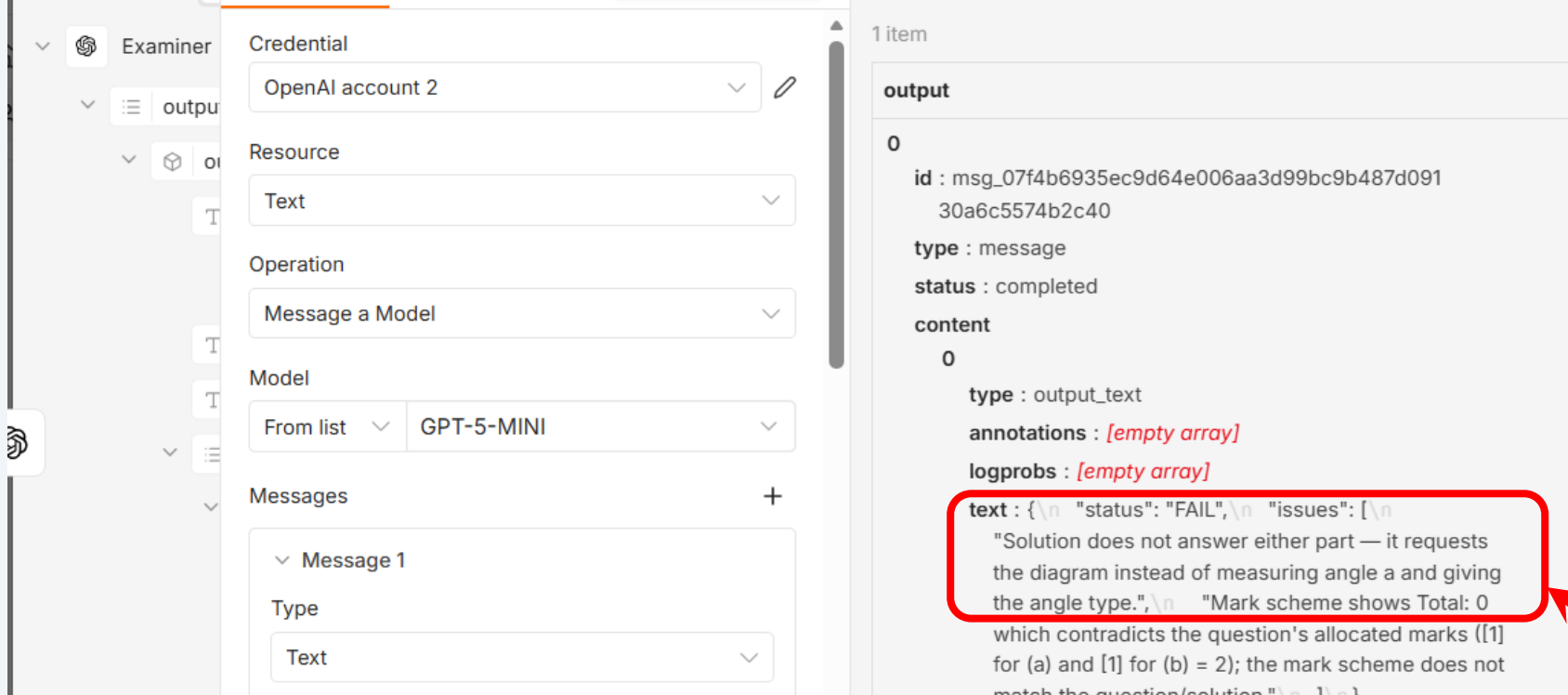
Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Failure Output – Moderator Node



The screenshot displays the OpenAI Studio interface for configuring a GPT-5-MINI model. The 'Parameters' tab is active, showing the following settings:

- Credential:** OpenAI account 2
- Resource:** Text
- Operation:** Message a Model
- Model:** From list (GPT-5-MINI)
- Messages:** A single message with Type 'Text' and Role 'System'.

The 'OUTPUT' tab shows the result of the step, which is a message with status 'completed' and content '0'. A red box highlights the 'text' field in the output, which contains a detailed explanation of the failure:

```
text : {\n  "status": "FAIL",\n  "issues": [\n    "Solution does not answer either part — it requests the diagram instead of measuring angle a and giving the angle type.",\n    "Mark scheme shows Total: 0 which contradicts the question's allocated marks ([1] for (a) and [1] for (b) = 2); the mark scheme does not match the question/solution."\n  ]\n}
```

Quality Gate Outcome

FAIL

FAIL

Failure Stage

Analyze Image

Analyze Image

Failure Detected by

Moderator

Moderator

LINE End-to-End Success

FAIL

FAIL

Final Answer Pass

N/A

N/A

Solution Completeness

N/A

N/A

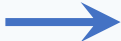
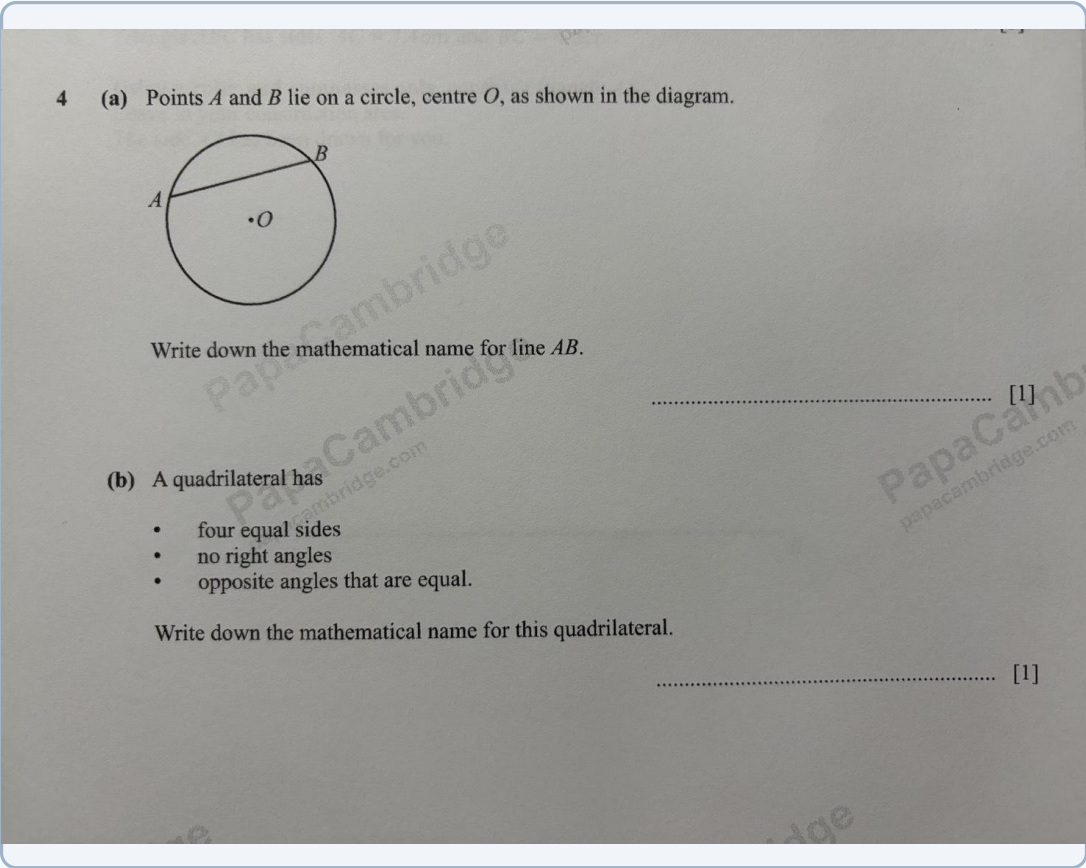
Assessment Alignment
N/A

N/A

REASON : The solution did not satisfy the task because the required diagram information was unavailable

Failure evidence rule: no delivered LINE output → Final Answer, Solution Completeness, Assessment Alignment and Sub-question Accuracy = N/A.

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 4 (a) Points A and B lie on a circle, centre O , as shown in the diagram.
Write down the mathematical name for line AB .
..... [1]
(b) A quadrilateral has
• four equal sides
• no right angles
• opposite angles that are equal.
Write down the mathematical name for this quadrilateral.
..... [1]

Image Quality

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MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_08e535fbec7e7190006aa3e031e08c87d0827a67ac59157708

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) Chord\n\n(b) Rhombus

role : assistant

Official Mark Scheme Answer

4(a)	chord	1
4(b)	rhombus	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Sub-question Final Answer Accuracy

100 %

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Geometry

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) Chord

(b) Rhombus

B1: (a) Chord

B1: (b) Rhombus

Total: 2

6:04 PM

Aa

Original Question Image

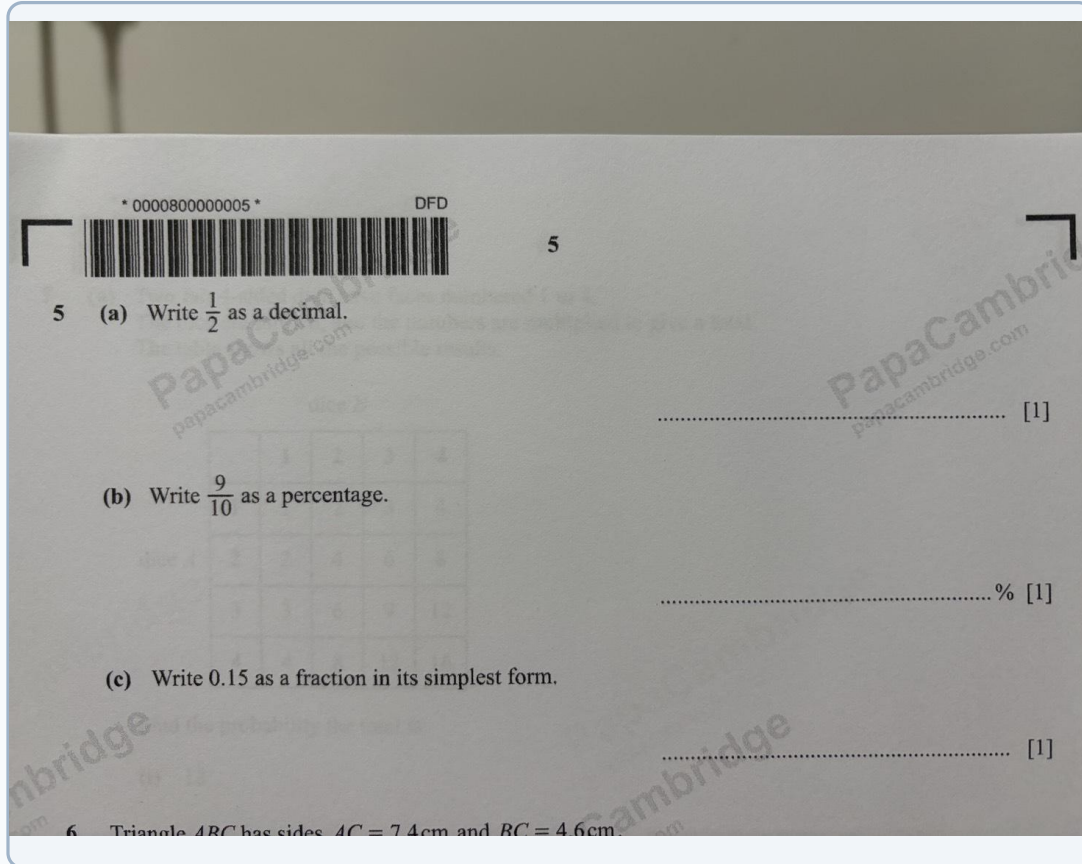


Image Quality

GOOD – All texts / symbols clearly readable

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POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 5 (a) Write $\frac{1}{2}$ as a decimal. \n[1] \n \n (b) Write $\frac{9}{10}$

as a

percentage. \n %

[1] \n \n (c) Write 0.15 as a fraction in its simplest form. \n

[1]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_09ddc7e886fba130006aa3ed2d565487d093222ab9fe81b661

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) 0.5\n\n(b) 90%\n\n(c) 3/20

role : assistant

Official Mark Scheme Answer

5(a)	[0].5	1
5(b)	90	1
5(c)	$\frac{3}{20}$ cao	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Sub-question Final Answer Accuracy

100 %

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Fractions, decimals and percentages

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) 0.5

(b) 90%

(c) 3/20

B1: (a) 0.5

B1: (b) 90%

B1: (c) 3/20

Total: 3

6:59 PM

Aa

Original Question Image

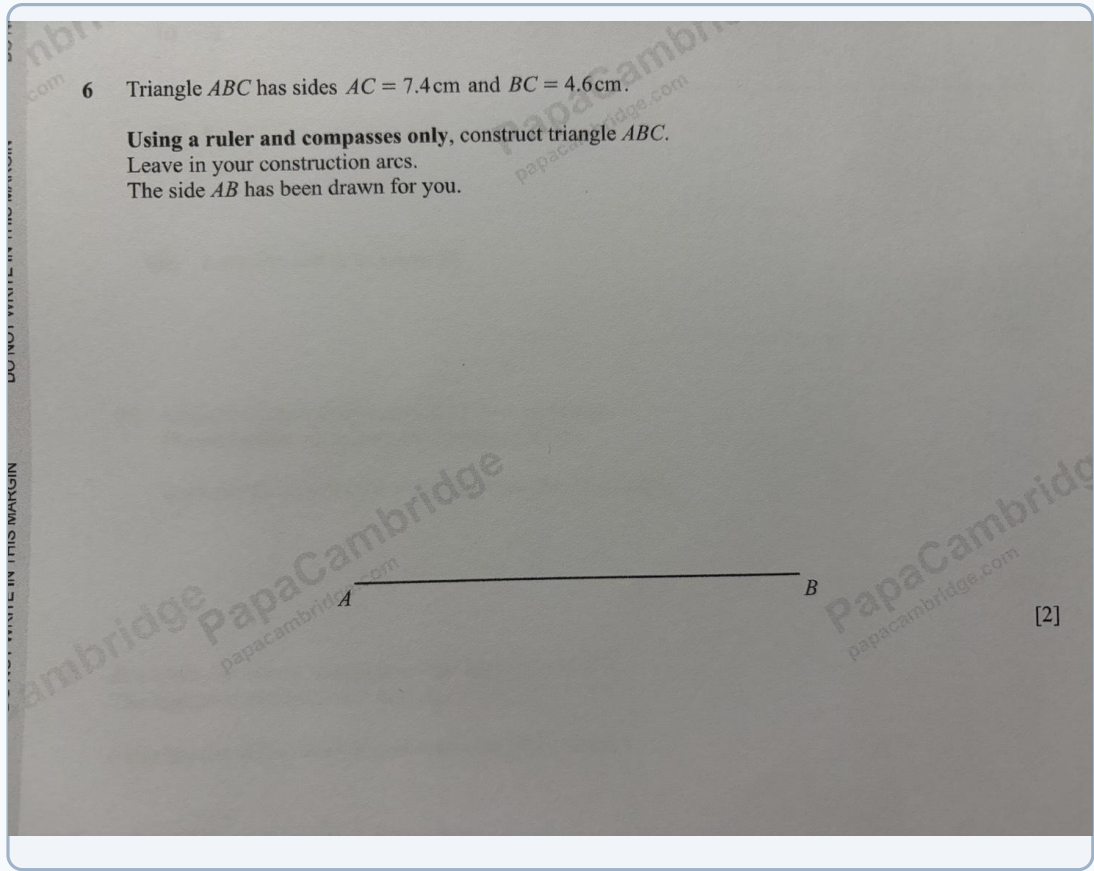


Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 6 Triangle ABC has sides $AC = 7.4\text{ cm}$ and $BC = 4.6\text{ cm}$.
Using a ruler and compasses only, construct triangle ABC .
Leave in your construction arcs.
The side AB has been drawn for you.

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0ecd7557f7d583a6006aa3f1f61b2487d0a5b501b5eac56c9e

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) Construction steps (ruler and compasses only; leave arcs visible):\n\n1) With the compass set to 7.4 cm, place the compass point at A and draw a large arc above the line AB.\n\n2) With the compass set to 4.6 cm, place the compass point at B and draw an arc to intersect the first arc. (There may be two intersections; choose either.)\n\n3) Mark an intersection of the arcs as C.\n\n4) Use a ruler to draw straight lines AC and BC.\n\n5) Leave all construction arcs. Label the triangle ABC.

role : assistant

Official Mark Scheme Answer

6	Correct triangle with correct arcs	2
B1 for correct triangle with incorrect or no arcs or for two correct arcs. If 0 scored, SC1 for accurate triangle with arcs but lines interchanged		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

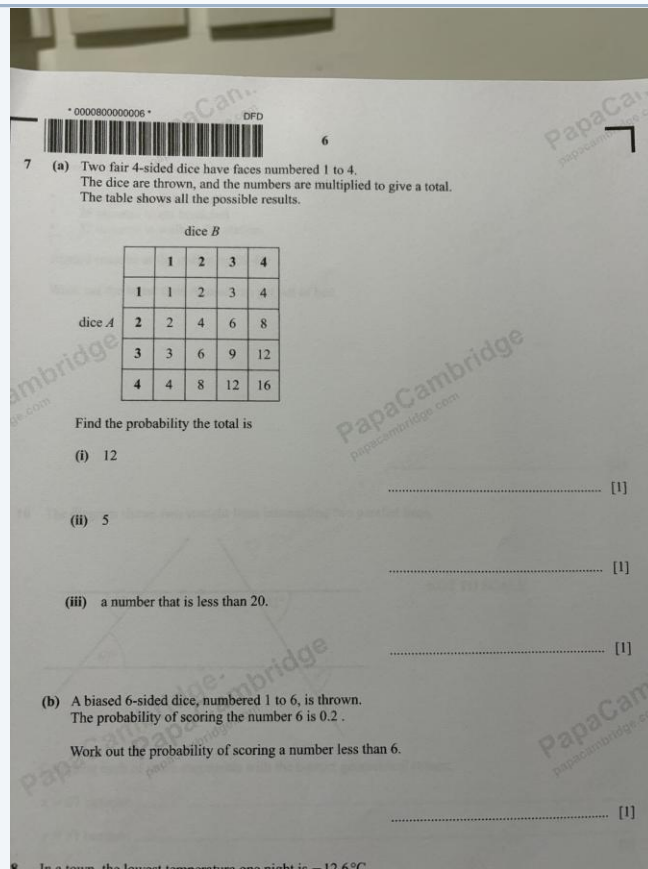
Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT ✓ ⓘ

1 item

1

content : 7 (a) Two fair 4-sided dice have faces numbered 1 to 4. \n The dice are thrown, and the numbers are multiplied to give a total. \n The table shows all the possible results. \n \n Find the probability the total is \n (i) 12 \n (ii) 5 \n (iii) a number that is less than 20. \n \n (b) A biased 6-sided dice, numbered 1 to 6, is thrown. \n The probability of scoring the number 6 is 0.2. \n \n Work out the probability of scoring a number less than 6.

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0166ca431a44a4a6006aa4252aada487d090fec3ad7d502565

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n\nStep 1:\nThere are $4 \times 4 = 16$ equally likely outcomes when two fair 4-sided dice are thrown.\n(ii)\nTotals of 12 occur for (3,4) and (4,3), so 2 outcomes.\nProbability = $2/16 = 1/8$ \n(ii)\nA total of 5 is impossible with numbers 1 to 4.\nProbability = $0/16 = 0$ \n(iii)\nAll products are at most 16, which is less than 20, so all 16 outcomes.\nProbability = $16/16 = 1$ \n(b)\nStep 1:\nP(number less than 6) = $1 - P(6) = 1 - 0.2 = 0.8$ \nFinal Answer:\n(a)(i) $1/8$ \n(ii) 0\n(iii) 1 \n(b) 0.8

role : assistant

Official Mark Scheme Answer

7(a)(i)	$\frac{1}{8}$ oe	1
7(a)(ii)	0	1
7(a)(iii)	1	1
7(b)	0.8	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Sub-question Final Answer Accuracy

100 %

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

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Success Layout – LINE Reply Message

Math Intellect

Topic: Probability
Difficulty: Easy
Assessment Objective: AO1

Solution

(a)

Step 1:
There are $4 \times 4 = 16$ equally likely outcomes when two fair 4-sided dice are thrown.

(i)
Totals of 12 occur for (3,4) and (4,3), so 2 outcomes.
Probability = $2/16 = 1/8$

(ii)
A total of 5 is impossible with numbers 1 to 4.
Probability = $0/16 = 0$

(iii)
All products are at most 16, which is less than 20, so all 16 outcomes.
Probability = $16/16 = 1$

(b)

Step 1:
 $P(\text{number less than 6}) = 1 - P(6) = 1 - 0.2 = 0.8$

Final Answer:
(a)(i) $1/8$ (ii) 0 (iii) 1
(b) 0.8

B1: (a)(i) $1/8$
B1: (a)(ii) 0
B1: (a)(iii) 1
B1: (b) 0.8

Total: 4

Math Intellect • Prototype Validation Evidence

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Original Question Image

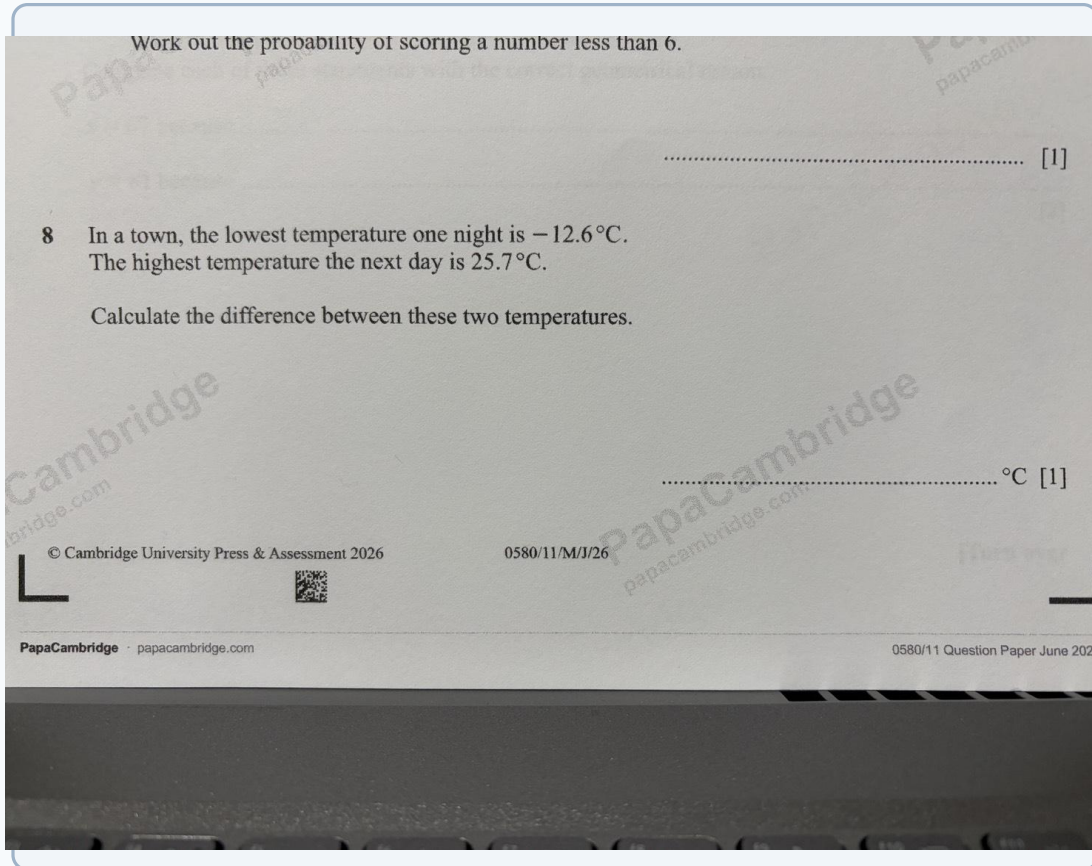


Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT ✓ ⓘ

1 item

1

content : 8 In a town, the lowest temperature one night is -12.6°C .
The highest temperature the next day is 25.7°C .
Calculate the difference between these two temperatures.
..... $^{\circ}\text{C}$ [1]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_045df707b19b77ea006aa4328e39bc87d0a0e173444a17911c

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\nDifference = 25.7 - (-12.6) = 25.7 + 12.6 = 38.3\n\nFinal Answer:\n38.3°C

role : assistant

Official Mark Scheme Answer

8	38.3	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Math Intellect • Prototype Validation Evidence

20

Original Question Image

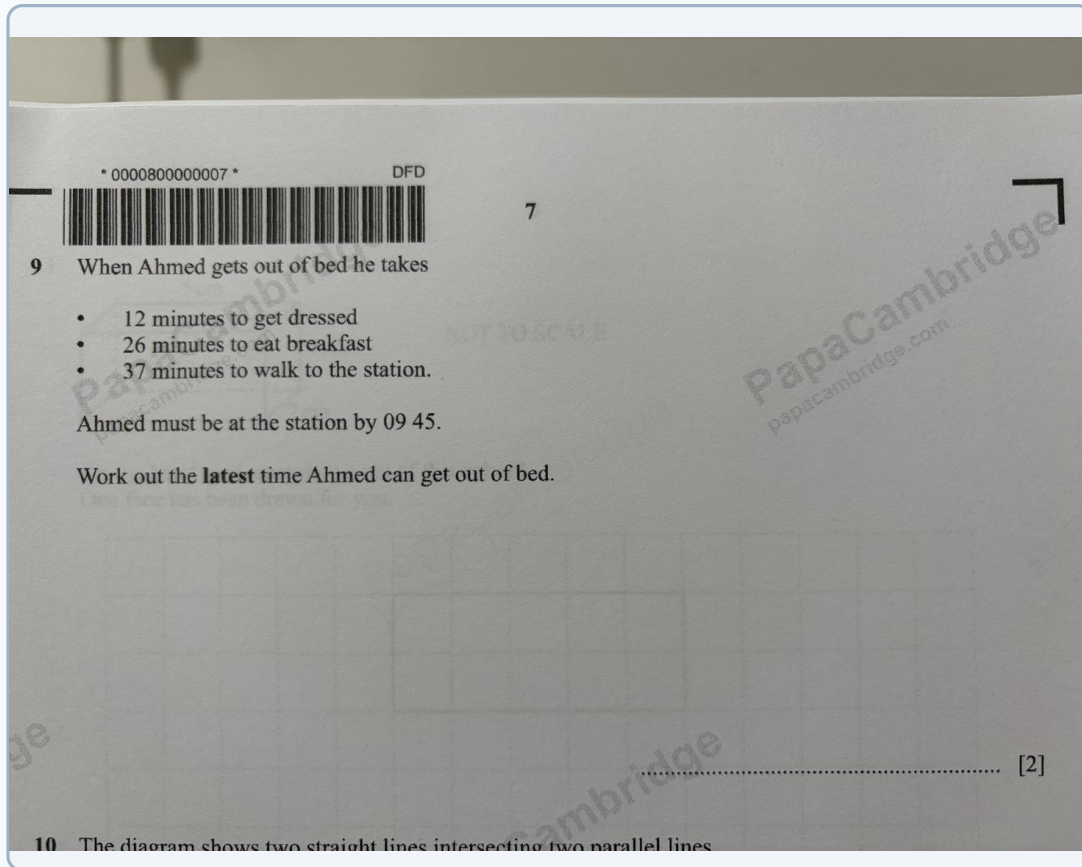


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 9 When Ahmed gets out of bed he takes \n \n • 12 minutes to get dressed \n • 26 minutes to eat breakfast \n • 37 minutes to walk to the station. \n \n Ahmed must be at the station by 09 45. \n \n Work out the latest time Ahmed can get out of bed.

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_074a12938db71656006aa43696d52c87d080185432924eec5a

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Total time needed = 12 + 26 + 37 = 75 minutes = 1 hour 15 minutes.\n \n Step 2:\n Latest time to get out of bed = 09 45 – 1 hour 15 minutes = 08 30.\n \n Final Answer:\n 08 30

role : assistant

Official Mark Scheme Answer

9	08 30 or 8 30am	2
M1 for 09 45 – 12 – 26 – 37 oe or B1 for 75 or 1[h] 15[m]		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Time
Difficulty: Easy
Assessment Objective: AO1

Solution

Step 1:
Total time needed = 12 + 26 + 37 = 75 minutes = 1 hour 15 minutes.

Step 2:
Latest time to get out of bed = 09 45 – 1 hour 15 minutes = 08 30.

Final Answer:
08 30

M1: Adds 12 + 26 + 37 = 75 minutes (1 hour 15 minutes)
A1: Gives latest wake-up time as 08:30

Total: 2

12:13 AM

Math Intellect • Prototype Validation Evidence

22

Original Question Image

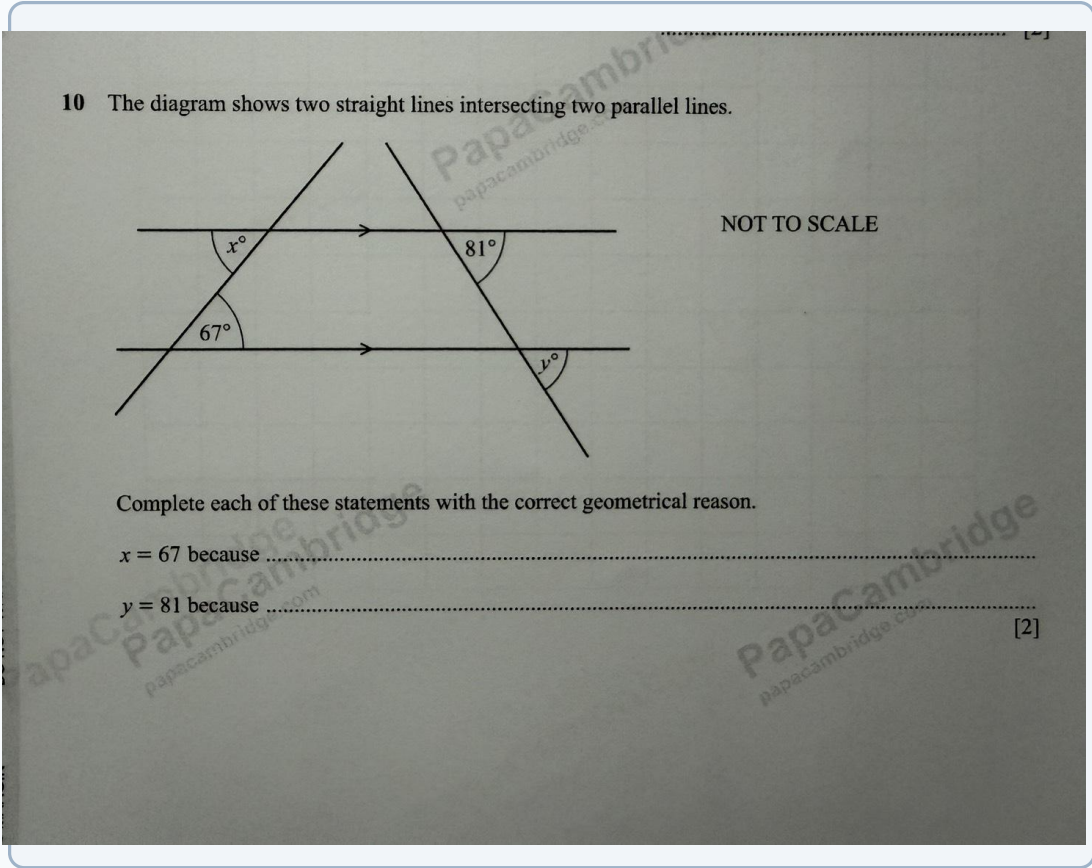


Image Quality

GOOD – All texts / symbols clearly readable
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 POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT

i

1 item

1

content

10 The diagram shows two straight lines intersecting two parallel lines.

NOT TO SCALE

Complete each of these statements with the correct geometrical reason.

$x = 67$ because

$y = 81$ because

[2]

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_024bfb078f309764006aa4402adaac87d099c46aa1f573c934

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : x = 67 because alternate angles are equal in parallel lines.\ny = 81 because corresponding angles are equal in parallel lines.

role : assistant

Official Mark Scheme Answer

10	Alternate angles are equal	2
	Corresponding angles are equal	
B1 for each		

LINE End-to-End

PASSFAIL

Final Answer

PASSFAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Parallel lines

Difficulty: Easy

Assessment Objective: AO1

Solution

x = 67 because alternate angles are equal in parallel lines.

y = 81 because corresponding angles are equal in parallel lines.

B1: x = 67 because alternate angles are equal in parallel lines.

B1: y = 81 because corresponding angles are equal in parallel lines.

Total: 2

12:54 AM



Aa



Original Question Image

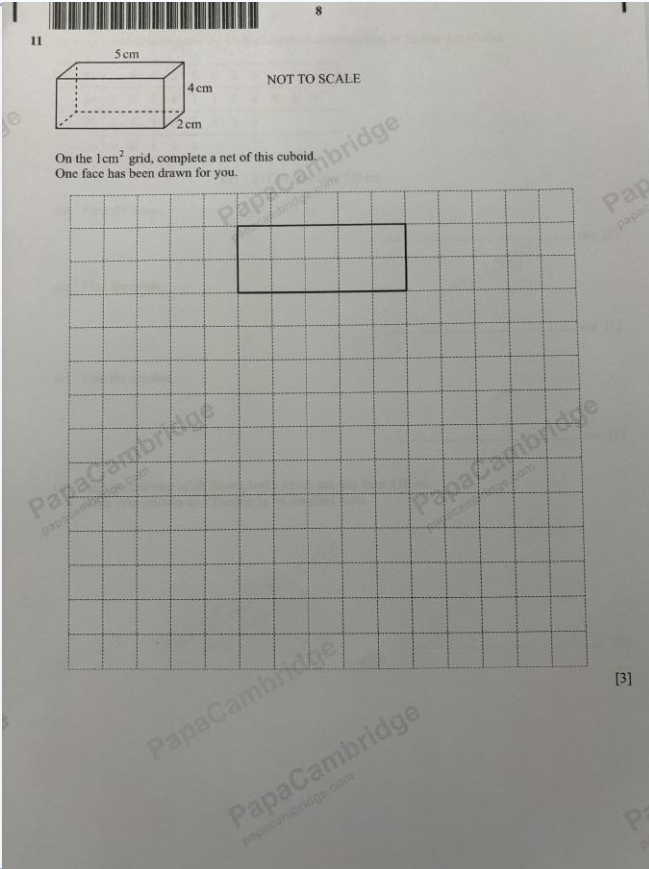


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT ✓ ⓘ

1 item

1

content : 11
On the 1 cm^2 grid, complete a net of this cuboid.
One face has been drawn for you.

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Drawing instructions to complete the net:\n\n - Identify the side lengths of the given rectangular face by counting grid squares along its two edges.\n\n - Identify the third dimension (the remaining edge length of the cuboid) by counting squares from the diagram/information for this cuboid.\n\n - From each of the four sides of the given face, draw a rectangle that shares the full side and has depth equal to the third dimension:\n\n - Along each side of the given face, construct a rectangle whose edge on the shared side is exactly the same length as that side, and whose other side equals the third dimension (counted in grid squares).\n\n - This gives four side faces attached around the original face.\n\n - Draw the sixth face (congruent to the original face) attached to the outer edge of any one of these four side rectangles, choosing a side where there is space so that it does not overlap any other part of the net.\n\n - Ensure all shared edges match in length on the grid and that no faces overlap.

role : assistant

Official Mark Scheme Answer

11	Fully correct net	3
B2 for 3 correct extra faces in the correct places B1 for 1 correct extra face in the correct place		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Math Intellect

Topic: 3D shapes

Difficulty: Easy

Assessment Objective: AO2

Solution

Drawing instructions to complete the net:

- Identify the side lengths of the given rectangular face by counting grid squares along its two edges.

- Identify the third dimension (the remaining edge length of the cuboid) by counting squares from the diagram/information for this cuboid.

- From each of the four sides of the given face, draw a rectangle that shares the full side and has depth equal to the third dimension:

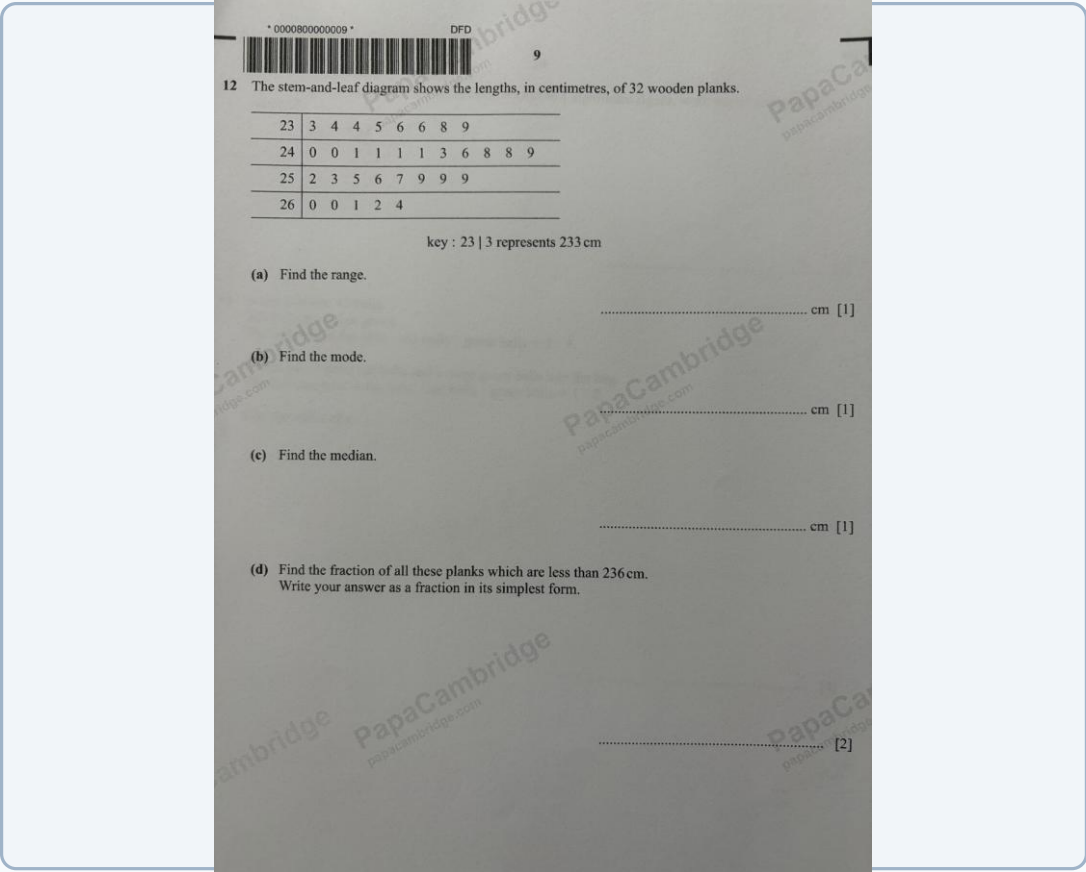
- Along each side of the given face, construct a rectangle whose edge on the shared side is exactly the same length as that side, and whose other side equals the third dimension (counted in grid squares).

- This gives four side faces attached around the original face.

- Draw the sixth face (congruent to the original face) attached to the outer edge of any one of these four side rectangles, choosing a side where there is space so that it does not overlap any other part of the net.

- Ensure all shared edges match in length on the grid and that no faces overlap.

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 12 The stem-and-leaf diagram shows the lengths, in centimetres, of 32 wooden planks.

$$\begin{array}{l} 23 \text{ \& } 3 \text{ \& } 4 \text{ \& } 4 \text{ \& } 5 \text{ \& } 6 \text{ \& } 6 \text{ \& } 8 \text{ \& } 9 \\ 24 \text{ \& } 0 \text{ \& } 1 \text{ \& } 1 \text{ \& } 1 \text{ \& } 3 \text{ \& } 6 \text{ \& } 8 \text{ \& } 8 \text{ \& } 9 \\ 25 \text{ \& } 2 \text{ \& } 3 \text{ \& } 5 \text{ \& } 6 \text{ \& } 7 \text{ \& } 9 \text{ \& } 9 \text{ \& } 9 \\ 26 \text{ \& } 0 \text{ \& } 1 \text{ \& } 2 \text{ \& } 4 \end{array}$$

key : 23 | 3 represents 233 cm

(a) Find the range.

..... cm [1]

(b) Find the mode.

..... cm [1]

(c) Find the median.

..... cm [1]

(d) Find the fraction of all these planks which are less than 236 cm. Write your answer as a fraction in its simplest form.

..... [2]



Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : missing data information while retrieving from original image

Failure Output – Moderator Node

Parameters Settings **Execute step**

Credential
OpenAI account 2

Resource
Text

Operation
Message a Model

Model
From list GPT-5-MINI

Messages
+
Message 1
Type: Text
Role: System

OUTPUT 1 item

output

0

id : msg_09f6e817729c7e16006aa502c59e7887d0892
caf34a10cf6d4

type : message
status : completed
content

0

type : output_text
annotations : [empty array]
logprobs : [empty array]

text : {\n "status": "FAIL",\n "issues": [\n "Inconsistency between question and data: the question states there are 32 planks, but the stem-and-leaf diagram contains 30 values. The solution and mark scheme use 30 values throughout (median taken as average of 15th and 16th, fraction out of 30). This discrepancy should be resolved (either correct the stated total to 30 or supply the missing two data values)."\n]\n }

role : assistant

Quality Gate Outcome

FAIL

Failure Stage

Analyze Image

Failure Detected by

Moderator

LINE End-to-End Success

FAIL

Final Answer Pass

N/A

Solution Completeness

N/A

Assessment Alignment

N/A

REASON : The solution did not satisfy the task because the required diagram information was unavailable.

Failure evidence rule: no delivered LINE output → Final Answer, Solution Completeness, Assessment Alignment and Sub-question Accuracy = N/A.

Original Question Image

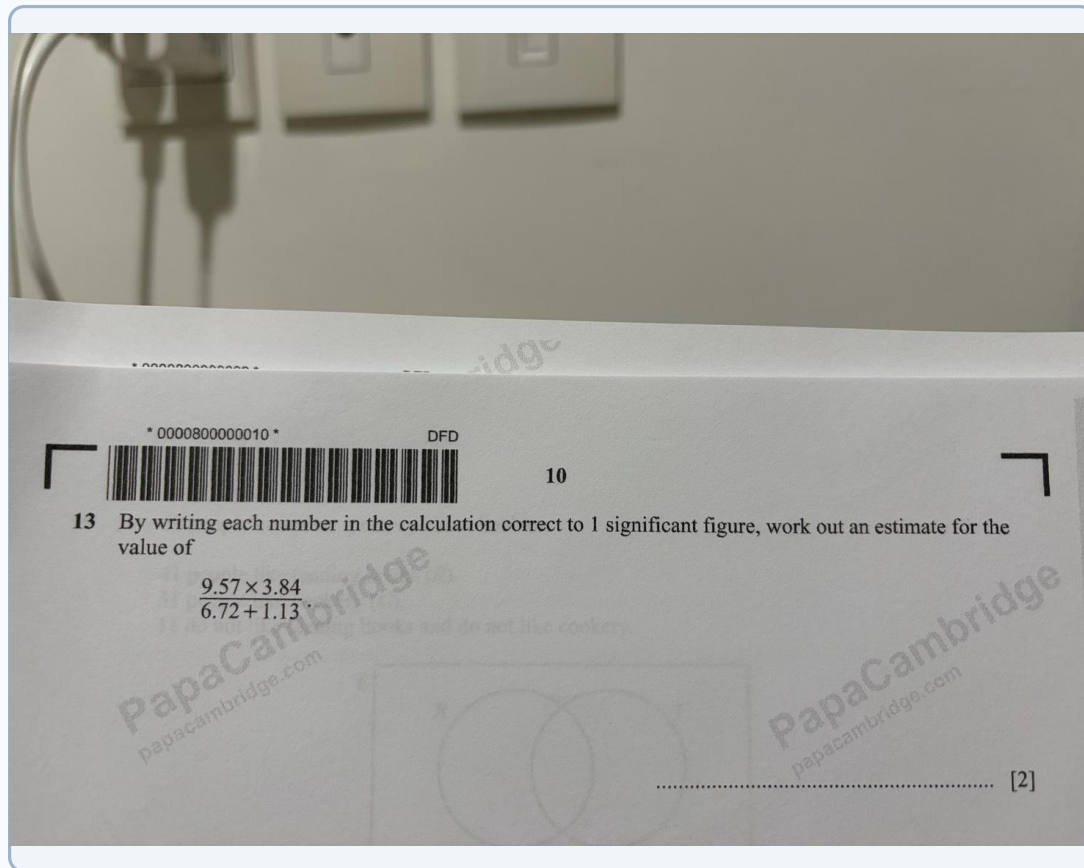


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 13 By writing each number in the calculation correct to 1 significant figure, work out an estimate for the value of $\frac{9.57 \times 3.84}{6.72 + 1.13}$

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0e2e9517b335315a006aa50750436887d0bd2939acf76151de

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Round each number to 1 significant figure:\n - 9.57 ≈ 10\n - 3.84 ≈ 4\n - 6.72 ≈ 7\n - 1.13 ≈ 1\n\n Step 2:\n Estimate:\n (10 × 4) / (7 + 1) = 40 / 8 = 5\n\n Final Answer:\n 5

role : assistant

Official Mark Scheme Answer

13	$\frac{10 \times 4}{7 + 1}$	M1
	5 final answer	A1

If 0 scored SC1 for 3 correct roundings or for all correct but with any trailing zeros

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Number

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Round each number to 1 significant figure:

- 9.57 ≈ 10

- 3.84 ≈ 4

- 6.72 ≈ 7

- 1.13 ≈ 1

Step 2:

Estimate:

(10 × 4) / (7 + 1) = 40 / 8 = 5

Final Answer:

5

M1: Rounds all numbers correctly to 1 s.f. 9.57→10, 3.84→4, 6.72→7, 1.13→1

A1: Correctly evaluates (10×4)/(7+1) = 5

Total: 2

3:03 PM

Original Question Image

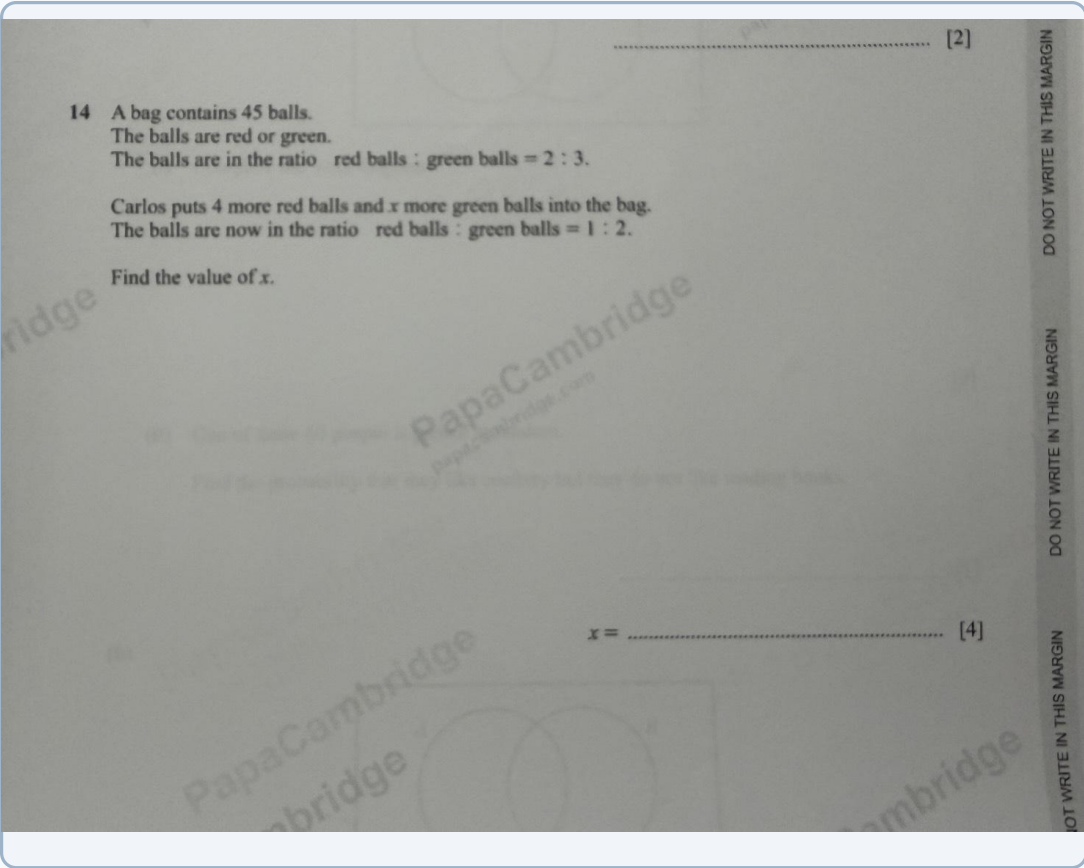


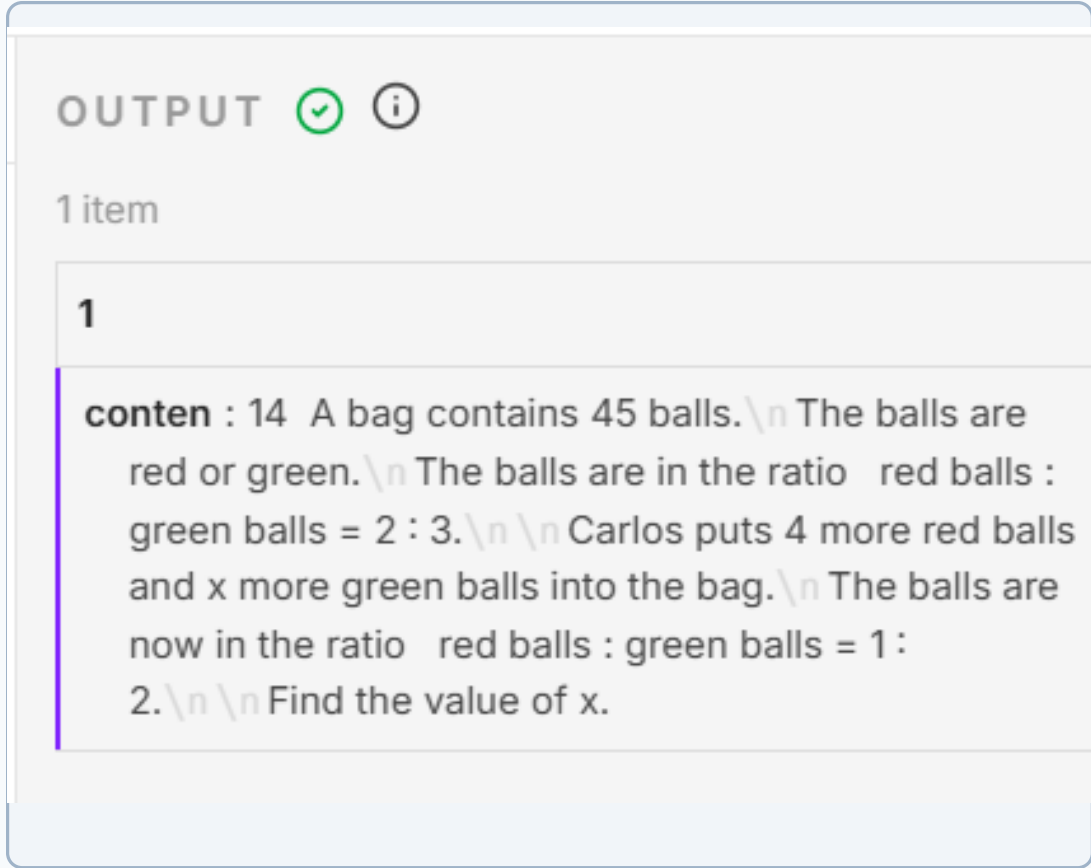
Image Quality	
GOOD	All texts / symbols clearly readable
MODERATE	minor blur / skew but all essential information recoverable
POOR	visible degradation affecting text / symbol yet question remains testable

MODERATE – minor blur / skew but all essential information recoverable

MODERATE – minor blur / skew but all essential information recoverable

POOR– visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node



Interpretation

Interpretation	Count
PASS	1
FAIL	0

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0435577e2eec915b006aa50ba8402887d09e09d53a17e509de

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\nInitial ratio red:green = 2:3 with 45 balls.\nTotal parts = 5 ⇒ 1 part = 45 ÷ 5 = 9.\nRed = 2 × 9 = 18, Green = 3 × 9 = 27.\n\nStep 2:\nAfter adding, Red = 18 + 4 = 22, Green = 27 + x.\nGiven new ratio red:green = 1:2 ⇒ Green = 2 × Red.\n27 + x = 2 × 22 = 44 ⇒ x = 44 - 27 = 17.\n\nFinal Answer:\n17

role : assistant

Official Mark Scheme Answer

14	17	4
B3 for [green =] 44 seen OR B2 for [red =] 18 and [green =] 27 seen OR M1 for $\frac{45}{2+3} \times k$ where $k = 1, 2$ or 3 seen M1 for $(their\ 18 + 4) \times 2$		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Ratio and proportion

Difficulty: Easy

Assessment Objective: AO2

Solution

Step 1:
Initial ratio red:green = 2:3 with 45 balls.
Total parts = 5 ⇒ 1 part = 45 ÷ 5 = 9.
Red = 2 × 9 = 18, Green = 3 × 9 = 27.

Step 2:
After adding, Red = 18 + 4 = 22, Green = 27 + x.
Given new ratio red:green = 1:2 ⇒ Green = 2 × Red.
27 + x = 2 × 22 = 44 ⇒ x = 44 - 27 = 17.

Final Answer:
17

M1: Divides 45 into 5 parts, obtains 1 part = 9
A1: Finds red = 18 and green = 27
M1: Forms equation 27 + x = 2 × (18 + 4)
A1: Solves equation to get x = 17

Total: 4

3:22 PM

Original Question Image

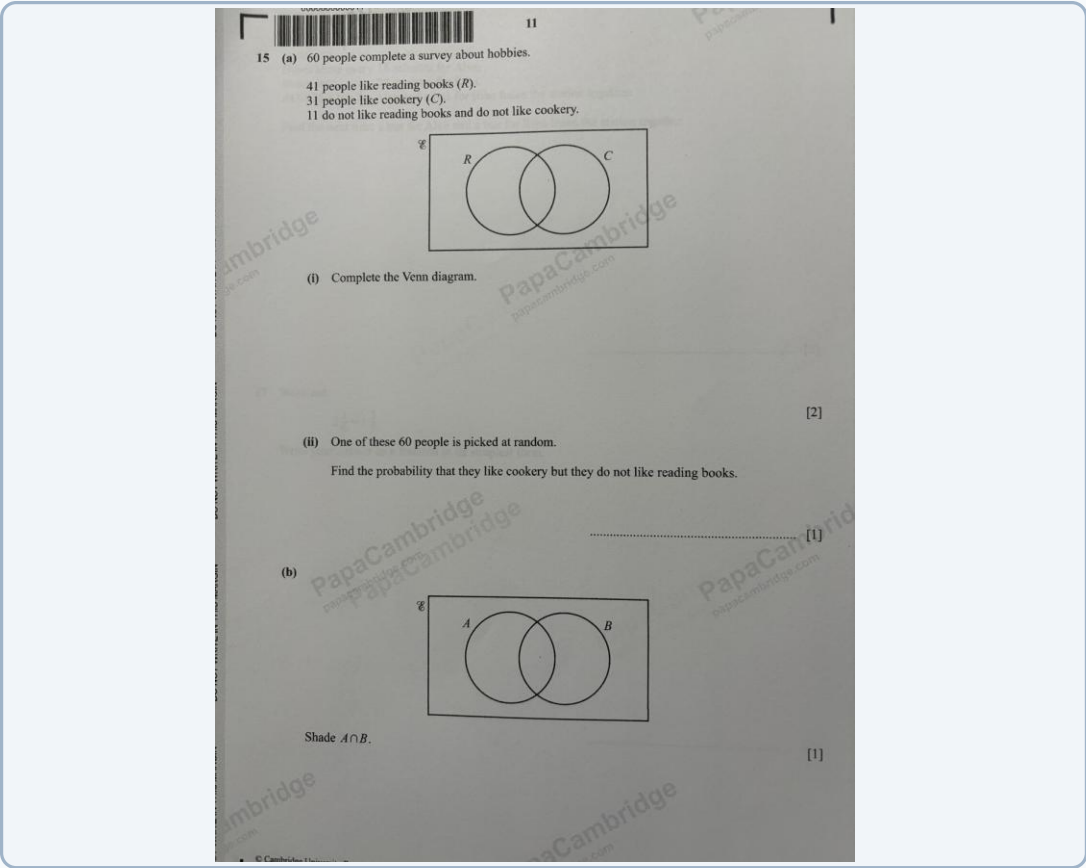


Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 15 (a) 60 people complete a survey about hobbies. \n \n 41 people like reading books (R). \n 31 people like cookery (C). \n 11 do not like reading books and do not like cookery. \n \n (i) Complete the Venn diagram. \n \n (ii) One of these 60 people is picked at random. \n Find the probability that they like cookery but they do not like reading books. \n [1] \n \n (b) \n Shade \ (A \cap B \). [1]

Interpretation

PASS FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_014b3200b6a50405006aa51268493087d0a7759f72f5ef9c1c

type : message

status : completed

content

0

type : output_text

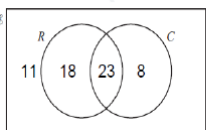
annotations : [empty array]

logprobs : [empty array]

text : (a)(i)\n Step 1:\n Number who like at least one = 60 - 11 = 49.\n\n Step 2:\n R \cap C = 41 + 31 - 49 = 23.\n\n Step 3:\n R only = 41 - 23 = 18.\n C only = 31 - 23 = 8.\n Neither (outside both) = 11.\n\n Final placement in the Venn diagram:\n - R only: 18\n - R \cap C: 23\n - C only: 8\n - Outside both: 11\n\n (a)(ii)\n Step 1:\n Cookery but not reading = C only = 8.\n\n Step 2:\n Probability = 8/60 = 2/15.\n\n Final Answer:\n 2/15\n\n (b)\n Shade only the region common to both sets A and B (the overlapping lens-shaped intersection A \cap B). Do not shade the parts that are only in A, only in B, or outside both.

role : assistant

Official Mark Scheme Answer

15(a)(i)		2
15(a)(ii)	$\frac{8}{60}$ oe	1
15(b)		1

FT $\frac{their\ 8}{60}$ provided $0 < their\ 8 < 31$

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

1

Topic: Venn diagrams and probability
Difficulty: Easy
Assessment Objective: AO1

Solution

(a)(i)
Step 1:
Number who like at least one = 60 - 11 = 49.

Step 2:
 $R \cap C = 41 + 31 - 49 = 23$.

Step 3:
 R only = 41 - 23 = 18.
 C only = 31 - 23 = 8.
Neither (outside both) = 11.

Final placement in the Venn diagram:
- R only: 18
- $R \cap C$: 23
- C only: 8
- Outside both: 11

(a)(ii)
Step 1:
Cookery but not reading = C only = 8.

Step 2:
Probability = $8/60 = 2/15$.

Final Answer:
2/15

(b)
Shade only the region common to both sets A and B (the overlapping lens-shaped intersection $A \cap B$). Do not shade the parts that are only in A, only in B, or outside both.



Original Question Image

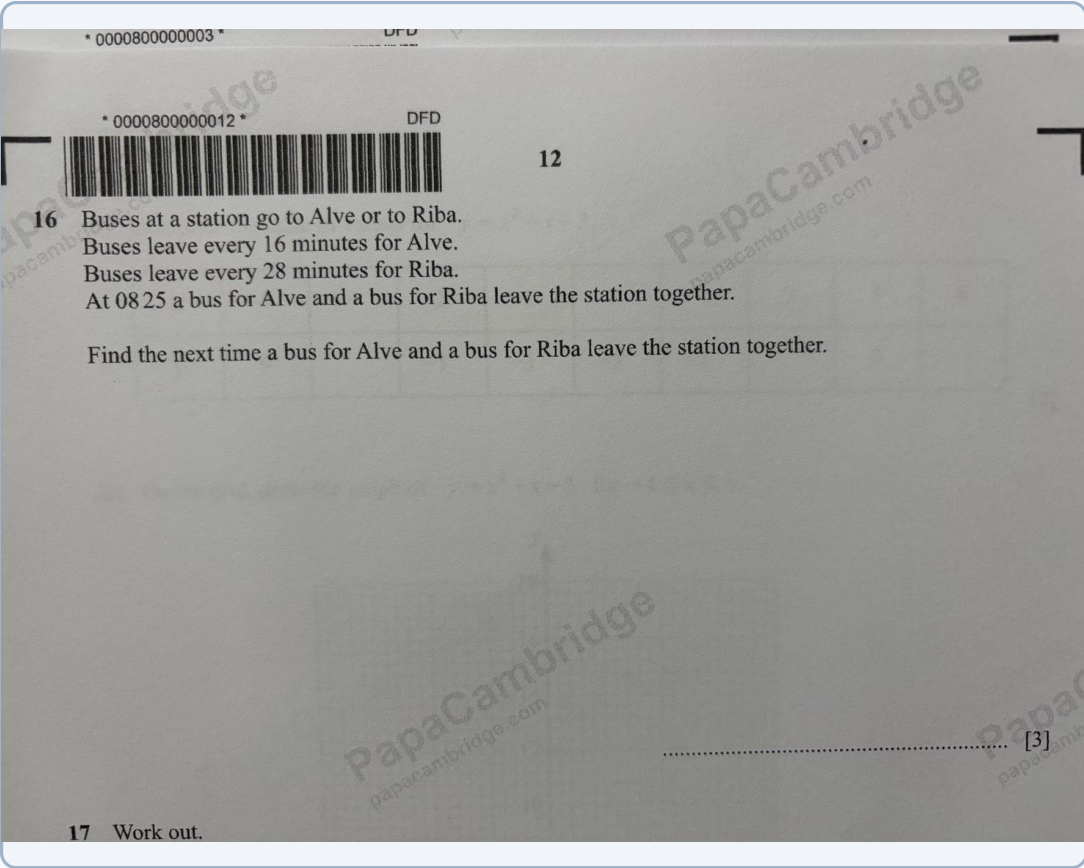


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 16 Buses at a station go to Alve or to Riba.
Buses leave every 16 minutes for Alve.
Buses leave every 28 minutes for Riba.
At 08 25 a bus for Alve and a bus for Riba leave the station together.
Find the next time a bus for Alve and a bus for Riba leave the station together.

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0d4b38d447fe2988006aa5173e8efc87d084176c3b21243b32

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Find LCM(16, 28).\n 16 = 2^4, 28 = 2^2 x 7\n LCM = 2^4 x 7 = 16 x 7 = 112 minutes\n \n Step 2:\n 112 minutes = 1 hour 52 minutes\n \n Step 3:\n 08 25 + 1 h 52 min = 10 17\n \n Final Answer:\n 10 17

role : assistant

Official Mark Scheme Answer

16	10 17	3
----	-------	---

B2 for 112 or 1h 52min

or **M1** for 112k oe

or for [16 =] 2 × 2 × 2 × 2 or [28 =] 2 × 2 × 7

or two correct factor trees/tables for both 16 and 28

OR

M2 for listing times/multiples for both 16 and 28 to at least 10 17 or 112[min]

or **M1** for listing at least 3 of each or one full list to 10 17 or 112[min].

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Number

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Find LCM(16, 28).

16 = 2^4, 28 = 2^2 × 7

LCM = 2^4 × 7 = 16 × 7 = 112 minutes

Step 2:

112 minutes = 1 hour 52 minutes

Step 3:

08 25 + 1 h 52 min = 10 17

Final Answer:

10 17

M1: Uses LCM of 16 and 28 to find common interval

A1: Correct LCM = 112 minutes

A1: Converts 112 minutes to 1 hour 52 minutes

A1: Adds 1h52 to 08:25 correctly to get 10:17

Total: 4

4:11 PM

Original Question Image

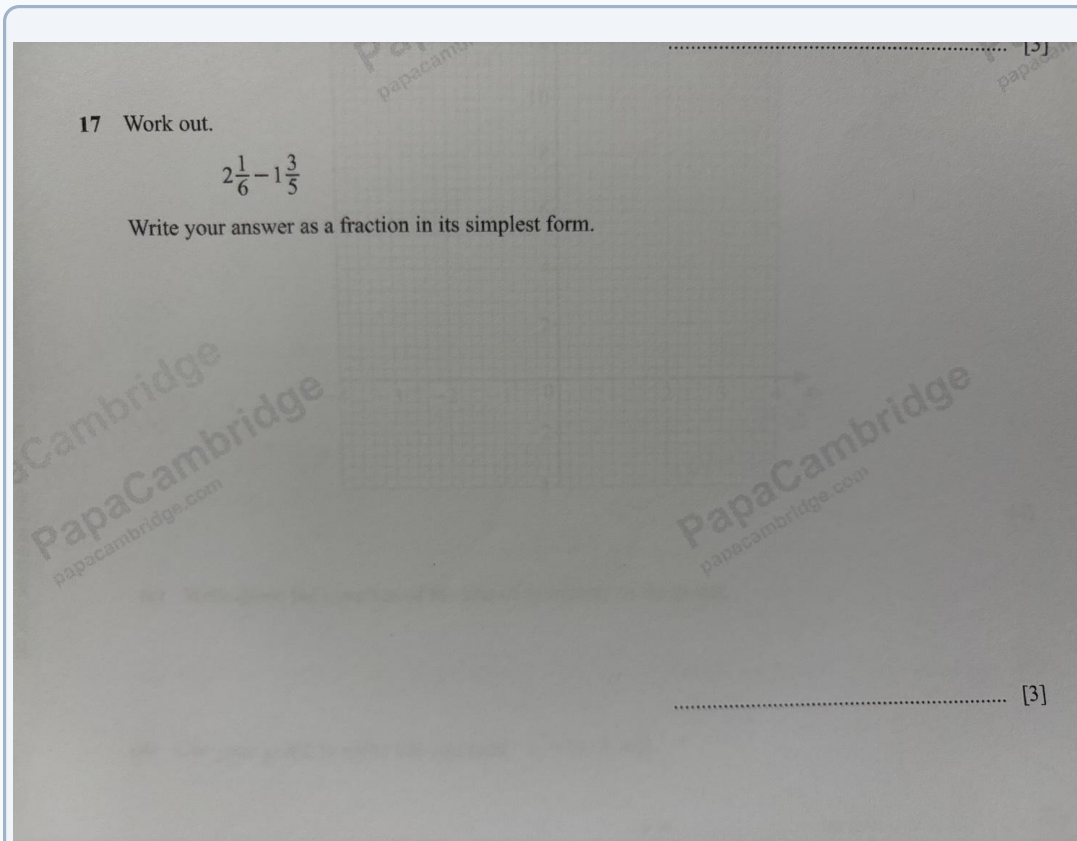


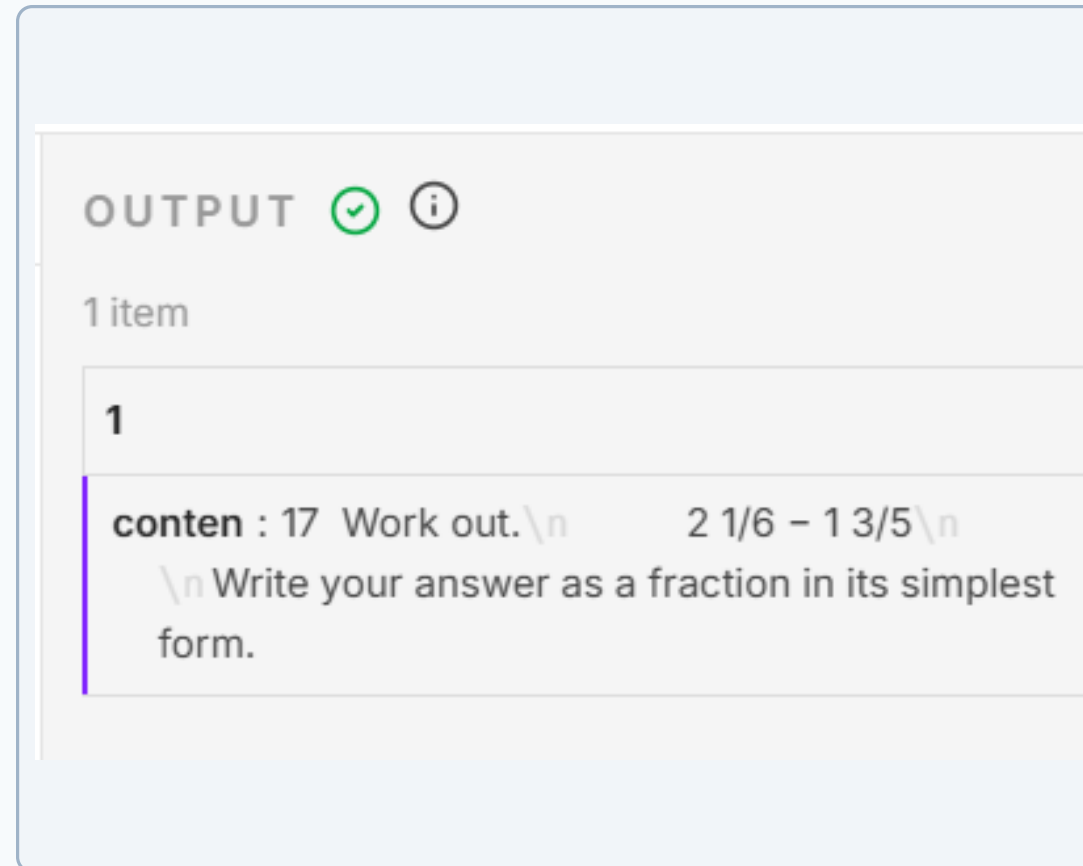
Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node



Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0097e51e4ca0e6e2006aa51a6d7b9c87d0a4c7a38e8f11a364

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Convert to improper fractions.\n 2 1/6 = 13/6, 1 3/5 = 8/5\n \n Step 2:\n Find a common denominator and subtract.\n 13/6 - 8/5 = 65/30 - 48/30 = 17/30\n \n Final Answer:\n 17/30

role : assistant

Official Mark Scheme Answer

17	$\frac{17}{30}$ cao final answer	3
----	----------------------------------	---

B1 for $\frac{13}{6}$ and $\frac{8}{5}$ or $1 + \frac{1}{6} - \frac{3}{5}$

M1 for $\frac{13 \times 5}{6 \times 5}$ and $\frac{8 \times 6}{5 \times 6}$ or better

or $[2] \frac{5[\times 1]}{6 \times 5}$ and $[1] \frac{3 \times 6}{5 \times 6}$

or for $\frac{their 13 \times 5}{6 \times 5}$ and $\frac{their 8 \times 6}{5 \times 6}$

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Fractions

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Convert to improper fractions.

2 1/6 = 13/6, 1 3/5 = 8/5

Step 2:

Find a common denominator and subtract.

13/6 - 8/5 = 65/30 - 48/30 = 17/30

Final Answer:

17/30

M1: Converts 2 1/6 and 1 3/5 to 13/6 and 8/5

A1: Subtracts to obtain 17/30 as final simplest fraction

Total: 2

4:25 PM

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Math Intellect • Prototype Validation Evidence

38

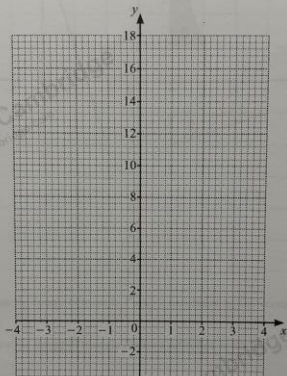
Original Question Image

18 (a) Complete the table of values for $y = x^2 + x - 3$.

x	-4	-3	-2	-1	0	1	2	3	4
y	9		-1	-3	-3	-1		9	

[2]

(b) On the grid, draw the graph of $y = x^2 + x - 3$ for $-4 \leq x \leq 4$.



[4]

(c) Write down the equation of the line of symmetry of the graph.

..... [1]

(d) Use your graph to solve the equation $x^2 + x - 3 = 0$.

x = or x = [2]

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Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 18 (a) Complete the table of values for $y = x^2 + x - 3$.
x | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
y | 9 | -1 | -3 | -3 | -1 | 3 | 9 |
(b) On the grid, draw the graph of $y = x^2 + x - 3$ for $-4 \leq x \leq 4$.
(c) Write down the equation of the line of symmetry of the graph.
.....
.....
(d) Use your graph to solve the equation $x^2 + x - 3 = 0$.
x =
or x =

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n Step 1:\n Calculate $y = x^2 + x - 3$ for each x .\n \n For example:\n $x = -4$: $y = 16 - 4 - 3 = 9$ \n $x = -3$: $y = 9 - 3 - 3 = 3$ \n $x = -2$: $y = 4 - 2 - 3 = -1$ \n $x = -1$: $y = 1 - 1 - 3 = -3$ \n $x = 0$: $y = 0 + 0 - 3 = -3$ \n $x = 1$: $y = 1 + 1 - 3 = -1$ \n $x = 2$: $y = 4 + 2 - 3 = 3$ \n $x = 3$: $y = 9 + 3 - 3 = 9$ \n $x = 4$: $y = 16 + 4 - 3 = 17$ \n \n Completed table (y-values in order for $x = -4, -3, -2, -1, 0, 1, 2, 3, 4$):
 $9, 3, -1, -3, -3, -1, 3, 9, 17$ \n \n (b)\n On the grid, plot the points:
 $(-4, 9), (-3, 3), (-2, -1), (-1, -3), (0, -3), (1, -1), (2, 3), (3, 9), (4, 17)$ \n \n Then draw a smooth U-shaped curve through the points for $-4 \leq x \leq 4$.\n \n (c)\n $x = -1/2$ \n \n (d)\n $x \approx -2.30$ or $x \approx 1.30$ \n \n Final Answer:
(a) $9, 3, -1, -3, -3, -1, 3, 9, 17$ (b) Plot the listed points and draw a smooth curve for $-4 \leq x \leq 4$ (c) $x = -1/2$ (d) $x \approx -2.30$ or $x \approx 1.30$

role : assistant

Official Mark Scheme Answer

18(a)	3 3 17	2
18(b)	Correct curve	4
18(c)	$x = -\frac{1}{2}$ oe	1
18(d)	-2.4 to -2.2 and 1.2 to 1.4	2

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Math Intellect

Difficulty: Easy

Assessment Objective: AO1

Solution

(a)

Step 1:

Calculate $y = x^2 + x - 3$ for each x .

For example:

$x = -4$: $y = 16 - 4 - 3 = 9$
 $x = -3$: $y = 9 - 3 - 3 = 3$
 $x = -2$: $y = 4 - 2 - 3 = -1$
 $x = -1$: $y = 1 - 1 - 3 = -3$
 $x = 0$: $y = 0 + 0 - 3 = -3$
 $x = 1$: $y = 1 + 1 - 3 = -1$
 $x = 2$: $y = 4 + 2 - 3 = 3$
 $x = 3$: $y = 9 + 3 - 3 = 9$
 $x = 4$: $y = 16 + 4 - 3 = 17$

Completed table (y-values in order for $x = -4, -3, -2, -1, 0, 1, 2, 3, 4$):
 $9, 3, -1, -3, -3, -1, 3, 9, 17$

(b)

On the grid, plot the points:
 $(-4, 9), (-3, 3), (-2, -1), (-1, -3), (0, -3), (1, -1), (2, 3), (3, 9), (4, 17)$

Then draw a smooth U-shaped curve through the points for $-4 \leq x \leq 4$.

(c)

$x = -1/2$

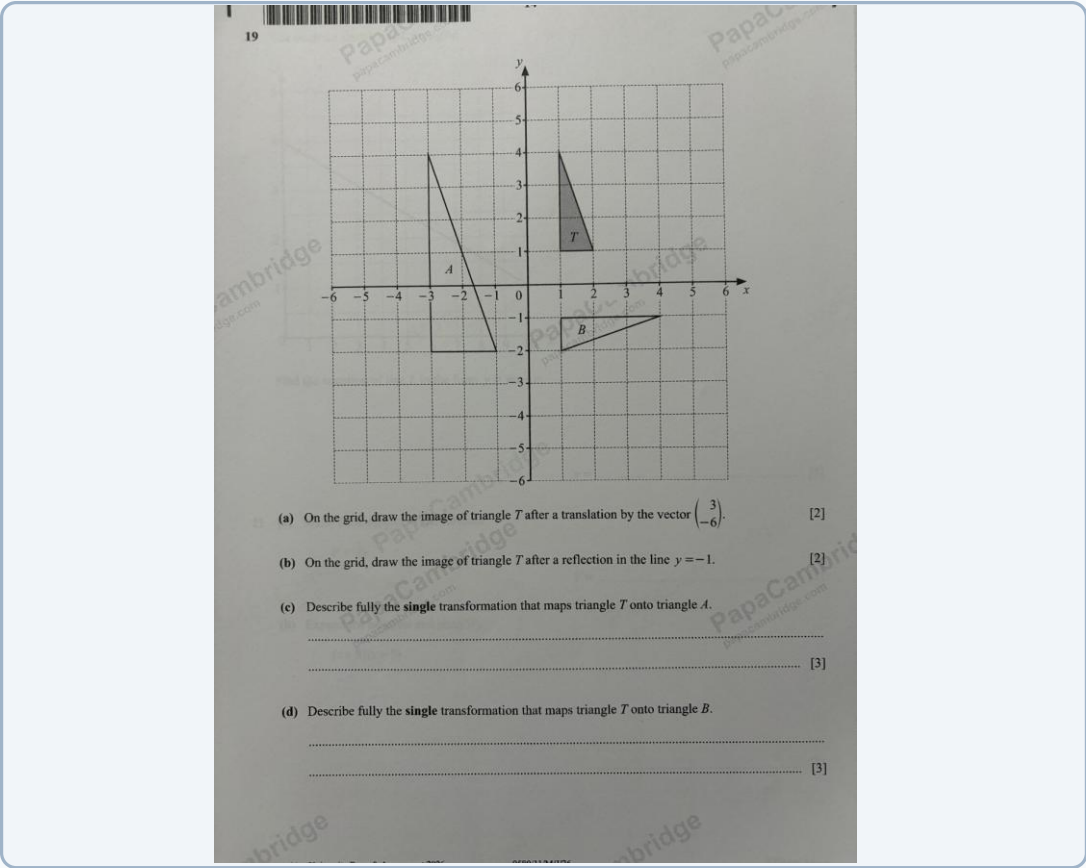
(d)

$x \approx -2.30$ or $x \approx 1.30$

Final Answer:

(a) $9, 3, -1, -3, -3, -1, 3, 9, 17$
(b) Plot the listed points and draw a smooth curve for $-4 \leq x \leq 4$
(c) $x = -1/2$
(d) $x \approx -2.30$ or $x \approx 1.30$

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT ☒ ⓘ

1 item

1

content : (a) On the grid, draw the image of triangle T after a translation by the vector $\begin{pmatrix} 3 \\ -6 \end{pmatrix}$. [2] (b) On the grid, draw the image of triangle T after a reflection in the line $y = -1$. [2] (c) Describe fully the single transformation that maps triangle T onto triangle A. [3] (d) Describe fully the single transformation that maps triangle T onto triangle B. [3]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS FAIL

REASON : missing data information while retrieving from original image

Failure Output – Moderator Node

INPUT

🔍

Examiner

output

on

T

T

...

...

Parameters

Settings

Execute step

Credential

OpenAI account 2

Resource

Text

Operation

Message a Model

Model

From list

GPT-5-MINI

Messages

> Message 1

> Message 2

+ Add Message

Simplify Output

OUTPUT

✓

i

1 item

output

0

id : msg_090d30805b04f669006aa53ba4791487d08757301ce03b7064

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : {\n "status": "FAIL",\n "issues": [\n "Parts (c) and (d) are not answered — missing the required descriptions of the single transformations mapping triangle T onto triangle A and onto triangle B.",\n "The provided mark scheme only covers parts (a) and (b); no scheme/marks are given for (c) and (d), so matching of mark scheme to those parts cannot be checked."\n]\n }

role : assistant

FAIL

Analyze Image

Moderator

FAIL

N/A

N/A

N/A

REASON : Solver could not complete the task because required diagram information was unavailable.

Failure evidence rule: no delivered LINE output → Final Answer, Solution Completeness, Assessment Alignment and Sub-question Accuracy = N/A.

Original Question Image

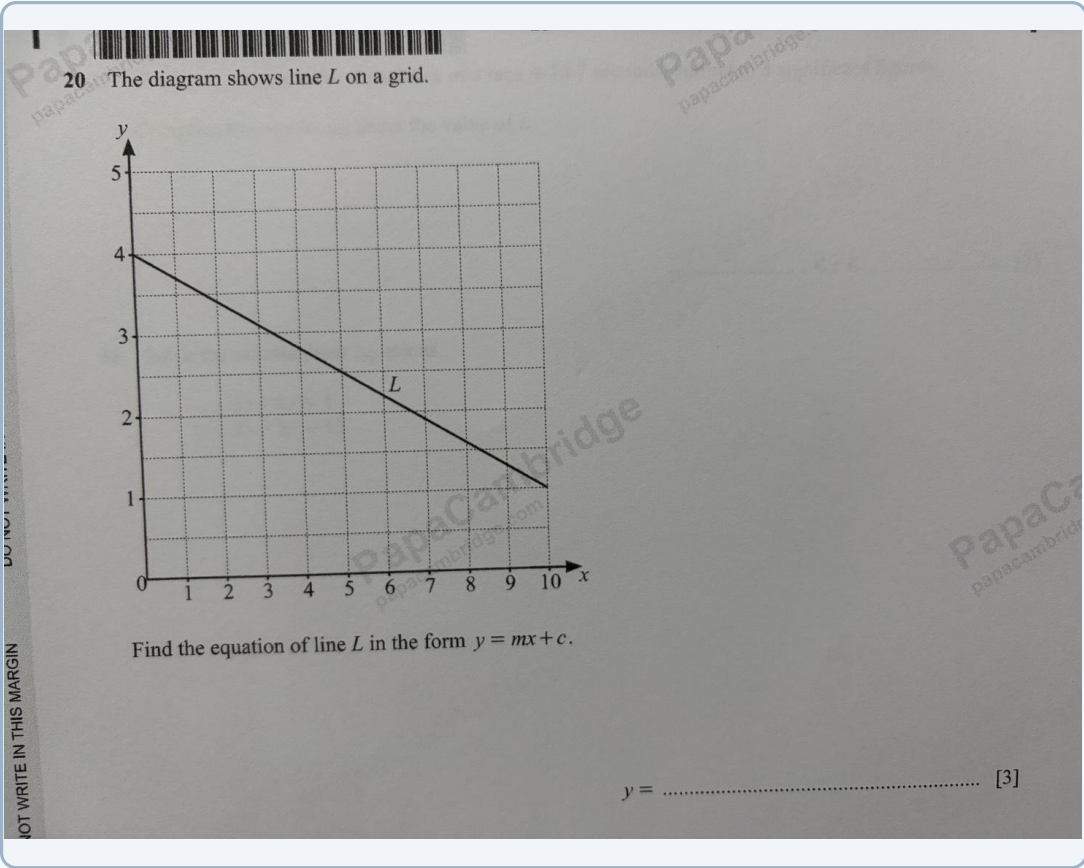


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 20 The diagram shows line L on a grid.
Find the equation of line L in the form $y = mx + c$.
 $y = \dots\dots\dots [3]$

Interpretation

PASS FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0e464755c8366709006aa53f625d3487d0a70a1ed5a5263a18

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Choose two lattice points on line L from the grid, for example (0, 2) and (2, 0).\n \n Step 2:\n Find the gradient.\n m = (0 - 2) / (2 - 0) = -2 / 2 = -1\n \n Step 3:\n Use the y-intercept from the graph. The line crosses the y-axis at 2, so c = 2.\n \n Final Answer:\n y = -x + 2

role : assistant

Official Mark Scheme Answer

20	$[y =] - 0.3x + 4$ oe	3
B2 for $-0.3x + c$ oe gradient or M1 for a correct $\frac{\text{rise}}{\text{run}}$ e.g. $\frac{1.5}{-5}$ or -0.3 oe B1 for $mx + 4, m \neq 0$		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Straight line graphs

Difficulty: Easy

Assessment Objective: AO2

Solution

Step 1:

Choose two lattice points on line L from the grid, for example (0, 2) and (2, 0).

Step 2:

Find the gradient.

$m = (0 - 2) / (2 - 0) = -2 / 2 = -1$

Step 3:

Use the y-intercept from the graph. The line crosses the y-axis at 2, so $c = 2$.

Final Answer:

$y = -x + 2$

M1: Uses two lattice points and gradient formula $(y_2 - y_1)/(x_2 - x_1)$

A1: Calculates gradient $m = -1$

B1: Gives correct equation $y = -x + 2$

Total: 3

7:03 PM

Math Intellect • Prototype Validation Evidence

44

Original Question Image

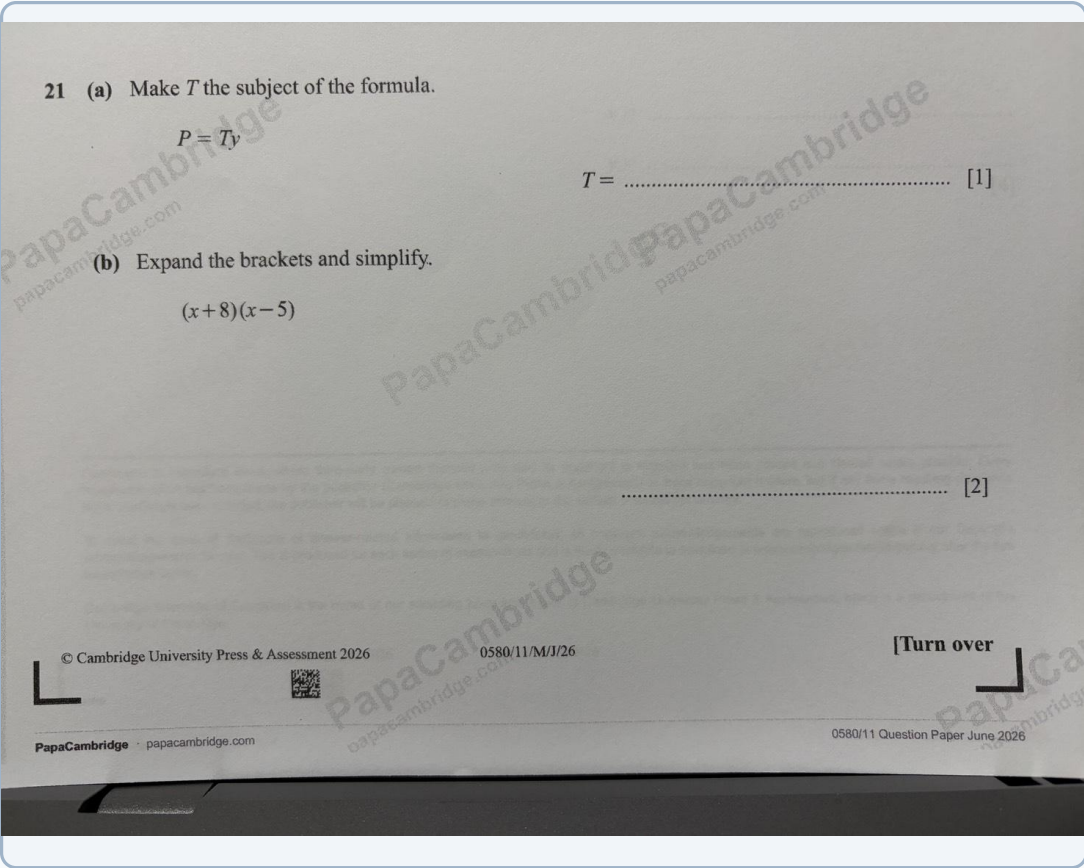


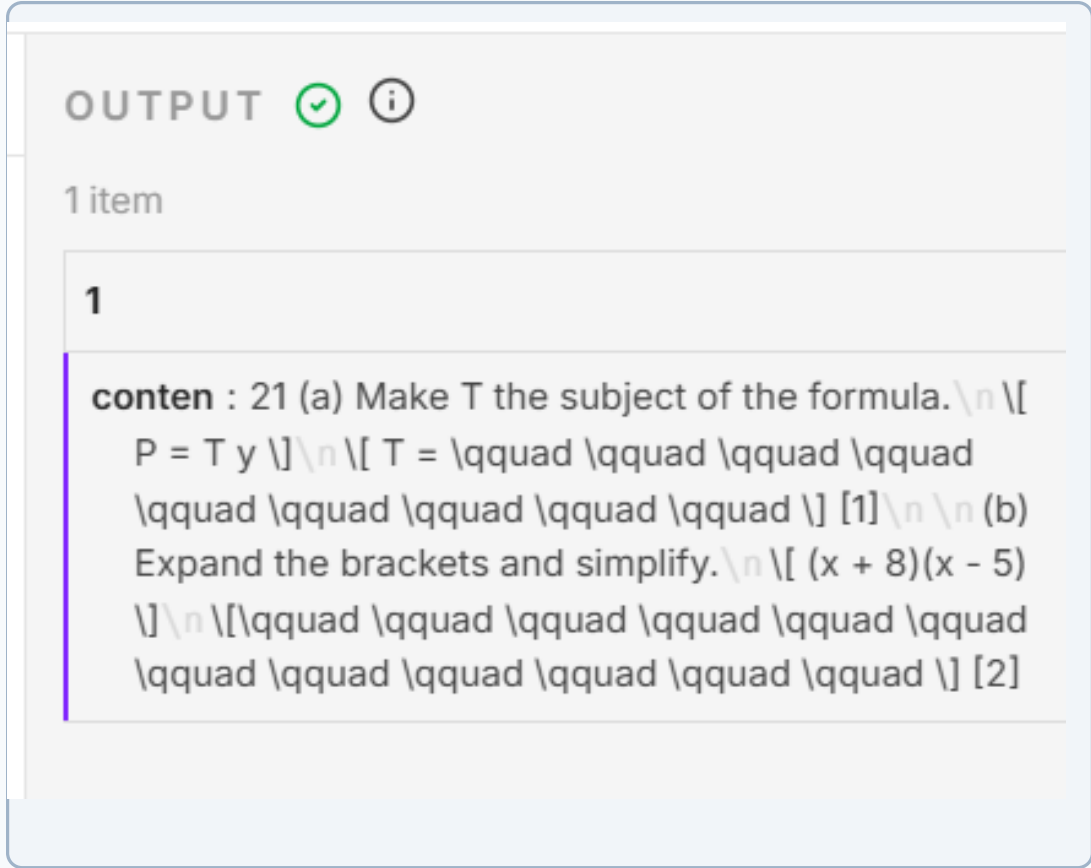
Image Quality
GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR– visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node



Interpretation

Interpretation	Score
PASS	70
FAIL	30

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0d1ef4d129869775006aa5431ecf8887d0b9e67a4f3236175d

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n T = P/y\n\n(b)\n Step 1:\n x(x - 5) + 8(x - 5) = x^2 - 5x + 8x - 40\n\n Step 2:\n x^2 + 3x - 40

role : assistant

Official Mark Scheme Answer

21(a)	$[T=] \frac{P}{y}$ final answer	1
21(b)	$x^2 + 3x - 40$	2
B1 for $x^2 + 8x - 5x - 40$ with three out of four terms correct		

LINE End-to-End

PASSFAIL

Final Answer

PASSFAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Algebra

Difficulty: Easy

Assessment Objective: AO1

Solution

(a)

T = P/y

(b)

Step 1:

$x(x - 5) + 8(x - 5) = x^2 - 5x + 8x - 40$

Step 2:

$x^2 + 3x - 40$

B1: (a) T = P/y

M1: Expands as $x(x-5)+8(x-5)=x^2-5x+8x-40$

A1: (b) Simplifies to $x^2 + 3x - 40$

Total: 3

7:18 PM

Math Intellect • Prototype Validation Evidence

46

Original Question Image

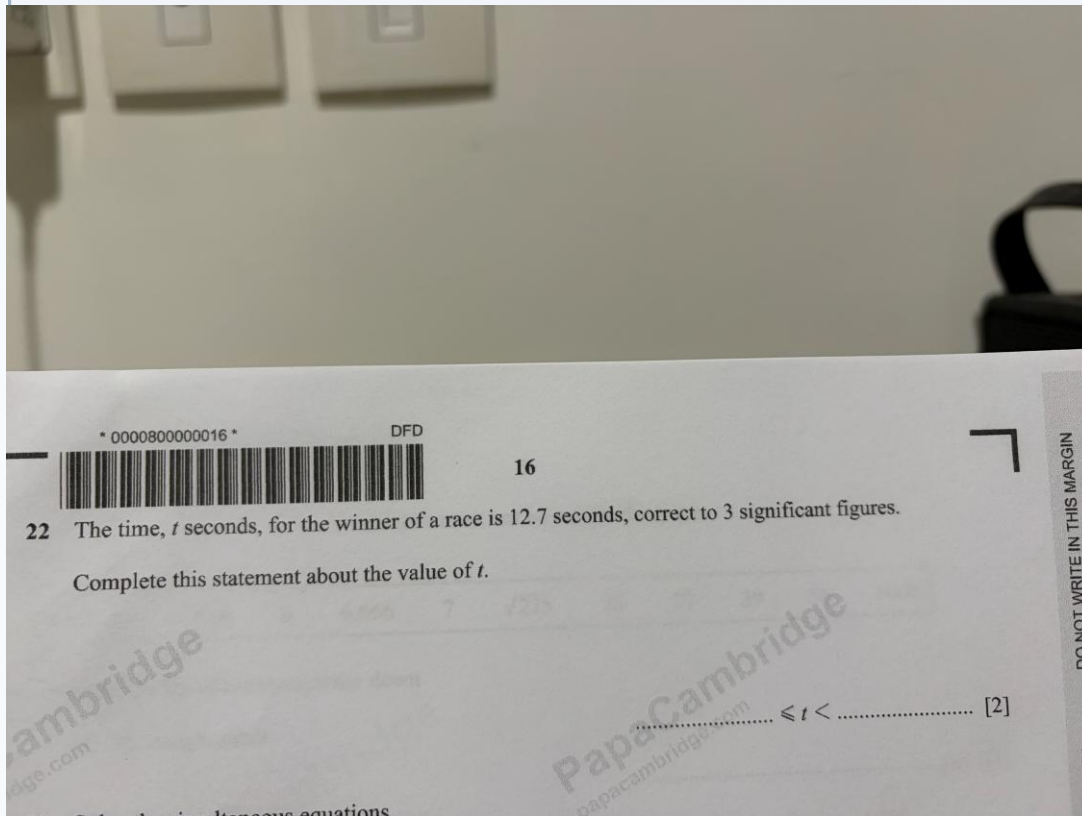


Image Quality

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POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 22 The time, t seconds, for the winner of a race is 12.7 seconds, correct to 3 significant figures. $\backslash n \backslash n$ Complete this statement about the value of t . $\backslash n \backslash n$ $< t <$ [2]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0309b0f27f702101006aa54ab4be3087d081ffe812bdfddc06

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : 12.65 < t < 12.75

role : assistant

Official Mark Scheme Answer

22	12.65 12.75	2
----	----------------	---

B1 for each

If 0 scored, SC1 for both correct but reversed

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Accuracy and bounds

Difficulty: Easy

Assessment Objective: AO1

Solution

12.65 < t < 12.75

B1: Lower bound 12.65 earns one mark

B1: Upper bound 12.75 earns one mark

Total: 2

7:51 PM

Aa

Original Question Image

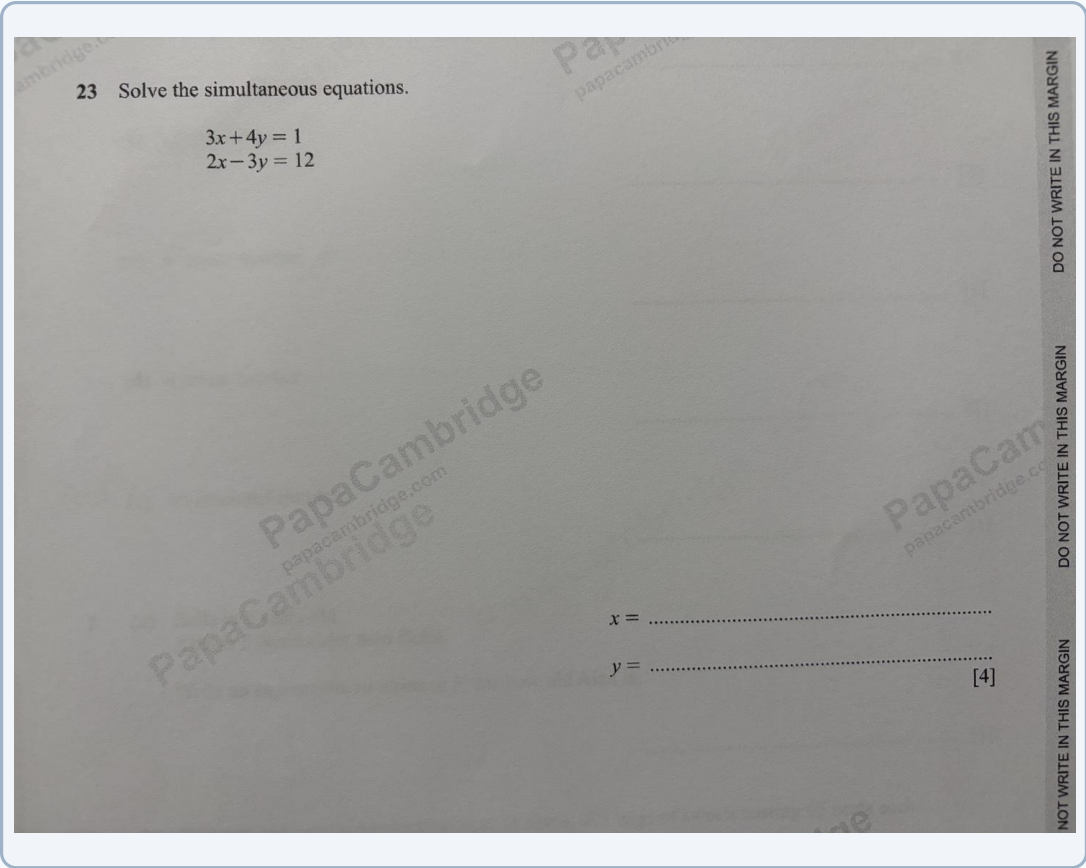


Image Quality

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POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 23 Solve the simultaneous equations. \n \n

3x + 4y = 1 \n 2x - 3y = 12 \n \n x =

..... \n y =

..... [4]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0ab8b40f9608ef1e006aa54cfc6a2c87d0a39a1921032e21c3

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:
 $3x + 4y = 1$
 $2x - 3y = 12$
Multiply the first equation by 3 and the second by 4:
 $9x + 12y = 3$
 $8x - 12y = 48$
Step 2:
Add the two equations:
 $(9x + 12y) + (8x - 12y) = 3 + 48$
 $17x = 51$
 $x = 3$
Step 3:
Substitute $x = 3$ into $3x + 4y = 1$
 $3(3) + 4y = 1$
 $9 + 4y = 1$
 $4y = -8$
 $y = -2$
Final Answer:
 $x = 3$
 $y = -2$

role : assistant

Official Mark Scheme Answer

23	$[x =] 3$ $[y =] -2$ nfw	4
----	-------------------------------	---

M1 for correctly equating one set of coefficients
M1 for correct method to eliminate one variable

OR

M1 for making x or y the subject of one equation
M1 for substitution of x or y from *their* rearranged equation

A1 for $x = 3$
A1 for $y = -2$

If at least one M mark awarded followed by A0, A0, then also give **SC1** for 2 values satisfying one of the original equations

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

8:00PM

8

Topic: Simultaneous equations

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

$3x + 4y = 1$
 $2x - 3y = 12$

Multiply the first equation by 3 and the second by 4:

$9x + 12y = 3$
 $8x - 12y = 48$

Step 2:

Add the two equations:

$(9x + 12y) + (8x - 12y) = 3 + 48$
 $17x = 51$
 $x = 3$

Step 3:

Substitute $x = 3$ into $3x + 4y = 1$:

$3(3) + 4y = 1$
 $9 + 4y = 1$
 $4y = -8$
 $y = -2$

Final Answer:

$x = 3$
 $y = -2$

M1: Eliminates y to obtain $17x = 51$.
A1: $x = 3$.
M1: Substitutes $x = 3$ into $3x + 4y = 1$ to get $4y = -8$.
A1: $y = -2$.

Total: 4

8:01PM

Math Intellect • Prototype Validation Evidence

50

Original Question Image

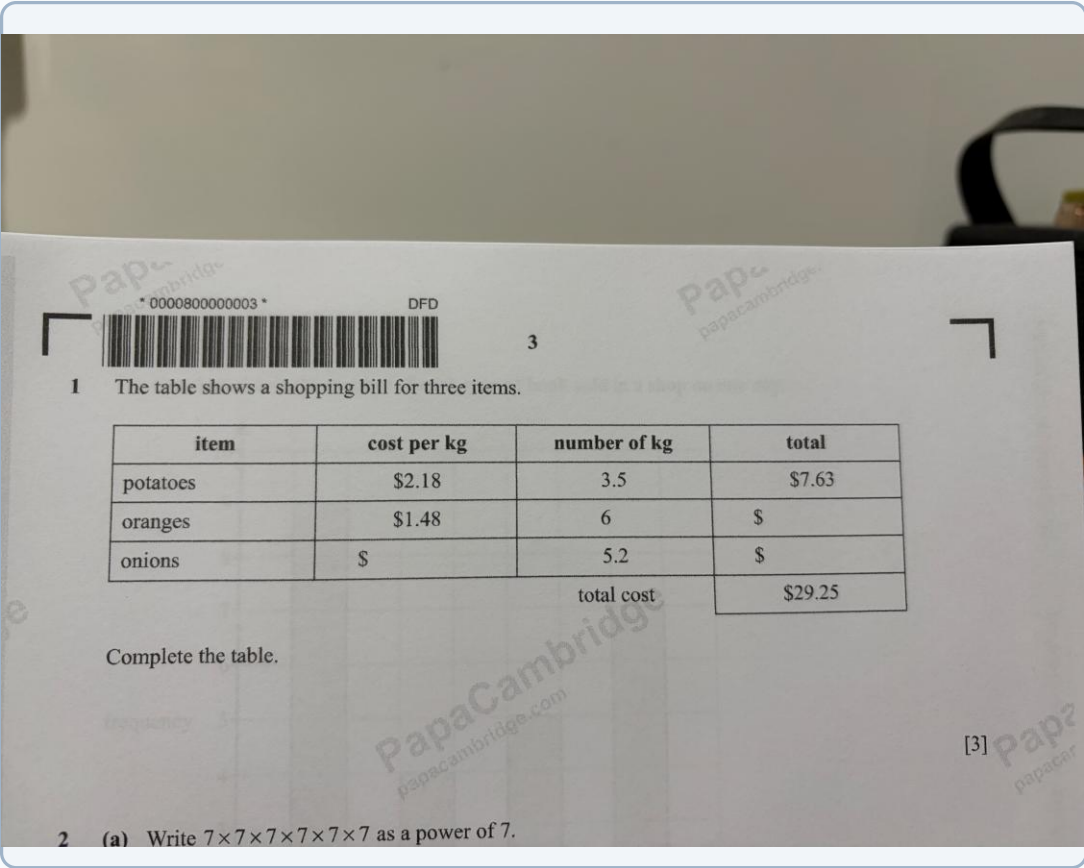


Image Quality
GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR– visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 1 The table shows a shopping bill for three items.

item	cost per kg	number of kg	total
potatoes	\$2.18	3.5	\$7.63
oranges	\$1.48	6	
onions	\$	5.2	\$
total cost			\$29.25

Complete the table.

1 item

1

conten : 1 The table shows a shopping bill for three items.

item	cost per kg	number of kg	total
potatoes	\$2.18	3.5	\$7.63
oranges	\$1.48	6	\$
onions	\$	5.2	\$
total cost			\$29.25

Complete the table.

Interpretation

Interpretation	Count
PASS	1
FAIL	0

PASS FAIL

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_037243f5a0dde5e7006aa583756b0087d0a48a9645dfa5f5f0

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Oranges total = 1.48 × 6 = 8.88\n\n Step 2:\n Onions total = 29.25 – (7.63 + 8.88) = 29.25 – 16.51 = 12.74\n\n Step 3:\n Onions cost per kg = 12.74 ÷ 5.2 = 2.45\n\n Final Answer:\n - Oranges total: \$8.88\n - Onions cost per kg: \$2.45\n - Onions total: \$12.74

role : assistant

Official Mark Scheme Answer

1	8.88 2.45 12.74	3
---	--------------------	---

B1 for 8.88
B1FT for 21.62 – *their* 8.88
B1FT for *their*12.74 ÷ 5.2

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Money and decimals
Difficulty: Easy
Assessment Objective: AO1

Solution

Step 1:
Oranges total = 1.48 × 6 = 8.88

Step 2:
Onions total = 29.25 – (7.63 + 8.88) = 29.25 – 16.51 = 12.74

Step 3:
Onions cost per kg = 12.74 ÷ 5.2 = 2.45

Final Answer:
- Oranges total: \$8.88
- Onions cost per kg: \$2.45
- Onions total: \$12.74

M1: Multiplies 1.48 by 6 for oranges total
A1: Oranges total = \$8.88

M1: Subtracts 7.63 and 8.88 from 29.25 to find onions total
A1: Onions total = \$12.74

M1: Divides 12.74 by 5.2 to find onions cost per kg
A1: Onions cost per kg = \$2.45

Total: 6

11:53 PM

Original Question Image

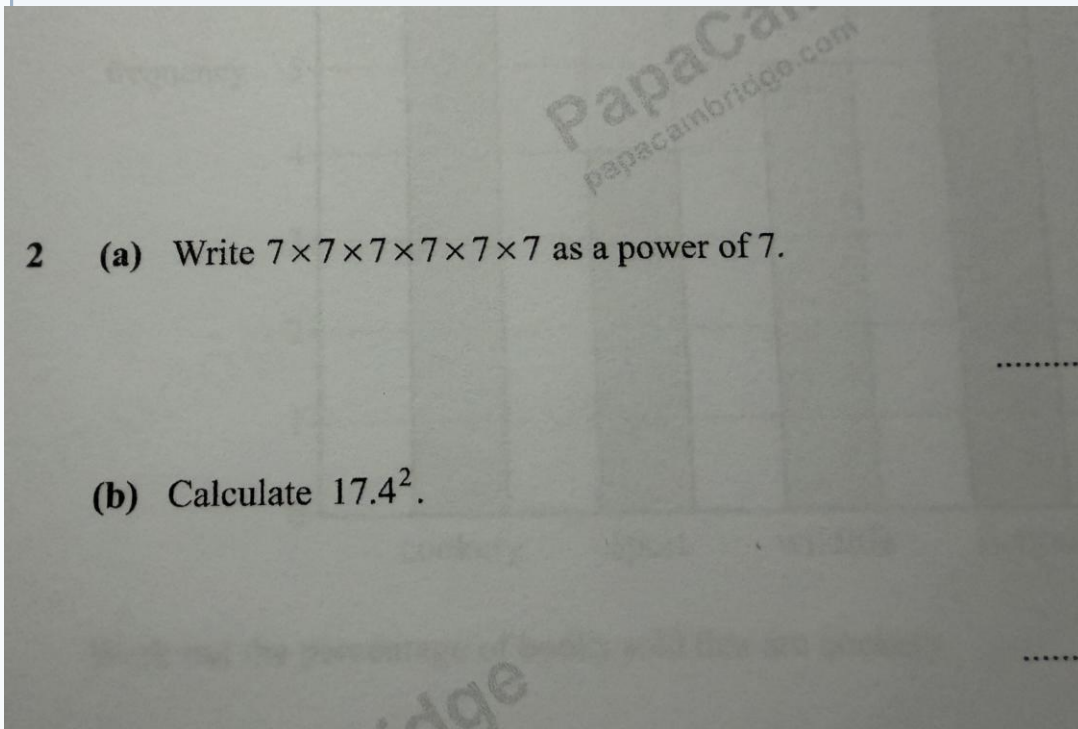


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 2 (a) Write $7 \times 7 \times 7 \times 7 \times 7 \times 7$ as a power of 7.
(b) Calculate 17.4^2 .

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_06dc2adb5b109e55006aa587564ed087d0a82f90f0c35bef57

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) 7^6 (b) Step 1: $17.4^2 = (17 + 0.4)^2 = 17^2 + 2 \times 17 \times 0.4 + 0.4^2$ Step 2: $= 289 + 13.6 + 0.16 = 302.76$ Final Answer: (a) 7^6 (b) 302.76

role : assistant

Official Mark Scheme Answer

2(a)	7^6	1
2(b)	302.76	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Powers

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) 7^6

(b)

Step 1:
 $17.4^2 = (17 + 0.4)^2 = 17^2 + 2 \times 17 \times 0.4 + 0.4^2$

Step 2:
 $= 289 + 13.6 + 0.16 = 302.76$

Final Answer:
(a) 7^6
(b) 302.76

B1: (a) 7^6

M1: (b) Uses expansion $(17+0.4)^2 = 17^2 + 2 \times 17 \times 0.4 + 0.4^2$

A1: (b) Obtains 302.76

Total: 3

12:10 AM

Math Intellect • Prototype Validation Evidence

54

Original Question Image

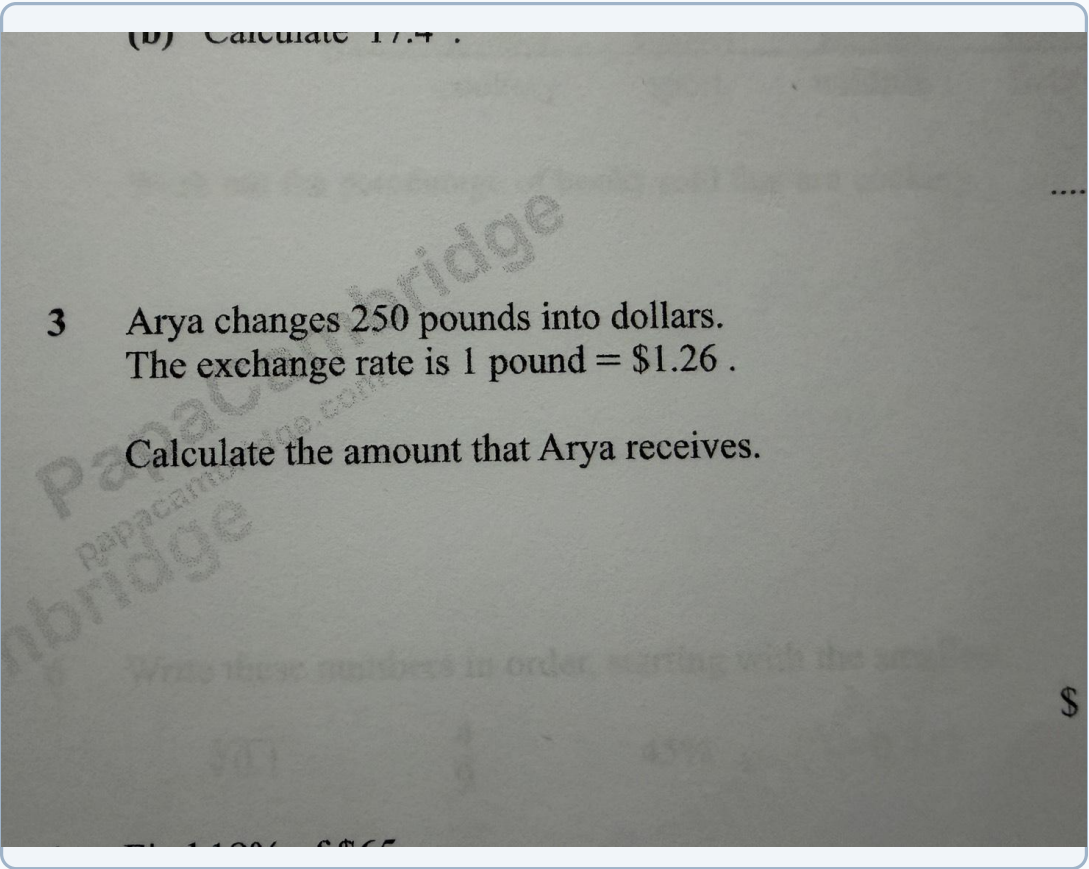


Image Quality

- GOOD – All texts / symbols clearly readable
- MODERATE – minor blur / skew but all essential information recoverable
- POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 3 Arya changes 250 pounds into dollars. \n The exchange rate is 1 pound = \$1.26. \n \n Calculate the amount that Arya receives.

Interpretation

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_00f0bd92553e946e006aa589ac20c087d09eae6fd2ea688ae5

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Amount in dollars = 250 × 1.26 = 315\n\n Final Answer:\n \$315

role : assistant

Official Mark Scheme Answer

3	315[.00]	1
---	----------	---

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Money

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:
Amount in dollars = 250 × 1.26 = 315

Final Answer:
\$315

M1: Uses multiplication 250 × 1.26 to convert pounds to dollars
A1: Obtains \$315

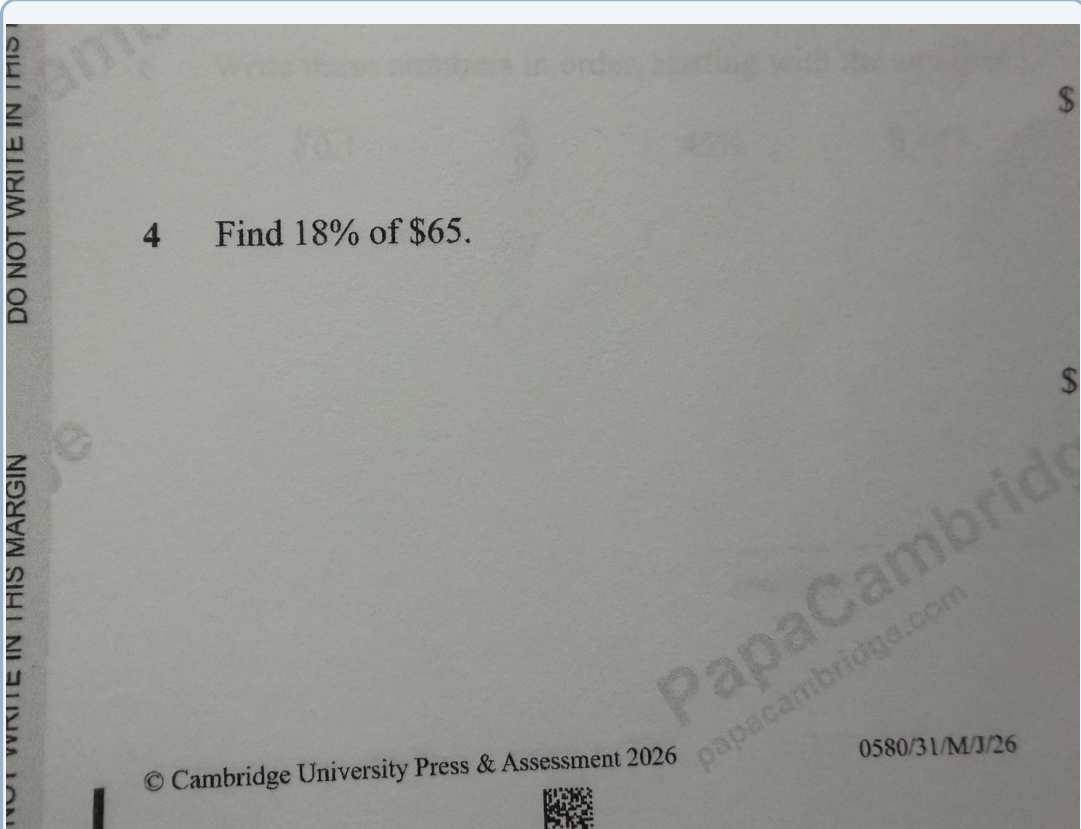
Total: 2

12:19 AM

Math Intellect • Prototype Validation Evidence

56

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 4 Find 18% of \$65.

Image Quality

- GOOD – All texts / symbols clearly readable
- MODERATE – minor blur / skew but all essential information recoverable
- POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0352031d5debd1a9006aa58c66027c87d0bf0ac0cc086825b6

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n18% of \$65 = (18/100) × 65\n\nStep 2:\n= 0.18 × 65 = 11.7\n\nFinal Answer:\n\$11.70

role : assistant

Official Mark Scheme Answer

4	11.7[0]	1
---	---------	---

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Percentages

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

18% of \$65 = (18/100) × 65

Step 2:

= 0.18 × 65 = 11.7

Final Answer:

\$11.70

B1: 18% of \$65 = \$11.70

Total: 1

12:31AM

Math Intellect • Prototype Validation Evidence

58

Original Question Image

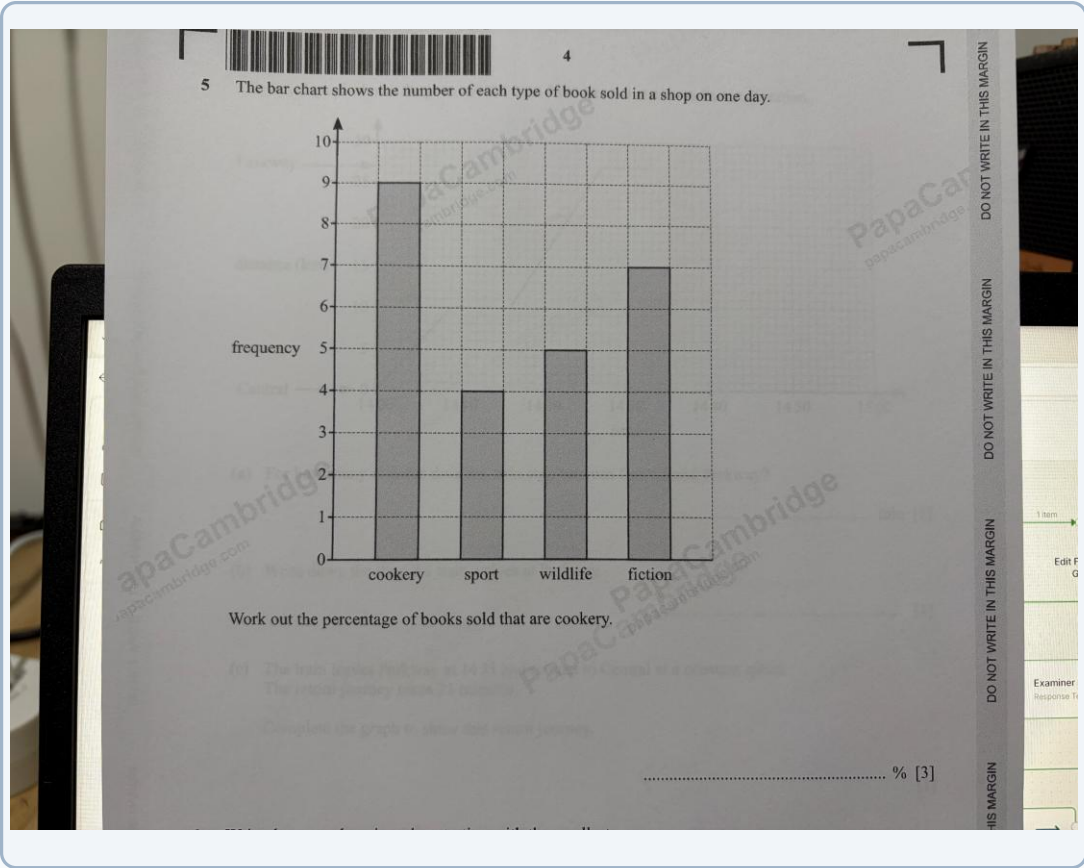


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT

1 item

1

content

: 5 The bar chart shows the number of each type of book sold in a shop on one day. \n \n Work out the percentage of books sold that are cookery. \n % [3]

Interpretation

PASS

FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0e7ad10bf9665418006aa59124ed1487d0b21ec65577f6adf5

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n From the bar chart, number of cookery books sold = 20.\n \n Step 2:\n Total number of books sold = 100.\n \n Step 3:\n Percentage of cookery books = (20 ÷ 100) × 100% = 20%.\n \n Final Answer:\n 20%

role : assistant

Official Mark Scheme Answer

5	36	3	M1 for $9 + 4 + 5 + 7$ or 25 oe M1 for $\frac{9}{\text{their } 25} [\times 100]$ oe
---	----	---	--

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Statistics
Difficulty: Easy
Assessment Objective: AO1

Solution

Step 1:
From the bar chart, number of cookery books sold = 20.

Step 2:
Total number of books sold = 100.

Step 3:
Percentage of cookery books = $(20 \div 100) \times 100\% = 20\%$.

Final Answer:
20%

B1: Reads number of cookery books sold as 20.
B1: Reads total number of books sold as 100.
A1: Correct percentage calculation $(20 \div 100 \times 100\% = 20\%)$.

Total: 3

Original Question Image

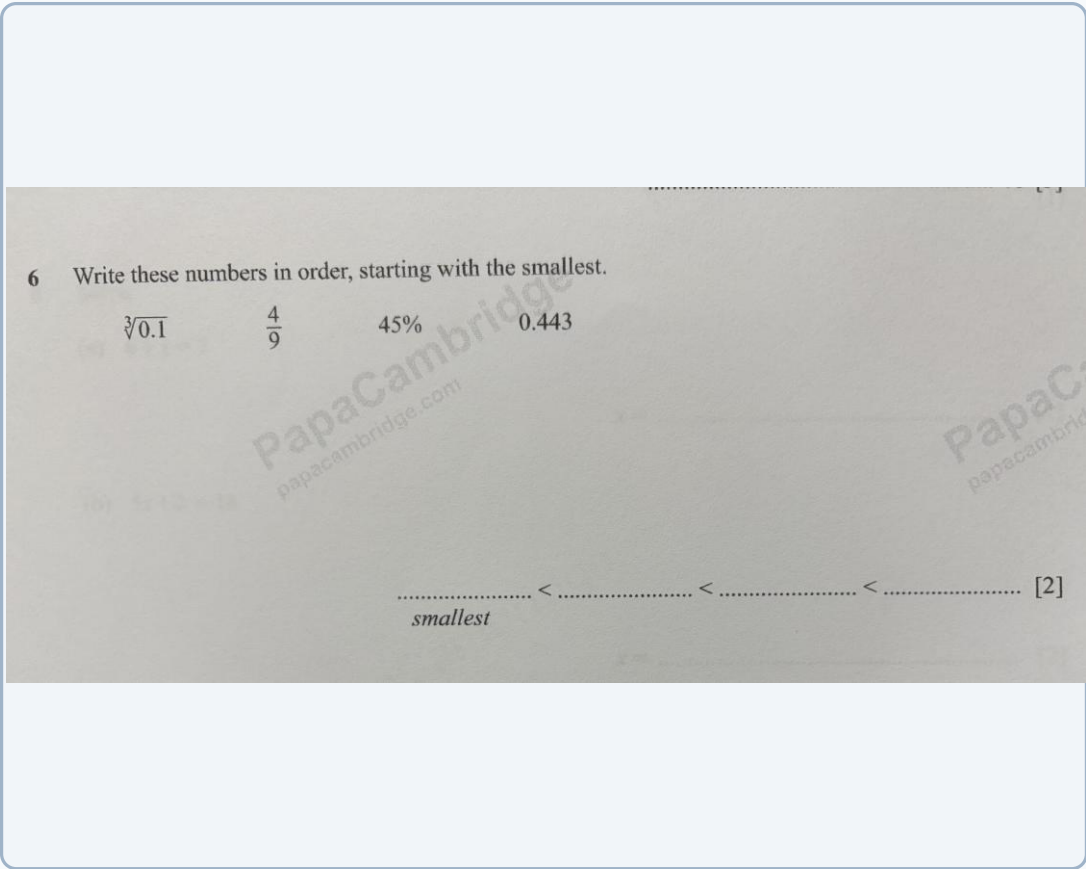


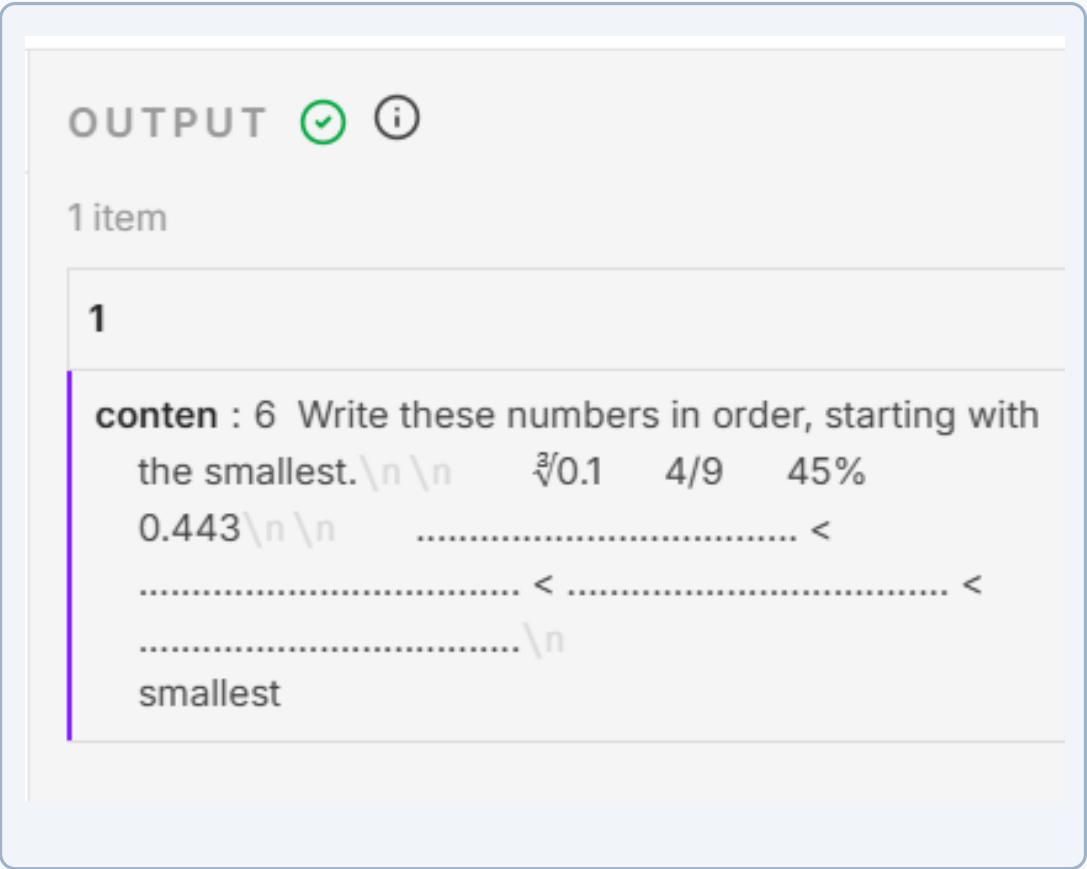
Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node



Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_01fd7a5d795cd70e006aa594aa5f8887d08e9c9608c448b2aa

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : 0.443 < 4/9 < 45% < $\sqrt[3]{0.1}$

role : assistant

Official Mark Scheme Answer

6	0.443 $\frac{4}{9}$ 45% $\sqrt[3]{0.1}$
2	B1 for 3 in correct order or M1 for all decimal equivalents to enable comparison 0.46..., 0.444, 0.45, [0.443]

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Ordering numbers

Difficulty: Medium

Assessment Objective: AO1 + AO2

Solution

$0.443 < 4/9 < 45\% < \sqrt[3]{0.1}$

B1: Correct order written as $0.443 < 4/9 < 45\% < \sqrt[3]{0.1}$

Total: 1

1:06 AM

Math Intellect • Prototype Validation Evidence

62

Original Question Image

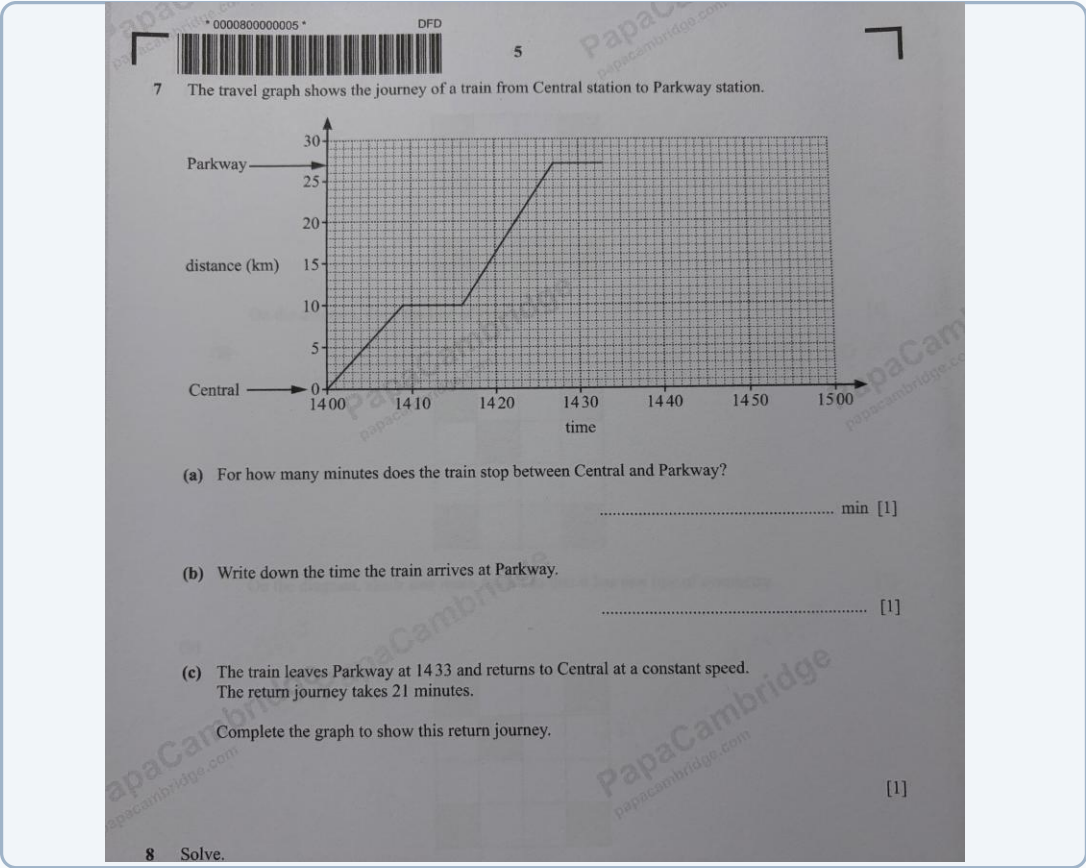


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT ✓ ⓘ

1 item

1

content : 7 The travel graph shows the journey of a train from Central station to Parkway station. (a) For how many minutes does the train stop between Central and Parkway? min [1] (b) Write down the time the train arrives at Parkway. [1] (c) The train leaves Parkway at 14 33 and returns to Central at a constant speed. The return journey takes 21 minutes. Complete the graph to show this return journey. [1]

Interpretation

PASS FAIL

REASON : Visual/diagram information was not fully captured from the source image

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0b7fba77d3851043006aa61db920cc87d0acd064878bd195a0

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) 4 min\n\n(b) 14 30\n\n(c) Draw a straight line of constant negative gradient from the point at Parkway at 14 33 down to Central at 14 54 (since 14 33 + 21 min = 14 54).

role : assistant

Official Mark Scheme Answer

7(a)	7	1
7(b)	14 27	1
7(c)	Ruled line from (1433, 27) to (1454, 0)	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Distance-time graph

Difficulty: Easy

Assessment Objective: AO1 + AO2

Solution

(a) 4 min

(b) 14 30

(c) Draw a straight line of constant negative gradient from the point at Parkway at 14 33 down to Central at 14 54 (since 14 33 + 21 min = 14 54).

B1: (a) Writes the stop time between stations as 4 minutes.

B1: (b) Writes the arrival time at Parkway as 14:30.

B1: (c) Draws straight line from Parkway at 14:33 to Central at 14:54.

Total: 3

10:51 AM

Original Question Image

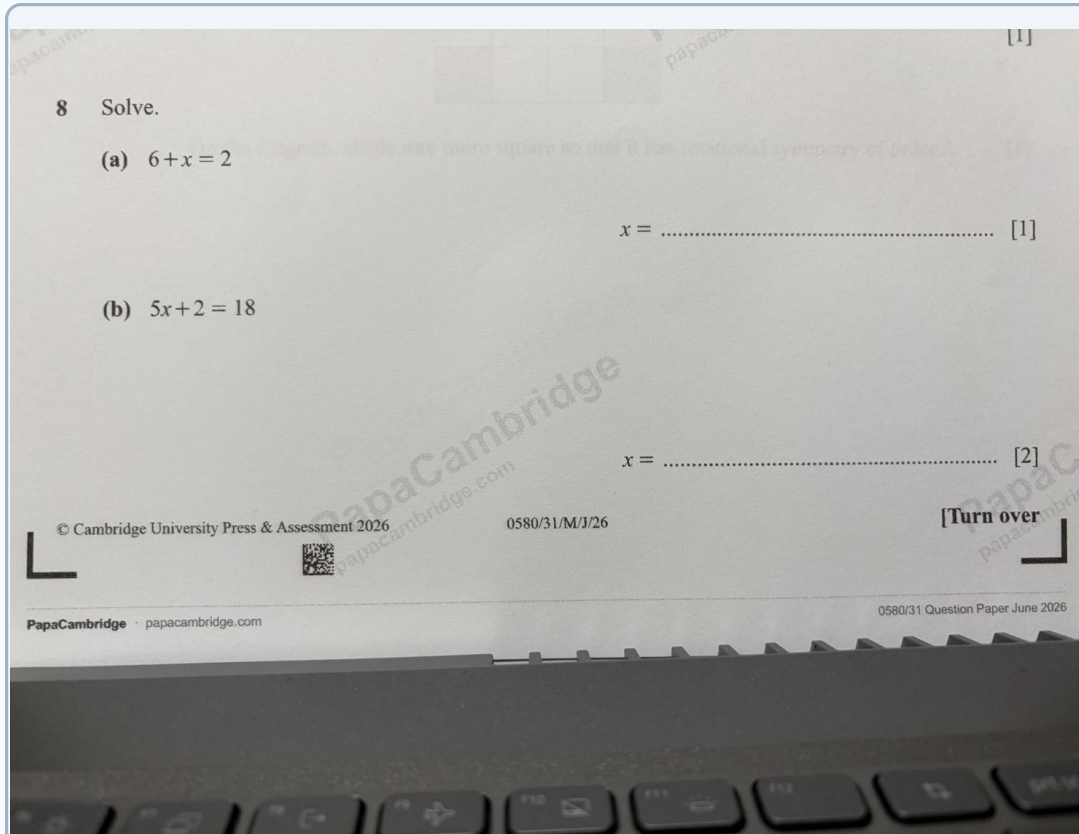


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 8 Solve.
2
..... [1]
= 18
..... [2]

(a) $6 + x =$
 $x =$
..... [1]
(b) $5x + 2$
 $x =$
..... [2]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0fa6e5fd5628f7e2006aa64cf5c75887d0879a
d28fc5aae36e

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) \n Step 1: \n 6 + x = 2 \n x = 2 - 6 =
-4 \n \n Final Answer: \n x = -4 \n \n (b) \n Step
1: \n 5x + 2 = 18 \n 5x = 18 - 2 = 16 \n \n Step 2: \n x
= 16 ÷ 5 = 16/5 \n \n Final Answer: \n x = 16/5 (3.2)

role : assistant

Official Mark Scheme Answer

8(a)	-4 final answer	1
8(b)	3.2 oe final answer	2
Accept $3\frac{1}{5}$ and $\frac{16}{5}$ for 2 marks.		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Algebra
Difficulty: Easy
Assessment Objective: AO1

Solution

(a)
Step 1:
6 + x = 2
x = 2 - 6 = -4

Final Answer:
x = -4

(b)
Step 1:
5x + 2 = 18
5x = 18 - 2 = 16

Step 2:
x = 16 ÷ 5 = 16/5

Final Answer:
x = 16/5 (3.2)

B1: (a) x = -4

M1: (b) obtains 5x = 18 - 2 = 16

A1: (b) x = 16/5

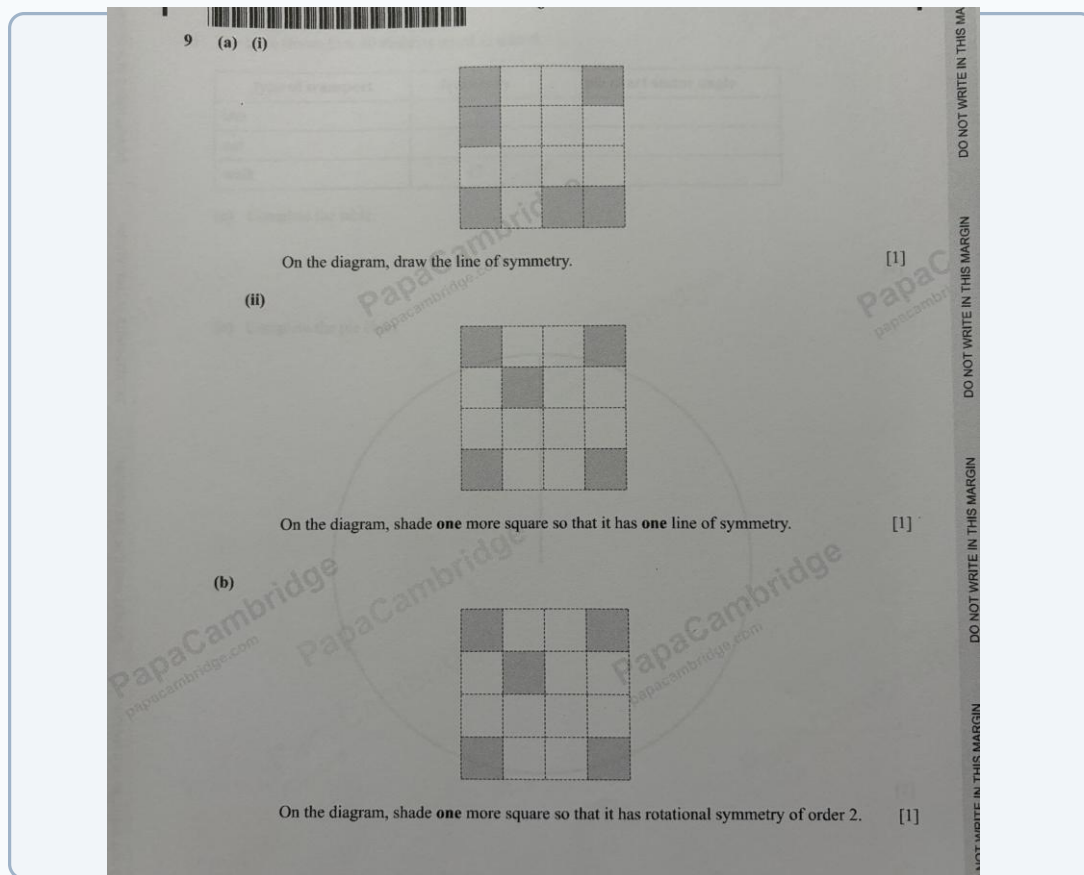
Total: 3

2:13 PM

Math Intellect • Prototype Validation Evidence

66

Original Question Image



Question Interpretation Output – Edit Fields Node

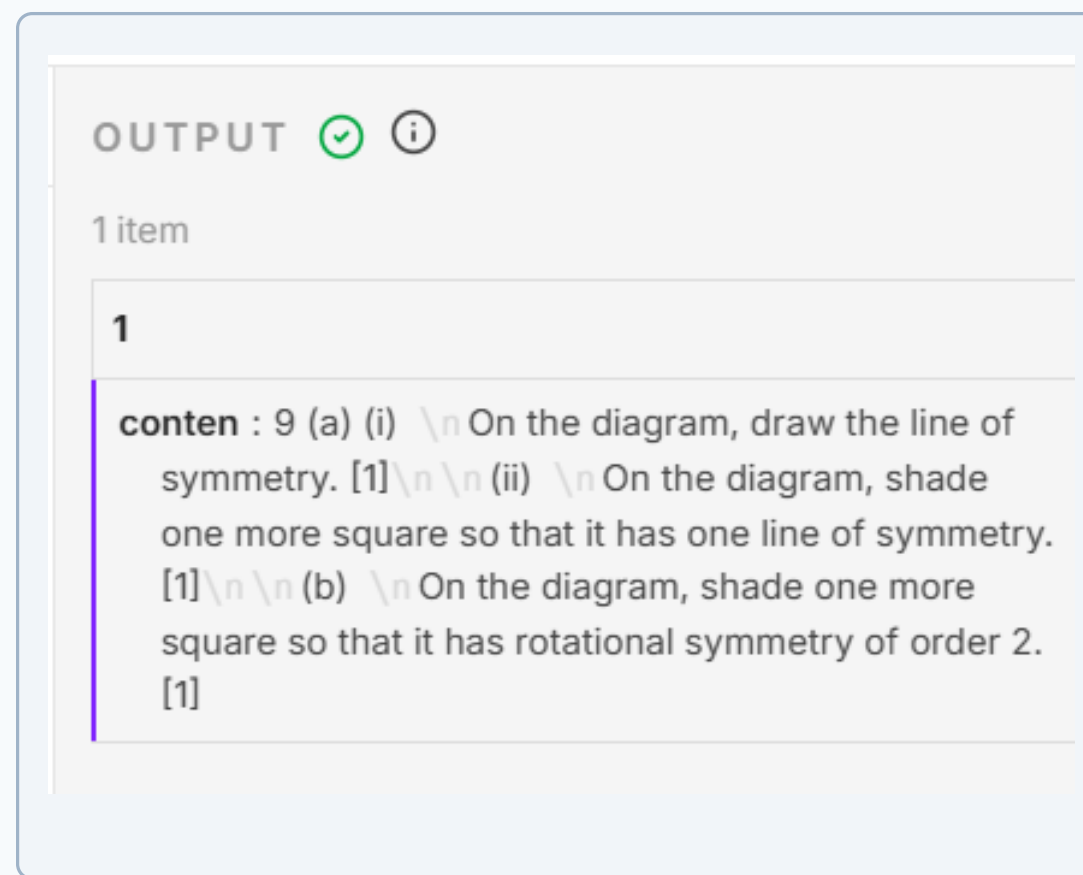


Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Failure Output – Moderator Node

INPUT **Parameters** **Settings** **Execute step**

Credential
OpenAI account 2

Resource
Text

Operation
Message a Model

Model
From list **GPT-5-MINI**

Messages
+
Message 1
Message 2
+ Add Message

OUTPUT **1 item**

output

0

id : msg_019580a34f542905006aa6520eb52887d0ac44de2cacca2c5f
type : message
status : completed
content

0

type : output_text
annotations : [empty array]
logprobs : [empty array]

text : {
 "status": "FAIL",
 "issues": [
 "No answers provided for parts (a)(i), (a)(ii) and (b); the solution only requests the diagram instead of drawing the required line and shading the required square(s).",
 "Important steps are missing: the line of symmetry is not drawn and the specific squares to be shaded are not indicated, so no mathematical work or justification is shown.",
 "The mark scheme is empty/incorrect (Total: 0) and does not correspond to the three required 1-mark parts of the question."]
}

role : assistant

Quality Gate Outcome

FAIL

Failure Stage

Analyze Image

Failure Detected by

Moderator

LINE End-to-End Success

FAIL

Final Answer Pass

N/A

Solution Completeness

N/A

Assessment Alignment

N/A

Failure evidence rule: no delivered LINE output → Final Answer, Solution Completeness, Assessment Alignment and Sub-question Accuracy = N/A.

REASON : Solver could not complete the task because required diagram information was unavailable.

Original Question Image

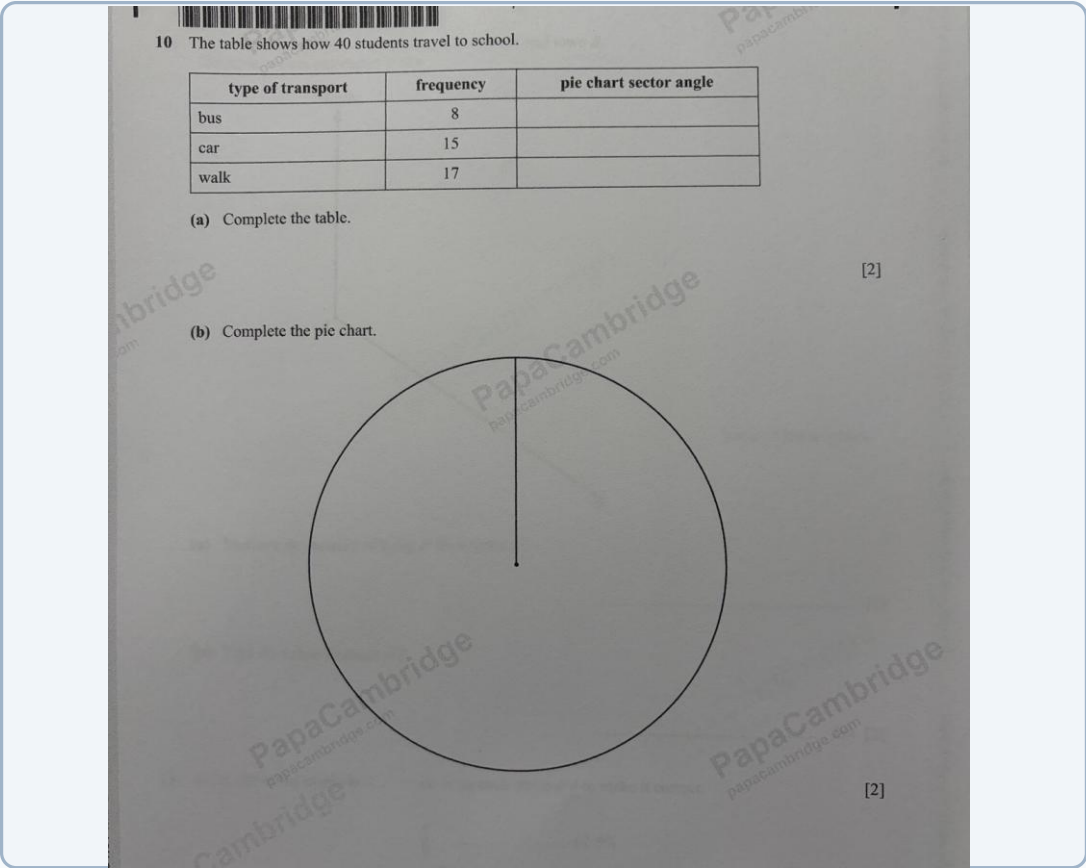


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 10 The table shows how 40 students travel to school. $\begin{array}{|c|c|c|} \hline \text{type of transport} & \text{frequency} & \text{pie chart sector angle} \\ \hline \text{bus} & 8 & \\ \text{car} & 15 & \\ \text{walk} & 17 & \\ \hline \end{array}$ (a) Complete the table. [2] (b) Complete the pie chart. [2]

Interpretation

PASS FAIL

REASON : Visual/diagram information was not fully captured from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0fcca8117f24d378006aa6587c228487d0a6f1675f88f6fe32

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n Step 1:\n Angle per student = $360^\circ \div 40 = 9^\circ$ \n \n Step 2:\n - Bus: $8 \times 9^\circ = 72^\circ$ \n - Car: $15 \times 9^\circ = 135^\circ$ \n - Walk: $17 \times 9^\circ = 153^\circ$ \n \n Final Answer:\n Bus 72° , Car 135° , Walk 153° \n \n (b)\n - From the centre of the given circle, use the starting radius shown on the diagram (or draw one vertical upwards if none is shown).\n - Measure and draw a 72° sector for Bus.\n - From the new radius, measure and draw a 135° sector for Car.\n - The remaining sector is 153° for Walk.\n - Label each sector accordingly.

role : assistant

Official Mark Scheme Answer

10(a)	72 135 153
10(b)	Accurate pie chart
2	B1 for 2 correct in correct place or M1 for $\frac{360}{40} \times k$ ($k = 1, 8, 15$ or 17)
2	B2FT if <i>their</i> angles add up to 360° or B1 for one accurate sector or B1FT for one accurate sector FT <i>their</i> table

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

3:01 PM

Topic: Statistics
Difficulty: Easy
Assessment Objective: AO1

Solution

(a)
Step 1:
Angle per student = $360^\circ \div 40 = 9^\circ$

Step 2:
- Bus: $8 \times 9^\circ = 72^\circ$
- Car: $15 \times 9^\circ = 135^\circ$
- Walk: $17 \times 9^\circ = 153^\circ$

Final Answer:
Bus 72° , Car 135° , Walk 153°

(b)
- From the centre of the given circle, use the starting radius shown on the diagram (or draw one vertical upwards if none is shown).
- Measure and draw a 72° sector for Bus.
- From the new radius, measure and draw a 135° sector for Car.
- The remaining sector is 153° for Walk.
- Label each sector accordingly.

M1: (a) angle per student $360^\circ \div 40 = 9^\circ$
A1: (a) Bus 72° , Car 135° , Walk 153°
M1: (b) uses protractor to draw 72° and 135° sectors
A1: (b) sectors correctly labelled Bus 72° , Car 135° , Walk 153°

Total: 4

3:02 PM

Math Intellect • Prototype Validation Evidence

70

Original Question Image

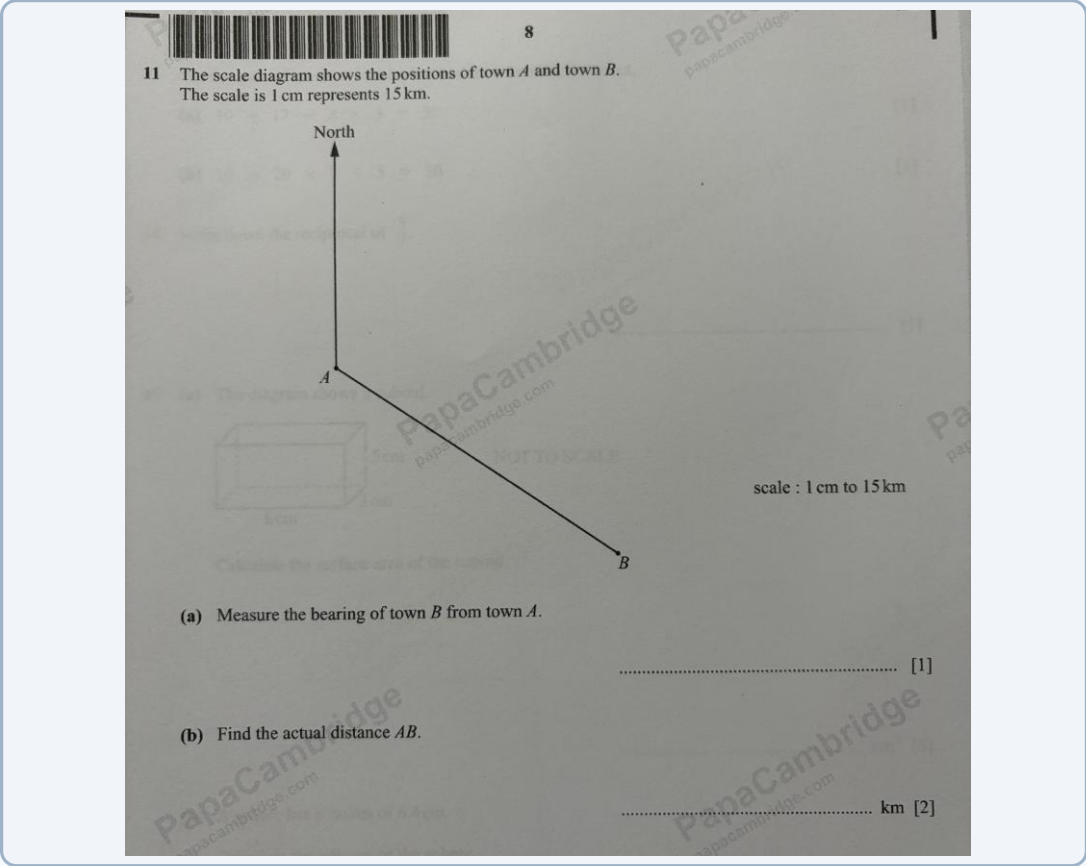


Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 11 The scale diagram shows the positions of town A and town B. The scale is 1 cm represents 15 km. \n \n (a) Measure the bearing of town B from town A.

[1]\n \n (b) Find the actual distance AB. km [2]

Interpretation

PASS FAIL

REASON : Required diagram information was not extracted from the source image.

Failure Output – Moderator Node

INPUT

Examiner

output

Message 1

Message 2

+ Add Message

Simplify Output

Connect your own custom n8n tools to this node on the canvas

Parameters

Settings

Execute step

Credential

OpenAI account 2

Resource

Text

Operation

Message a Model

Model

From list

GPT-5-MINI

Messages

Message 1

Message 2

+ Add Message

Simplify Output

Connect your own custom n8n tools to this node on the canvas

OUTPUT

1 item

output

6dfa051ac73da

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : {\n "status": "FAIL",\n "issues": [\n "No answer provided: the solution only requests the diagram and does not give the bearing of B from A or the actual distance AB.",\n "Missing mathematical steps: there is no measurement of the bearing (angle clockwise from North at A), no measurement of AB on the diagram, and no conversion of diagram length to actual distance.",\n "Formula not used: the scale conversion (actual distance = measured length in cm \times 15 km/cm) is not applied or stated.",\n "Mark scheme incorrect/incomplete: Mark Scheme shows 'Total: 0' while the question parts carry 1 mark (part a) and 2 marks (part b), total 3 marks; the provided mark scheme does not match the required allocation." \n]\n }

role : assistant

Quality Gate Outcome

FAIL

FAIL

Failure Stage

Analyze Image

Analyze Image

Failure Detected by

Moderator

Moderator

LINE End-to-End Success

FAIL

FAIL

Final Answer Pass

N/A

N/A

Solution Completeness

N/A

N/A

Assessment Alignment
N/A

N/A

Failure evidence rule: no delivered LINE output → Final Answer, Solution Completeness, Assessment Alignment and Sub-question Accuracy = N/A.

REASON : Solver could not complete the task because required diagram information was unavailable

Original Question Image

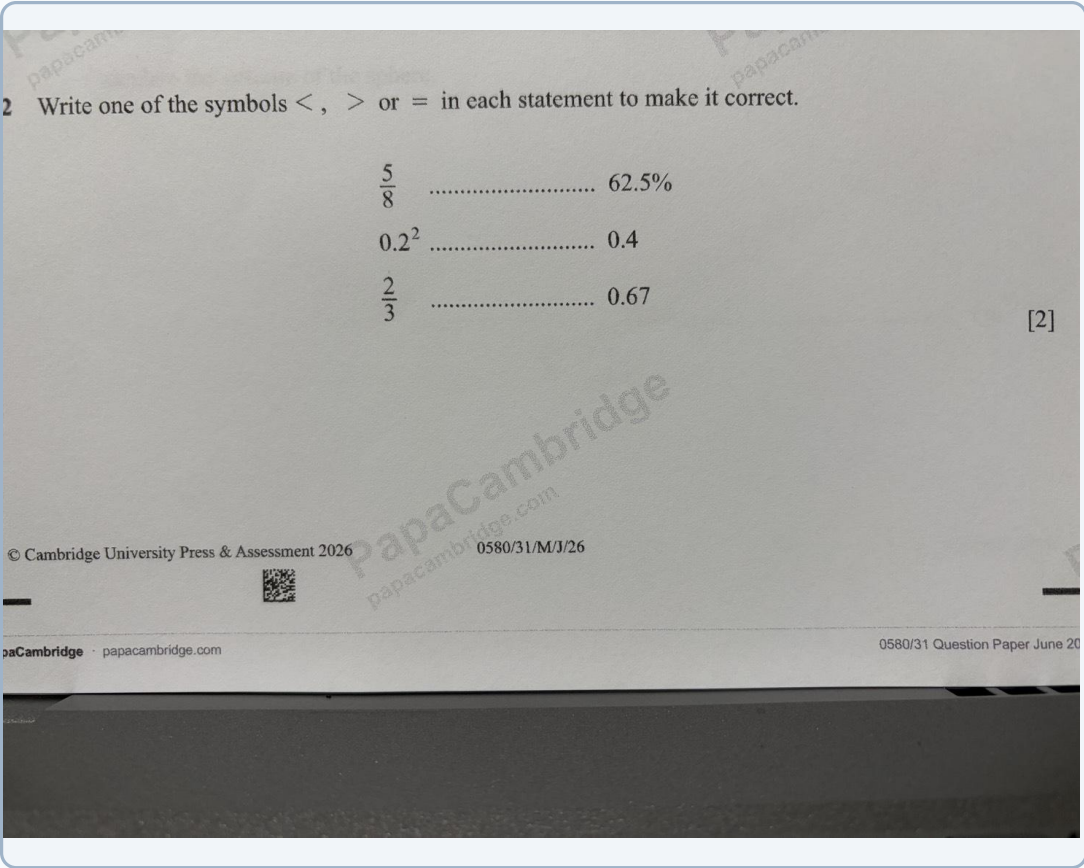


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 2 Write one of the symbols < , > or = in each statement to make it correct. $\frac{5}{8}$ 62.5% 0.2^2 0.4 $\frac{2}{3}$ 0.67

Interpretation

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_05e3394349833738006aa665a1290887d0a267ca7dd15a74d5

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : 5/8 = 62.5%\n\n0.2^2 < 0.4\n\n2/3 < 0.67

role : assistant

Official Mark Scheme Answer

12	=	2	B1 for 2 correct
	<		
	<		

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Fractions, decimals and percentages

Difficulty: Easy

Assessment Objective: AO1

Solution

5/8 = 62.5%

0.2^2 < 0.4

2/3 < 0.67

B1: (a) 5/8 = 62.5% correct equality

B1: (b) 0.2^2 < 0.4 correct inequality

B1: (c) 2/3 < 0.67 correct inequality

Total: 3

3:58 PM

Math Intellect • Prototype Validation Evidence

74

Original Question Image

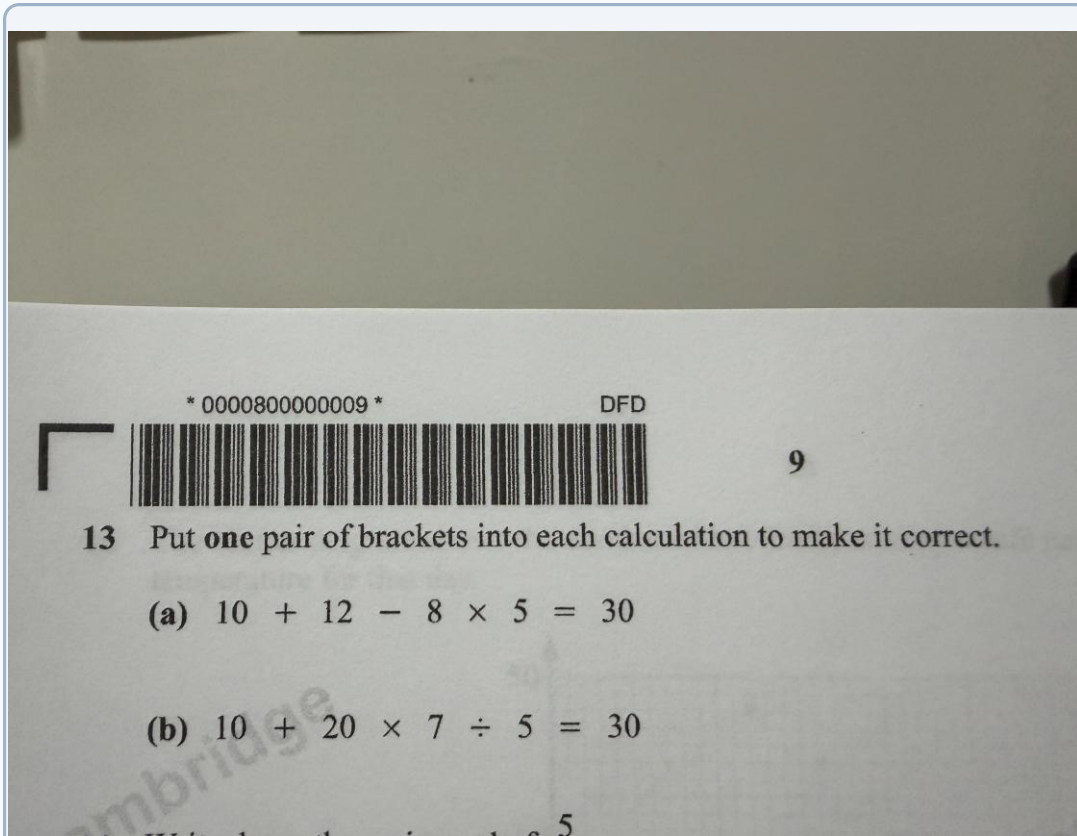


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 13 Put one pair of brackets into each calculation to make it correct. \n \n (a) $10 + 12 - 8 \times 5 = 30$ \n \n (b) $10 + 20 \times 7 \div 5 = 30$

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_06e8c03a9211f933006aa66a606b1887d0ade5e27375440587

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) $10 + (12 - 8) \times 5 = 30$ \n \n (b) $(10 + 20 \times 7) \div 5 = 30$

role : assistant

Official Mark Scheme Answer

13(a)	$10 + (12 - 8) \times 5 = 30$	1
13(b)	$(10 + 20 \times 7) \div 5 = 30$	1

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Order of operations

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) $10 + (12 - 8) \times 5 = 30$

(b) $(10 + 20 \times 7) \div 5 = 30$

B1: (a) $10 + (12 - 8) \times 5 = 30$

B1: (b) $(10 + 20 \times 7) \div 5 = 30$

Total: 2

4:18 PM

Math Intellect • Prototype Validation Evidence

76

Original Question Image

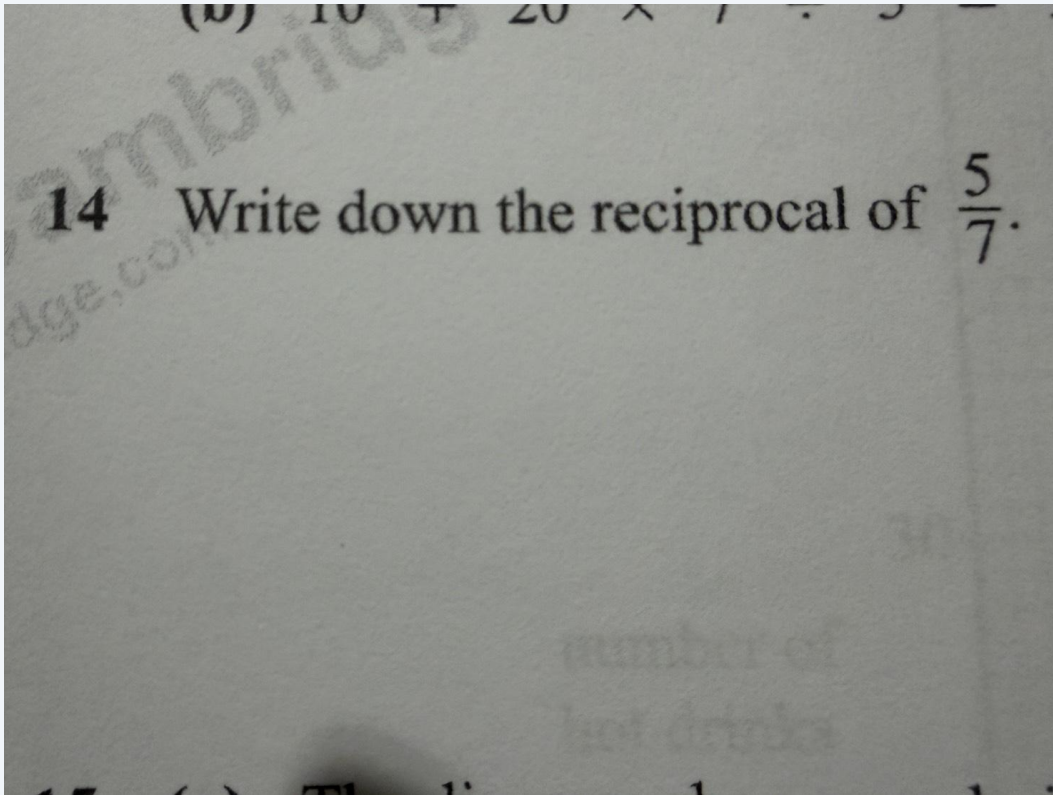


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 14 Write down the reciprocal of $\left(\frac{5}{7}\right)$.

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0b1740cced23338a006aa66d500a6c87d08b
bcc2421d695583

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : 7/5

role : assistant

Official Mark Scheme Answer

14	$\frac{7}{5}$ or $1\frac{2}{5}$ or 1.4	1
----	--	---

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Fractions

Difficulty: Easy

Assessment Objective: AO1

Solution

$\frac{7}{5}$

B1: Reciprocal of $\frac{5}{7}$ is $\frac{7}{5}$

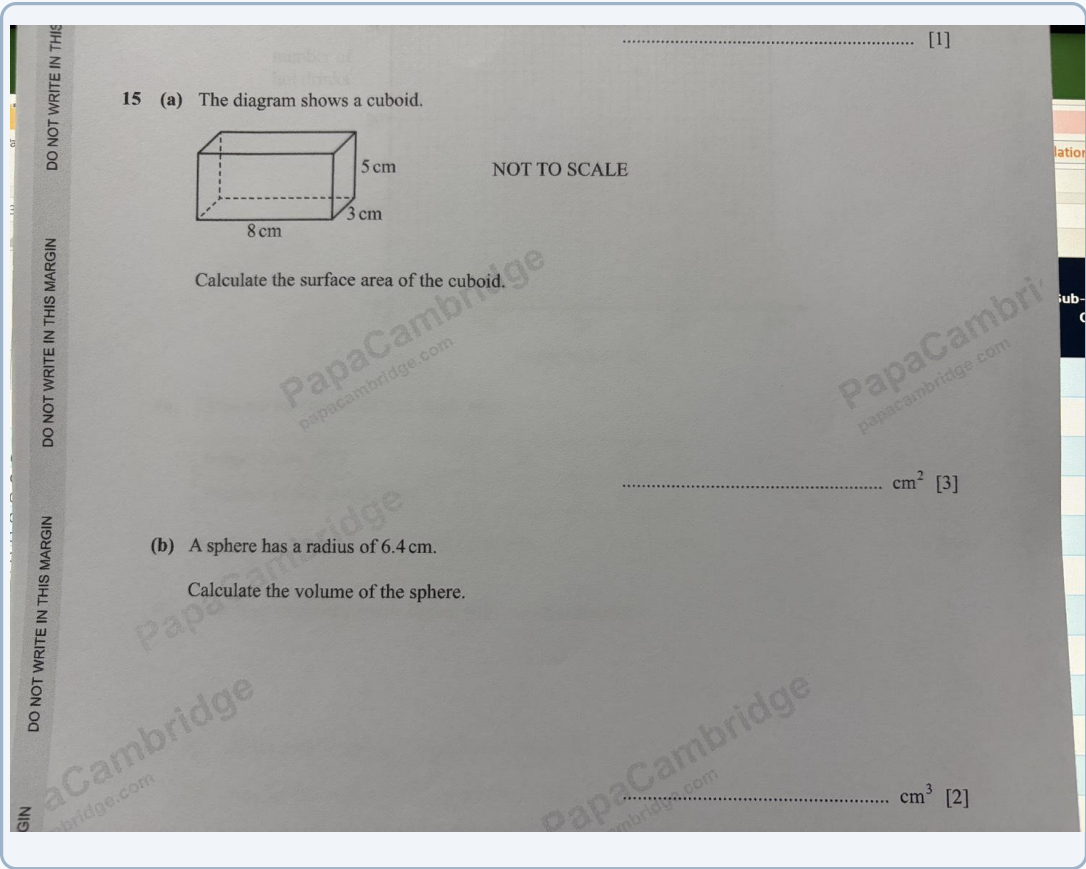
Total: 1

4:31 PM

Math Intellect • Prototype Validation Evidence

78

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 15 (a) The diagram shows a cuboid.
[Diagram of a cuboid with dimensions 5 cm, 3 cm, and 8 cm]
Calculate the surface area of the cuboid. cm² [3]
(b) A sphere has a radius of 6.4 cm.
Calculate the volume of the sphere. cm³ [2]

Image Quality

- GOOD – All texts / symbols clearly readable
- MODERATE – minor blur / skew but all essential information recoverable
- POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0db4085231095b4e006aa6725ff43c87d0bdbb252af647e0f3

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n Step 1:\n Surface area of a cuboid = 2(lw + lh + wh)\n \n Step 2:\n = 2(5x3 + 5x8 + 3x8)\n = 2(15 + 40 + 24)\n = 2x79\n \n Step 3:\n = 158 cm²\n \n Final Answer:\n 158 cm²\n \n (b)\n Step 1:\n Volume of a sphere = (4/3)πr³\n \n Step 2:\n = (4/3)π(6.4)³\n = (4/3)πx262.144\n ≈ 1098.066...\n \n Final Answer:\n 1100 cm³ (3 s.f.)

role : assistant

Official Mark Scheme Answer

15(a)	158
15(b)	1100 or 1098 or 1098.0 to 1098.2...
3	M2 for $[2 \times](3 \times 5 + 3 \times 8 + 8 \times 5)$ oe or M1 for $[2 \times] 3 \times 5$ or $[2 \times] 3 \times 8$ or $[2 \times] 8 \times 5$
2	M1 for $\frac{4}{3} \times \pi \times 6.4^3$ oe

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Mensuration

Difficulty: Easy

Assessment Objective: AO1

Solution

(a)

Step 1:
Surface area of a cuboid = $2(lw + lh + wh)$

Step 2:
 $= 2(5 \times 3 + 5 \times 8 + 3 \times 8)$
 $= 2(15 + 40 + 24)$
 $= 2 \times 79$

Step 3:
 $= 158 \text{ cm}^2$

Final Answer:
 158 cm^2

(b)

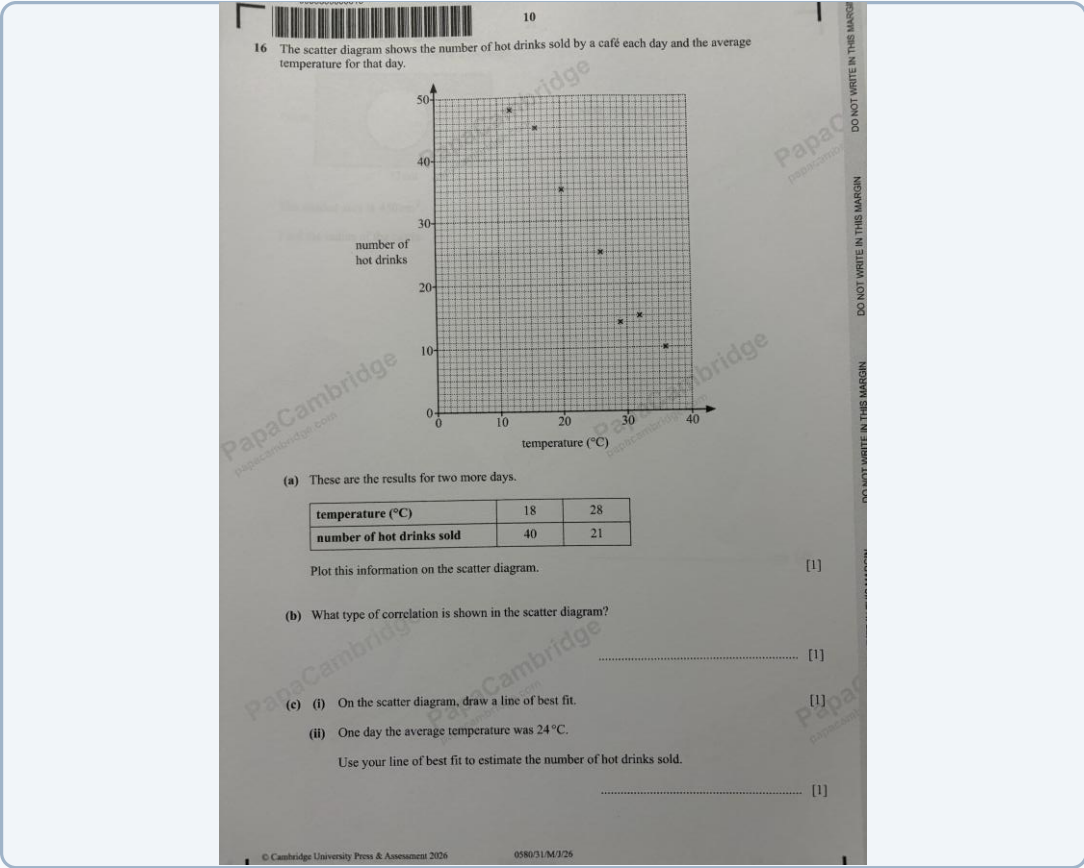
Step 1:
Volume of a sphere = $(4/3)\pi r^3$

Step 2:
 $= (4/3)\pi(6.4)^3$
 $= (4/3)\pi \times 262.144$
 $\approx 1098.066...$

Final Answer:
 1100 cm^3 (3 s.f.)

M1: (a) Uses $SA = 2(lw + lh + wh)$ (student Step 1)
M1: (a) Substitutes values and simplifies to 2×79 (Step 2)
A1: (a) Final answer 158 cm^2 (student Step 3)
M1: (b) Uses $V = (4/3)\pi r^3$ (student Step 1)
A1: (b) Final answer 1100 cm^3 (rounded to 3 s.f.) (Step 2)

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 16 The scatter diagram shows the number of hot drinks sold by a café each day and the average temperature for that day. (a) These are the results for two more days.

temperature (°C)	18	28
number of hot drinks sold	40	21

 Plot this information on the scatter diagram. [1] (b) What type of correlation is shown in the scatter diagram? [1] (c) (i) On the scatter diagram, draw a line of best fit. [1] (ii) One day the average temperature was 24°C. Use your line of best fit to estimate the number of hot drinks sold. [1]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_00468b7981e71e55006aa67ef2288487d0b1fe5f467d189ac

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n - Plot the points (18, 40) and (28, 21) on the scatter diagram.\n\n(b)\n - Negative correlation.\n\n(c)\n(i)\n - Draw a straight, downward-sloping line of best fit through the cluster of points so that the points are evenly spread above and below the line.\n\n(ii)\n - From 24°C on the horizontal axis, draw a vertical line up to your line of best fit, then across to the vertical axis. Estimated number of hot drinks ≈ 29.\n\nFinal Answer:\n\n(a) Points at (18, 40) and (28, 21)\n\n(b) Negative correlation\n\n(c)(i) Line of best fit drawn (downward slope)\n\n(c)(ii) 29

role : assistant

Official Mark Scheme Answer

16(a)	(18, 40) and (28, 21) plotted	1
16(b)	Negative	1
16(c)(i)	Acceptable ruled line of best fit	1
16(c)(ii)	<i>their</i> reading at 24	1

Strict FT *their* straight line with negative gradient

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Statistics

Difficulty: Easy

Assessment Objective: AO1 + AO2

Solution

(a)

- Plot the points (18, 40) and (28, 21) on the scatter diagram.

(b)

- Negative correlation.

(c)

(i)

- Draw a straight, downward-sloping line of best fit through the cluster of points so that the points are evenly spread above and below the line.

(ii)

- From 24°C on the horizontal axis, draw a vertical line up to your line of best fit, then across to the vertical axis. Estimated number of hot drinks ≈ 29.

Final Answer:

(a) Points at (18, 40) and (28, 21)

(b) Negative correlation

(c)(i) Line of best fit drawn (downward slope)

(c)(ii) 29

B1: (a) Points plotted at (18, 40) and (28, 21).

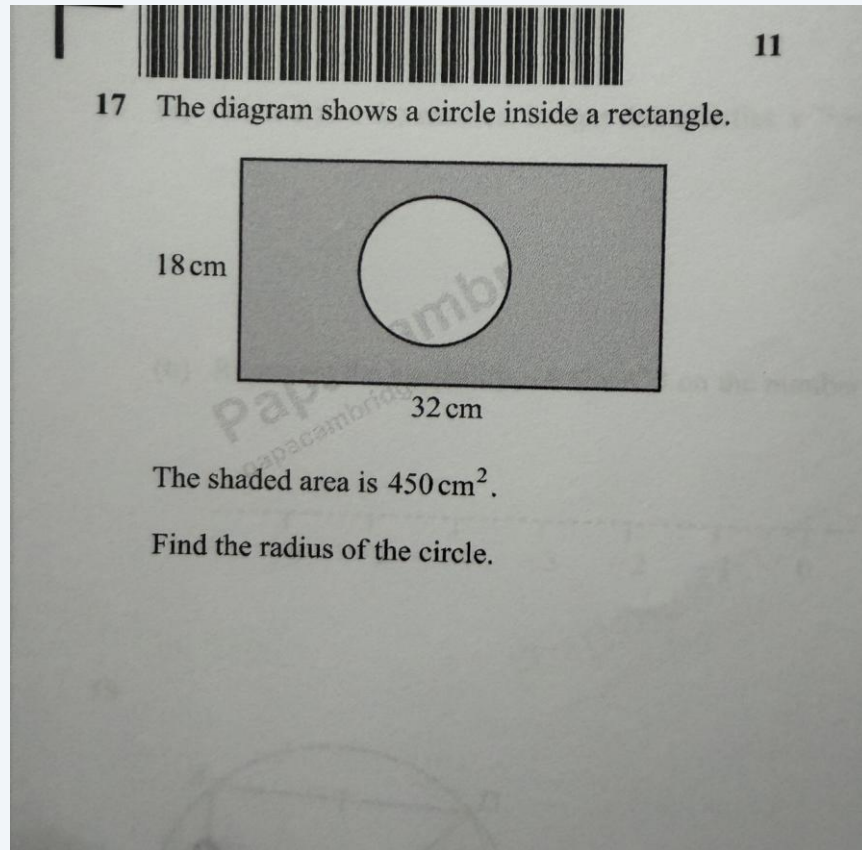
B1: (b) Negative correlation.

B1: (c)(i) Straight line of best fit drawn through data points.

A1: (c)(ii) From 24°C estimate ≈ 29 hot drinks.

Total: 4

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 17 The diagram shows a circle inside a rectangle. \n \n The shaded area is 450 cm^2 . \n \n Find the radius of the circle.

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0fb504f76d0dfd3e006aa6e1ddea1487d094cb3ff764226056

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n From the diagram, area of rectangle = 30 × 20 = 600 cm².\n \n Step 2:\n Shaded area = rectangle area – circle area\n 450 = 600 – πr²\n \n Step 3:\n πr² = 150\n r² = 150/π\n r = √(150/π)\n \n Step 4:\n r ≈ √(150 ÷ 3.142) ≈ √47.74 ≈ 6.91 cm\n \n Final Answer:\n 6.91 cm

role : assistant

Official Mark Scheme Answer

17	6.33 or 6.332 to 6.333...
4	M3 for $\sqrt{\frac{18 \times 32 - 450}{\pi}}$ oe or M2 for $\pi r^2 = 18 \times 32 - 450$ oe or M1 for 18×32 oe or better

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Area

Difficulty: Medium

Assessment Objective: AO2

Solution

Step 1:

From the diagram, area of rectangle = 30 × 20 = 600 cm².

Step 2:

Shaded area = rectangle area – circle area

450 = 600 – πr²

Step 3:

πr² = 150

r² = 150/π

r = √(150/π)

Step 4:

r ≈ √(150 ÷ 3.142) ≈ √47.74 ≈ 6.91 cm

Final Answer:

6.91 cm

M1: Rectangle area calculated 30 × 20 = 600 cm²

M1: Uses shaded = rectangle – circle, so πr² = 600 – 450 = 150

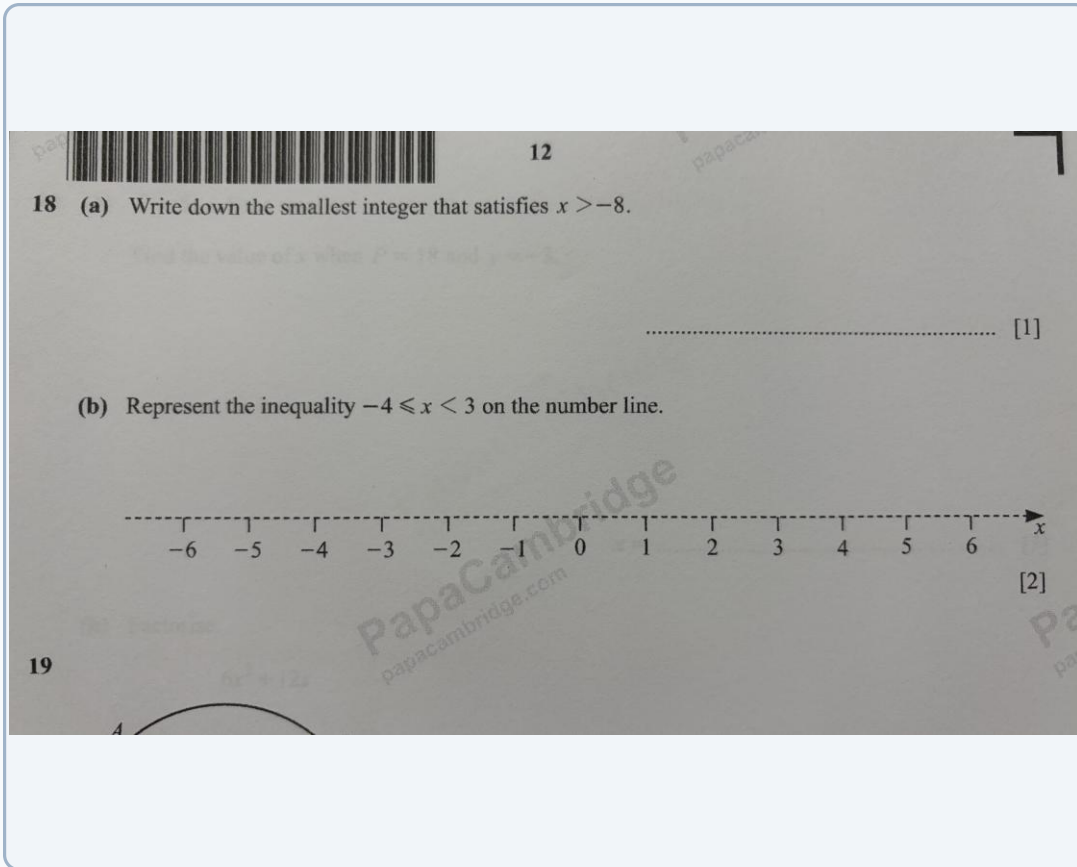
A1: r = √(150/π) = 6.91 cm (correct numerical value)

Total: 3

Math Intellect • Prototype Validation Evidence

84

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 18 (a) Write down the smallest integer that satisfies $(x \geq -8)$.
(b) Represent the inequality $(-4 \leq x < 3)$ on the number line.

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0ab2f9e405d485ec006aa6e9b0922887d08c566fd95ed824fc

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a) -8\n\n(b) Number line instructions:\n- Draw a horizontal number line with integer tick marks from at least -5 to 4.\n- Label the ticks at -4 and 3 clearly.\n- Place a solid (filled) circle at -4.\n- Place an open (hollow) circle at 3.\n- Draw a thick line segment joining -4 to 3.

role : assistant

Official Mark Scheme Answer

18(a)	-7
18(b)	

1	
2	B1 for correct line with one end correct or both ends correct and no/incorrect line

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a text-based response.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Inequalities

Difficulty: Easy

Assessment Objective: AO1

Solution

(a) -8

(b) Number line instructions:

- Draw a horizontal number line with integer tick marks from at least -5 to 4.
- Label the ticks at -4 and 3 clearly.
- Place a solid (filled) circle at -4.
- Place an open (hollow) circle at 3.
- Draw a thick line segment joining -4 to 3.

B1: (a) -8

B1: (b) number line with integer tick marks at least -5 to 4

B1: (b) solid (filled) circle at -4 and open at 3

B1: (b) thick line segment joining -4 to 3

Total: 4

1:21 AM

Math Intellect • Prototype Validation Evidence

86

Original Question Image

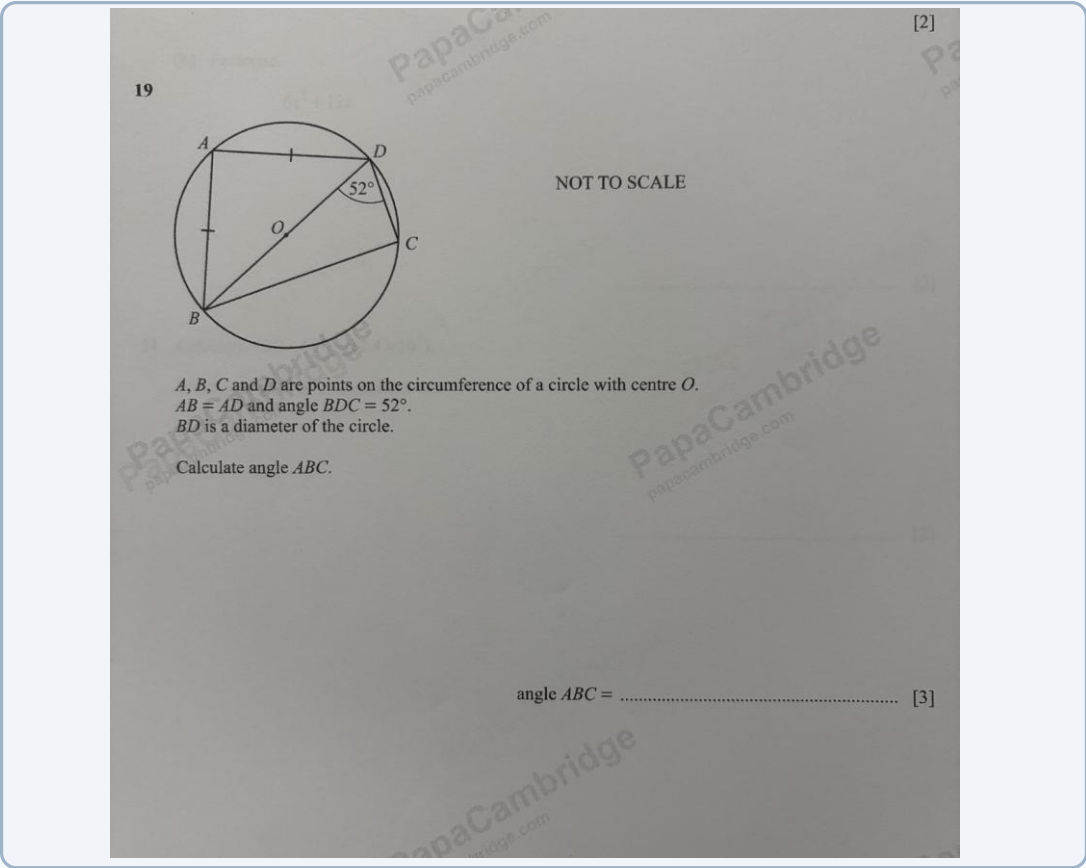


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 19
A, B, C and D are points on the circumference of a circle with centre O.
AB = AD and angle BDC = 52°.
BD is a diameter of the circle.
Calculate angle ABC.
angle ABC = [3]

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_05d92730858c7e45006aa6ef43817087d0bc893648a41205c0

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n BD is a diameter $\Rightarrow$ angle BAD = 90° and angle BCD = 90° (angle in a semicircle).\n \n Step 2:\n AB = AD $\Rightarrow$ triangle ABD is isosceles with vertex angle 90° at A.\n So angle ABD = angle ADB = $(180^\circ - 90^\circ) \div 2 = 45^\circ$.\n \n Step 3:\n In triangle BDC, angle BCD = 90° and angle BDC = 52° .\n So angle DBC = $180^\circ - 90^\circ - 52^\circ = 38^\circ$.\n \n Step 4:\n angle ABC = angle ABD + angle DBC = $45^\circ + 38^\circ = 83^\circ$.\n \n Final Answer:\n 83^\circ

role : assistant

Official Mark Scheme Answer

19	83
3	M2 for any fully correct method OR B1 for $DBC = 38$ B1 for $ABD = 45$ OR M1 for either DAB or $A = 90$ or DCB or $C = 90$

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

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Success Layout – LINE Reply Message



Topic: Circles

Difficulty: Medium

Assessment Objective: AO1 + AO2

Solution

Step 1:

BD is a diameter $\Rightarrow$ angle BAD = 90° and angle BCD = 90° (angle in a semicircle).

Step 2:

AB = AD $\Rightarrow$ triangle ABD is isosceles with vertex angle 90° at A.
So angle ABD = angle ADB = $(180^\circ - 90^\circ) \div 2 = 45^\circ$.

Step 3:

In triangle BDC, angle BCD = 90° and angle BDC = 52° .
So angle DBC = $180^\circ - 90^\circ - 52^\circ = 38^\circ$.

Step 4:

angle ABC = angle ABD + angle DBC = $45^\circ + 38^\circ = 83^\circ$.

Final Answer:

83^\circ

M1: Uses AB = AD and BAD = 90° to deduce ABD = 45° .
M1: Uses BCD = 90° and BDC = 52° to deduce DBC = 38° .
A1: Adds ABD and DBC to obtain ABC = 83° .

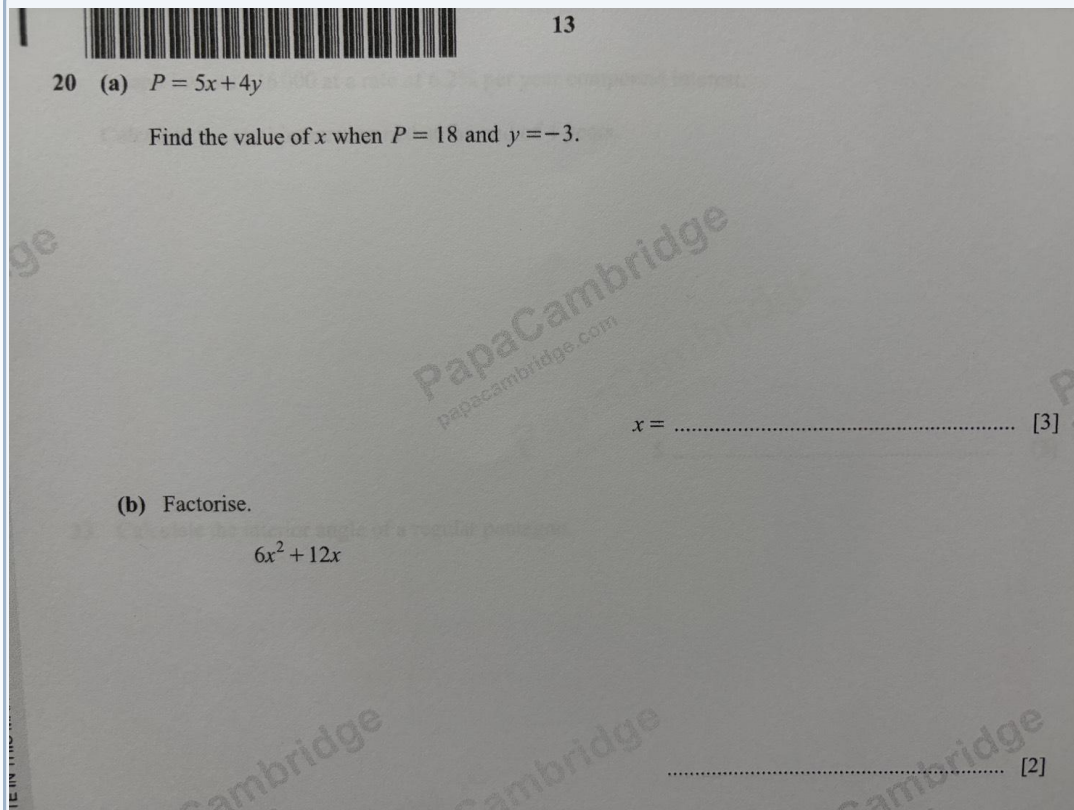
Total: 3

1:45 AM

Math Intellect • Prototype Validation Evidence

88

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 20 (a) $(P = 5x + 4y)$ Find the value of (x) when $(P = 18)$ and $(y = -3)$.
 $\dots\dots\dots$
[3] (b) Factorise. $(6x^2 + 12x)$
 $\dots\dots\dots$
[2]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0bcf45c37504a70b006aa762258a9c87d0a43a877a5765e67e

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)\n Step 1:\n Substitute P = 18 and y = -3 into P = 5x + 4y:\n 18 = 5x + 4(-3) = 5x - 12\n\n Step 2:\n 5x - 12 = 18\n 5x = 30\n x = 6\n\n Final Answer:\n x = 6\n\n (b)\n 6x^2 + 12x = 6x(x + 2)\n\n Final Answer:\n 6x(x + 2)

role : assistant

Official Mark Scheme Answer

20(a)	6 final answer
20(b)	6x(x + 2) final answer
3	M1 for $18 = 5x + 4 \times -3$ oe M1FT for a correct first step e.g. $18 + 12 = 5x$ or $x + \frac{-12}{5} = \frac{18}{5}$ oe
2	B1 for $6(x^2 + 2x)$ or $x(6x + 12)$ or $2x(3x + 6)$ or $3x(2x + 4)$ or $6x(x + 2)$ seen then spoilt

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

8

Topic: Algebra

Difficulty: Easy

Assessment Objective: AO1

Solution

(a)

Step 1:
Substitute $P = 18$ and $y = -3$ into $P = 5x + 4y$:
 $18 = 5x + 4(-3) = 5x - 12$

Step 2:
 $5x - 12 = 18$
 $5x = 30$
 $x = 6$

Final Answer:
 $x = 6$

(b)

$6x^2 + 12x = 6x(x + 2)$

Final Answer:
 $6x(x + 2)$

M1: (a) Substitutes $P = 18$, $y = -3$ into $5x + 4y$, gets $18 = 5x - 12$
M1: (a) Rearranges to $5x = 30$ (adds 12 to both sides)
A1: (a) Writes $x = 6$

M1: (b) Factors out common factor $6x$, obtaining $6x(x + 2)$
A1: (b) Final answer $6x(x + 2)$

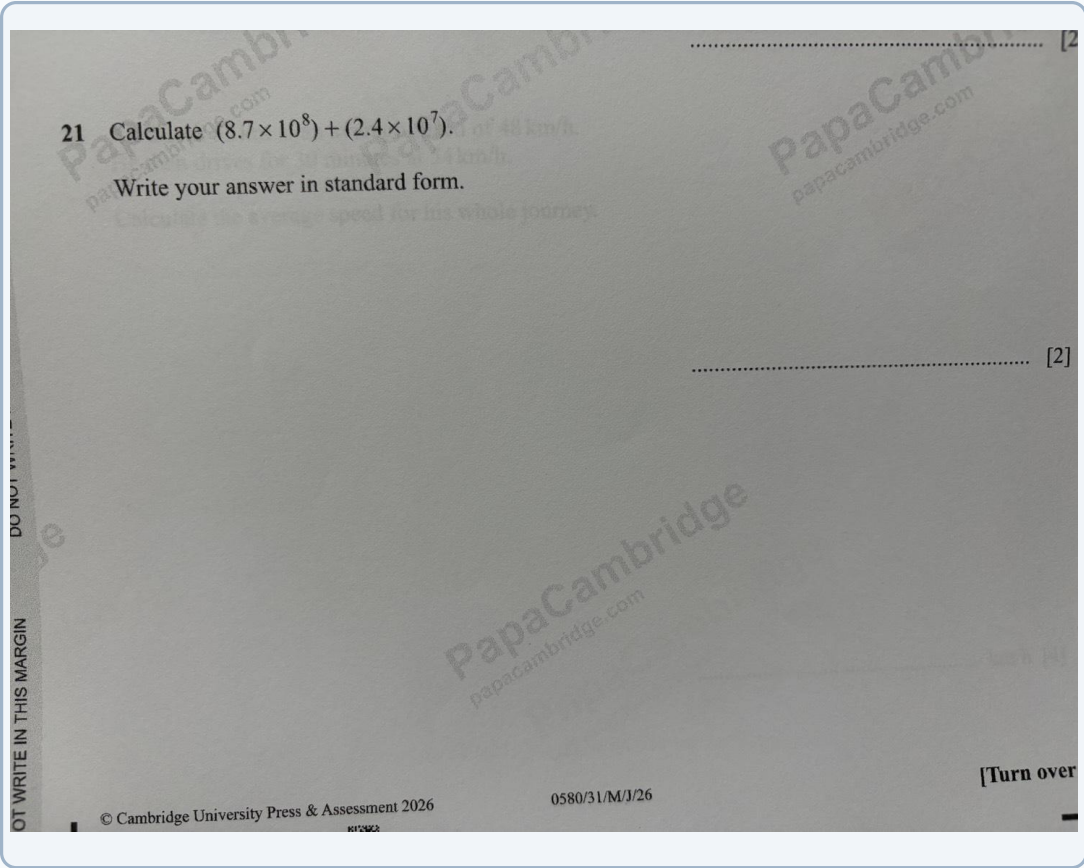
Total: 5

9:55 AM

Math Intellect • Prototype Validation Evidence

90

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 21 Calculate $(8.7 \times 10^8) + (2.4 \times 10^7)$.
Write your answer in standard form.

Image Quality

GOOD – All texts / symbols clearly readable
MODERATE – minor blur / skew but all essential information recoverable
POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0ddd8ea8b1bad911006aa76469419c87d08b18ae2e7b3a0e11

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Write both numbers with the same power of 10.\n 2.4 × 10^7 = 0.24 × 10^8\n \n Step 2:\n Add the coefficients.\n 8.7 × 10^8 + 0.24 × 10^8 = (8.7 + 0.24) × 10^8 = 8.94 × 10^8\n \n Final Answer:\n 8.94 × 10^8

role : assistant

Official Mark Scheme Answer

21	8.94×10^8
2	B1 for 894 000 000 oe

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Standard form

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Write both numbers with the same power of 10.

$2.4 \times 10^7 = 0.24 \times 10^8$

Step 2:

Add the coefficients.

$8.7 \times 10^8 + 0.24 \times 10^8 = (8.7 + 0.24) \times 10^8 = 8.94 \times 10^8$

Final Answer:

8.94×10^8

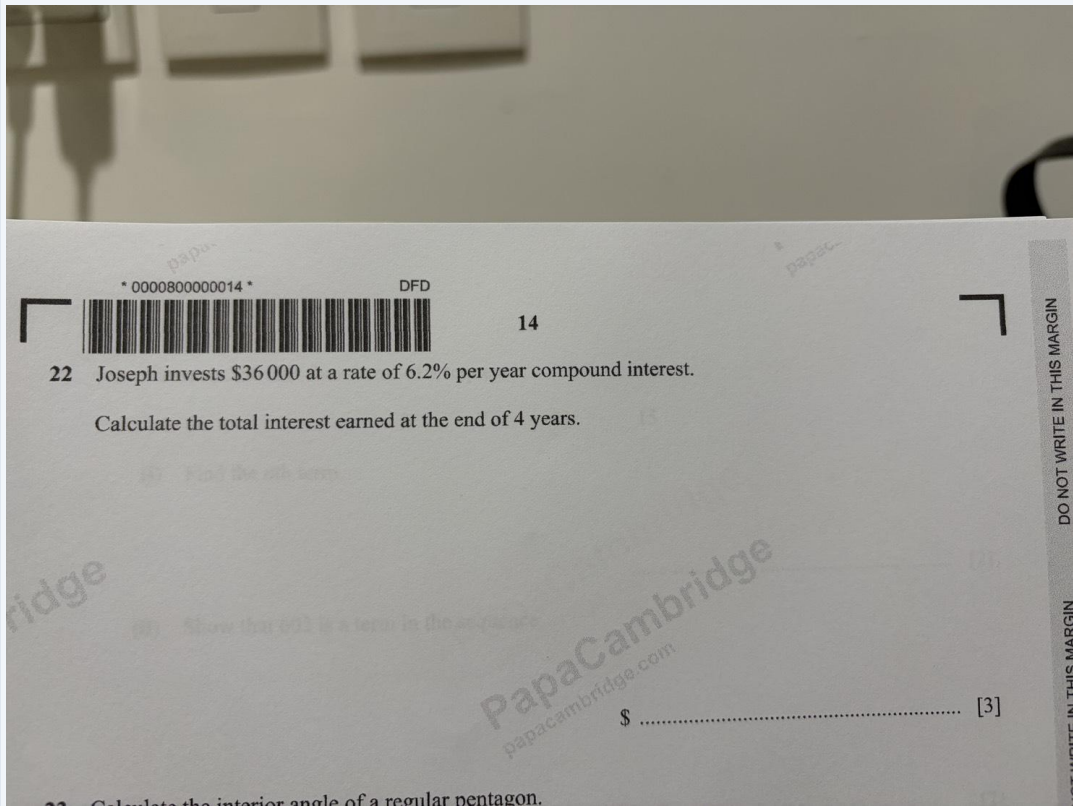
M1: Rewrites 2.4×10^7 as 0.24×10^8

A1: Final answer 8.94×10^8

Total: 2

10:05 AM

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 22 Joseph invests \$36 000 at a rate of 6.2%
per year compound interest. \n \n Calculate the total
interest earned at the end of 4 years. \n \n \$
..... [3]

Image Quality

- GOOD – All texts / symbols clearly readable
- MODERATE – minor blur / skew but all essential information recoverable
- POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_Od0d6ff5f6928a03006aa766de194487d08110fe66e58b13c1

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Amount after 4 years:\n A = 36000 × (1 + 0.062)^4 = 36000 × (1.062)^4 = 36000 × 1.272032088... = 45793.155...\n \n Step 2:\n Total interest earned:\n Interest = A – P = 45793.155... – 36000 = 9793.155...\n \n Final Answer:\n \$9793.16

role : assistant

Official Mark Scheme Answer

22	9790 or 9793[.....]
3	B2 for 45 800 or 45 790 or 45 793[....] or M2 for $36\,000 \times \left(1 + \frac{6.2}{100}\right)^4 - 36\,000$ or M1 for $36\,000 \times \left(1 + \frac{6.2}{100}\right)^4$ oe

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Compound interest

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Amount after 4 years:

$A = 36000 \times (1 + 0.062)^4 = 36000 \times (1.062)^4 = 36000 \times 1.272032088... = 45793.155...$

Step 2:

Total interest earned:

$\text{Interest} = A - P = 45793.155... - 36000 = 9793.155...$

Final Answer:

\$9793.16

M1: Uses $A = 36000 \times (1.062)^4$ (compound interest formula)

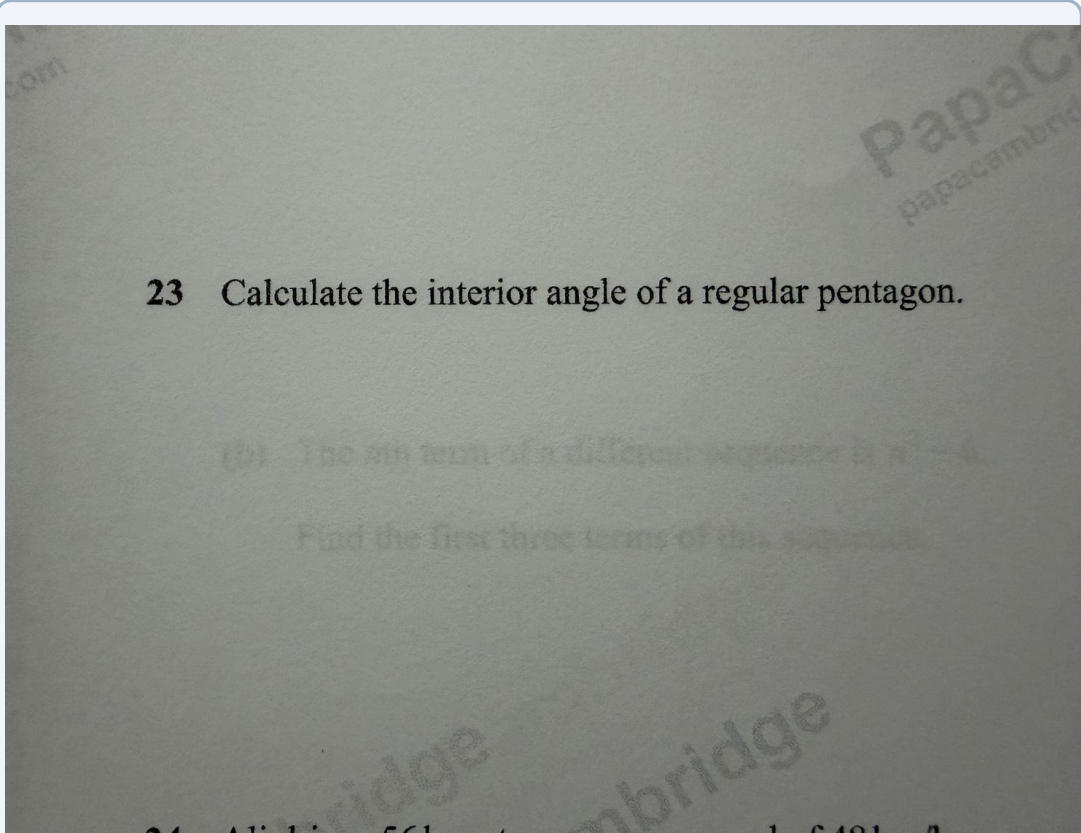
A1: Obtains amount A = $36000 \times 1.062^4 = 45793.155...$

A1: Finds interest = $45793.155... - 36000 = 9793.155... \rightarrow 9793.16$

Total: 3

10:16 AM

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 23 Calculate the interior angle of a regular pentagon.

Image Quality

- GOOD – All texts / symbols clearly readable
- MODERATE – minor blur / skew but all essential information recoverable
- POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

1 item

output

0

id : msg_0d89839015a81a79006aa76b35dc3087d08ac39b8c96466cbd

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Sum of interior angles of an n-gon = (n - 2) × 180°\n For a pentagon, (5 - 2) × 180° = 3 × 180° = 540°.\n \n Step 2:\n Each interior angle (regular pentagon) = 540° ÷ 5 = 108°.\n \n Final Answer:\n 108°

role : assistant

Official Mark Scheme Answer

	23	108
2	M1 for $180 - (360 \div 5)$ oe or $\frac{(5 - 2) \times 180}{5}$ oe	

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Polygons

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Sum of interior angles of an n-gon = (n - 2) × 180°

For a pentagon, (5 - 2) × 180° = 3 × 180° = 540°.

Step 2:

Each interior angle (regular pentagon) = 540° ÷ 5 = 108°.

Final Answer:

108°

M1: Uses (n - 2) × 180 = (5 - 2) × 180 and divides 540° by 5

A1: Obtains 108°

Total: 2

10:34 AM

Original Question Image

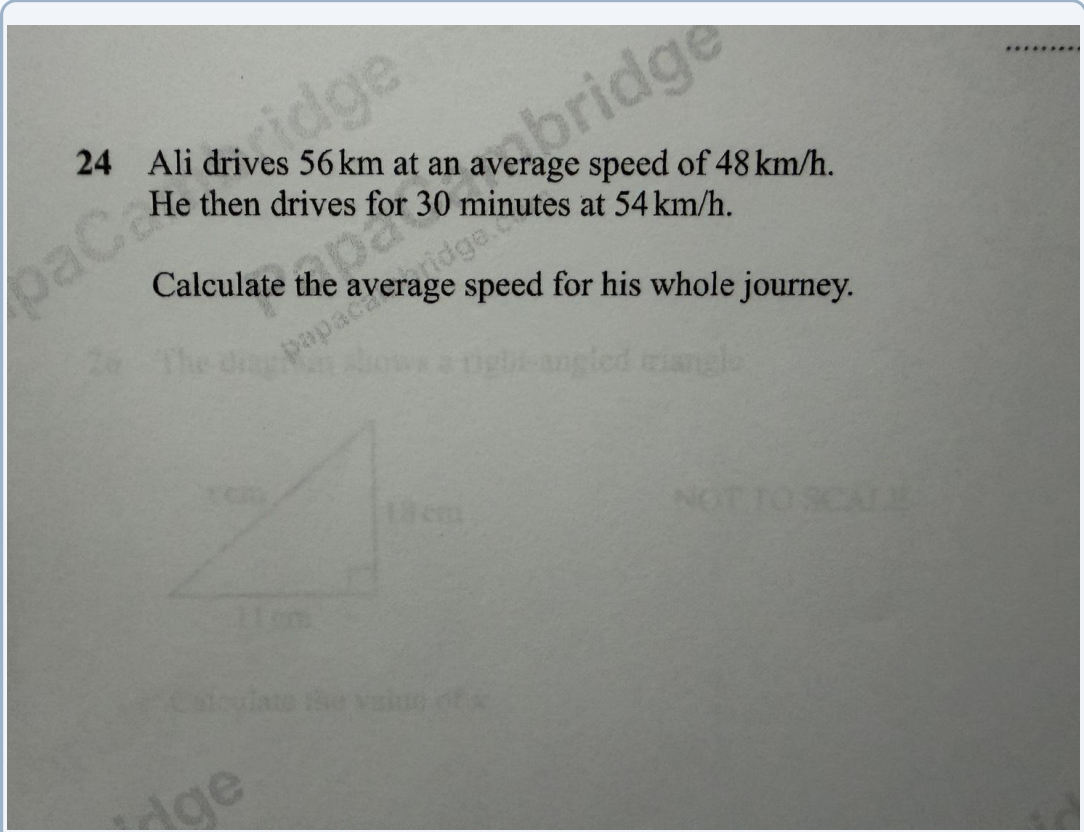


Image Quality	
GOOD	All texts / symbols clearly readable
MODERATE	minor blur / skew but all essential information recoverable
POOR	visible degradation affecting text / symbol yet question remains testable

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR– visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT



1 item

1

conten : 24 Ali drives 56 km at an average speed of 48 km/h. He then drives for 30 minutes at 54 km/h. \n \n Calculate the average speed for his whole journey.

OUTPUT



1 item

1

conten : 24 Ali drives 56 km at an average speed of 48 km/h. He then drives for 30 minutes at 54 km/h. \n \n Calculate the average speed for his whole journey.

OUTPUT



1 item

1

conten : 24 Ali drives 56 km at an average speed of 48 km/h. He then drives for 30 minutes at 54 km/h. \n \n Calculate the average speed for his whole journey.

OUTPUT



1 item

1

conten : 24 Ali drives 56 km at an average speed of 48 km/h. He then drives for 30 minutes at 54 km/h. \n \n Calculate the average speed for his whole journey.

Interpretation

Interpretation	Score
PASS	70
FAIL	30

PASS FAIL

PASS FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

output

0

```
id : msg_0f0b5ce398227865006aa772fb5fe887d08bc
    2e5eb9a389b61
```

```
type : message
```

```
status : completed
content
```

0

```
type : output_text
```

```
annotations : [empty array]
```

```
logprobs : [empty array]
```

56 ÷ 48 = 7/6 h. \n \n Step 2: \n Distance for second

part = speed $\times$ time = $54 \times 0.5 = 27$ km.
Time for second part = 0.5 h.
Step 3: Total distance = $56 + 27 = 83$ km.
Total time = $7/6 + 1/2 = 7/6 + 3/6 = 10/6 = 5/3$ h.
Step 4: Average speed = total distance $\div$ total time = $83 \div (5/3) = 83 \times 3/5 = 249/5 = 49.8$ km/h.
Final Answer: 49.8 km/h

role : assistant

Official Mark Scheme Answer

24	49.8 nfw
4	M3 for $\frac{30}{60} \times 54 + 56$ $\frac{56}{48} + 0.5 \quad \text{oe}$ OR M1 for $\frac{56}{48}$ oe $\frac{56}{48} \times 60$ oe or better M1 for $\frac{30}{60} \times 54$ oe or better

LINE End-to-End

PASS

FAIL

PASS FAIL

Final Answer

PASS

FAIL

PASS FAIL

Response	Percentage
Appropriate	55%
Not appropriate	45%

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score— [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]
0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

11:07 AM

11:07 AM

Topic: Distance, speed and time
Difficulty: Medium
Assessment Objective: AO2

Solution

Step 1:
Time for first part = distance $\div$ speed =
 $56 \div 48 = 7/6$ h.

Step 2:
Distance for second part = speed $\times$
time = $54 \times 0.5 = 27$ km.
Time for second part = 0.5 h.

Step 3:
Total distance = $56 + 27 = 83$ km.
Total time = $7/6 + 1/2 = 7/6 + 3/6 = 10/6$
 $= 5/3$ h.

Step 4:
Average speed = total distance $\div$ total
time = $83 \div (5/3) = 83 \times 3/5 = 249/5 =$
 49.8 km/h.

Final Answer:
49.8 km/h

M1: Time for first part correctly found
 $56 \div 48 = 7/6$ h.
M1: Second part distance and time
correctly $54 \times 0.5 = 27$ km, 0.5 h.
M1: Total distance and total time
correctly $56 + 27 = 83$ km, $7/6 + 1/2 =$
 $5/3$ h.
A1: Average speed correctly $83 \div (5/3)$
 $= 249/5 = 49.8$ km/h.

Total: 4

11:07 AM

Original Question Image

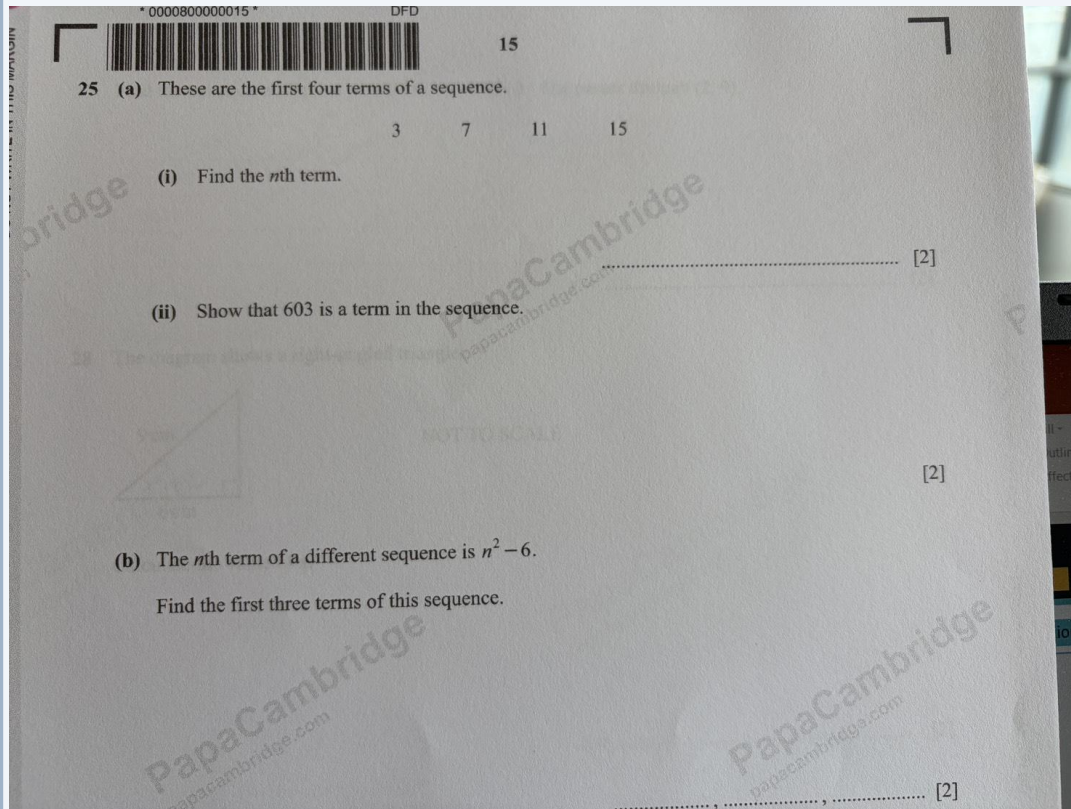


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 25 (a) These are the first four terms of a
sequence. 3 7
11 15 (i) Find the n th term. [2] (ii)
Show that 603 is a term in the sequence.
[2] (b) The n th term of a different sequence is $\backslash$
($n^2 - 6$). Find the first three terms of this
sequence. [2]

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_01e174ae7042b730006aa77a1de01887d0b7f685889dea033b

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : (a)(i)\n Step 1:\n Common difference d = 7 - 3 = 4.\n \n Step 2:\n nth term = 3 + (n - 1)×4 = 4n - 1.\n \n Answer: 4n - 1\n \n (a)(ii)\n Step 1:\n Set 4n - 1 = 603.\n \n Step 2:\n 4n = 604 ⇒ n = 151 (an integer).\n \n Therefore, 603 is the 151st term and is in the sequence.\n \n (b)\n Step 1:\n For n = 1: 1^2 - 6 = -5\n For n = 2: 2^2 - 6 = -2\n For n = 3: 3^2 - 6 = 3\n \n Answer: -5, -2, 3\n \n Final Answer:\n (a) (i) 4n - 1\n (a)(ii) Yes; n = 151\n (b) -5, -2, 3

role : assistant

Official Mark Scheme Answer

25(a)(i)	$4n - 1$ oe final answer
25(a)(ii)	$4n - 1 = 603$ $n = \frac{603 + 1}{4} = 151$
25(b)	- 5 - 2 3
2	B1 for $4n + j$ or $kn - 1$ ($k \neq 0$), or $4n - 1$ seen then spoilt
M1	<i>FT their (a)(i)</i>
M1	Accept any correct method e.g. 604 is a multiple of 4 and this sequence is one less than the multiple of 4s or counting in 4's
2	B1 for 2 correct and in the correct places or if 0 scored SC1 for - 6 - 5 - 2

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Sequences
Difficulty: Easy
Assessment Objective: AO1 + AO2

Solution

(a)(i)
Step 1:
Common difference $d = 7 - 3 = 4$.

Step 2:
nth term $= 3 + (n - 1) \times 4 = 4n - 1$.

Answer: $4n - 1$

(a)(ii)
Step 1:
Set $4n - 1 = 603$.

Step 2:
 $4n = 604 \Rightarrow n = 151$ (an integer).

Therefore, 603 is the 151st term and is in the sequence.

(b)
Step 1:
For $n = 1$: $1^2 - 6 = -5$
For $n = 2$: $2^2 - 6 = -2$
For $n = 3$: $3^2 - 6 = 3$

Answer: -5, -2, 3

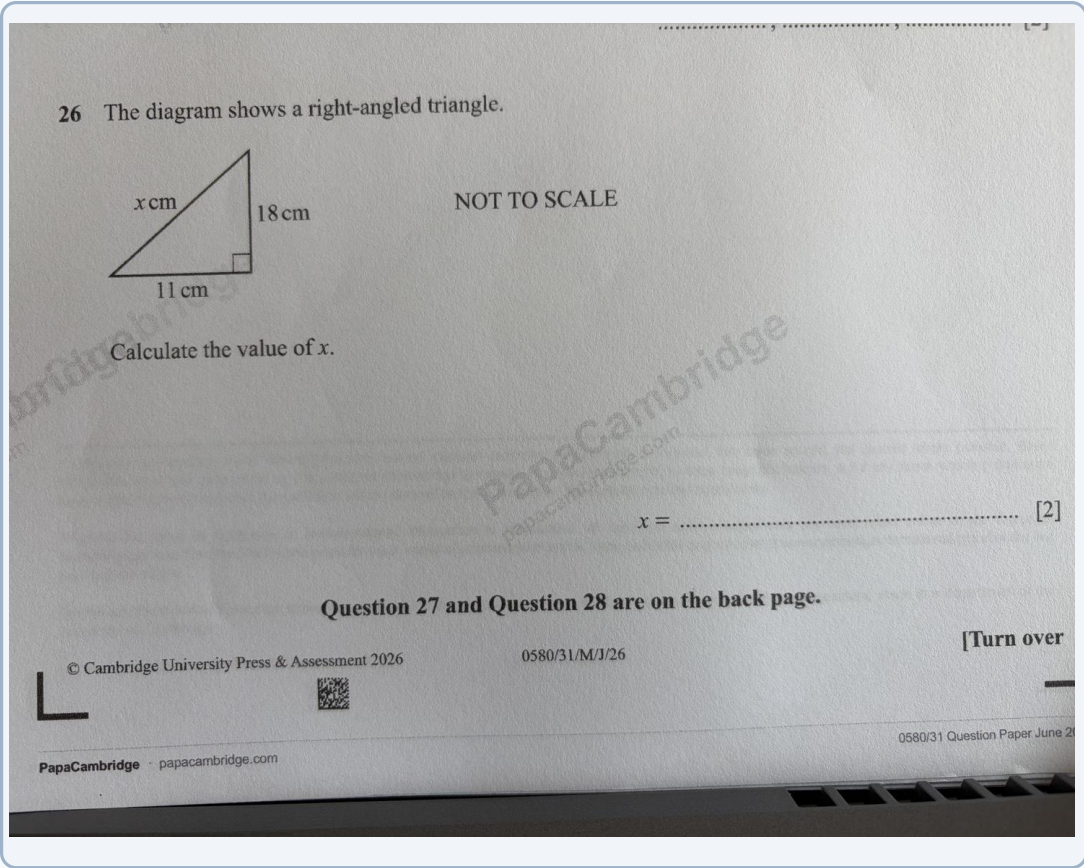
Final Answer:
(a)(i) $4n - 1$
(a)(ii) Yes; $n = 151$
(b) -5, -2, 3

M1: Uses $a_n = a + (n-1)d$ with $a = 3$, $d = 4$
A1: Obtains nth term $4n - 1$

Math Intellect • Prototype Validation Evidence

100

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 26 The diagram shows a right-angled triangle.
 \n \n [Diagram: right-angled triangle with sides labelled x cm, 18 cm, and 11 cm]
 \n \n Calculate the value of x .
 \n \n $x = \dots\dots\dots$ [2]

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

id : msg_0281c6f4423e4af2006aa7814df76887d0a5c6f39408fef0d2

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\nUsing Pythagoras' theorem with 18 cm as the hypotenuse:\n18^2 = 11^2 + x^2\n\nStep 2:\nx^2 = 18^2 - 11^2 = 324 - 121 = 203\n\nStep 3:\nx = \sqrt{203} = 14.247...\n\nFinal Answer:\nx = 14.2 cm (3 s.f.)

role : assistant

Official Mark Scheme Answer

26	21.1 or 21.09... to 21.10
2	M1 for $11^2 + 18^2$ or better

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a wrong reasoning.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Pythagoras

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Using Pythagoras' theorem with 18 cm as the hypotenuse:
 $18^2 = 11^2 + x^2$

Step 2:

$x^2 = 18^2 - 11^2 = 324 - 121 = 203$

Step 3:

$x = \sqrt{203} = 14.247...$

Final Answer:

$x = 14.2 \text{ cm (3 s.f.)}$

M1: Uses Pythagoras with 18 as hypotenuse: $18^2 = 11^2 + x^2$

A1: Calculates $x = \sqrt{203} = 14.2 \text{ cm (3 s.f.)}$

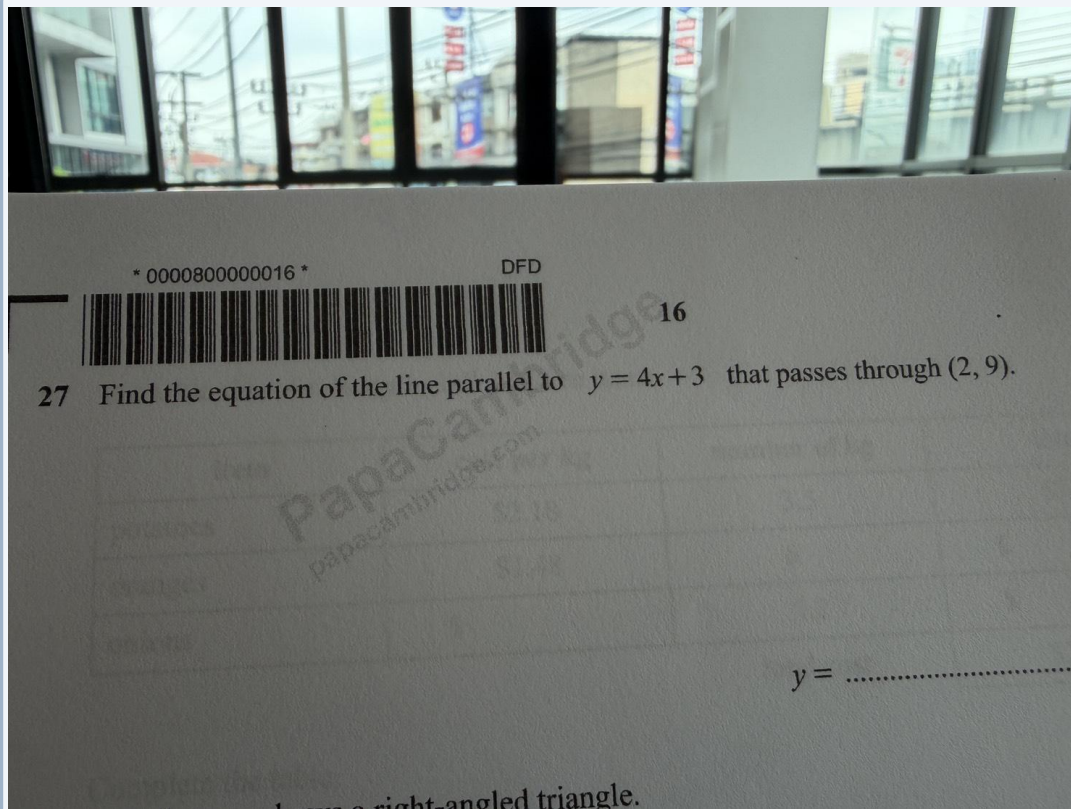
Total: 2

12:08 PM

Math Intellect • Prototype Validation Evidence

102

Original Question Image



Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

conten : 27 Find the equation of the line parallel to $y = 4x + 3$ that passes through $(2, 9)$.
.....

Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Interpretation

PASS

FAIL

Mathematical Reasoning Output – Solver Node

OUTPUT

i

1 item

output

0

id : msg_095fb28461fc4f6d006aa7862b626487d098949177eb85b22b

type : message

status : completed

content

0

type : output_text

annotations : [empty array]

logprobs : [empty array]

text : Step 1:\n Gradient of $y = 4x + 3$ is 4. A parallel line has the same gradient, 4.\n\n Step 2:\n Use point-slope form with point (2, 9):\n $y - 9 = 4(x - 2)$ \n\n Step 3:\n $y - 9 = 4x - 8$ \n $y = 4x + 1$ \n\n Final Answer:\n $y = 4x + 1$

role : assistant

Official Mark Scheme Answer

27	$[y =] 4x + 1$ oe final answer
2	B1 for $4x + c$ ($c \neq 3$) oe or M1 for $9 = 4 \times 2 + c$ oe

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message

Topic: Straight line

Difficulty: Easy

Assessment Objective: AO1

Solution

Step 1:

Gradient of $y = 4x + 3$ is 4. A parallel line has the same gradient, 4.

Step 2:

Use point-slope form with point (2, 9):
 $y - 9 = 4(x - 2)$

Step 3:

$y - 9 = 4x - 8$
 $y = 4x + 1$

Final Answer:

$y = 4x + 1$

M1: Uses gradient 4 and point-slope $y - 9 = 4(x - 2)$
A1: Obtains $y = 4x + 1$

Total: 2

12:29 PM

Math Intellect • Prototype Validation Evidence

104

Original Question Image

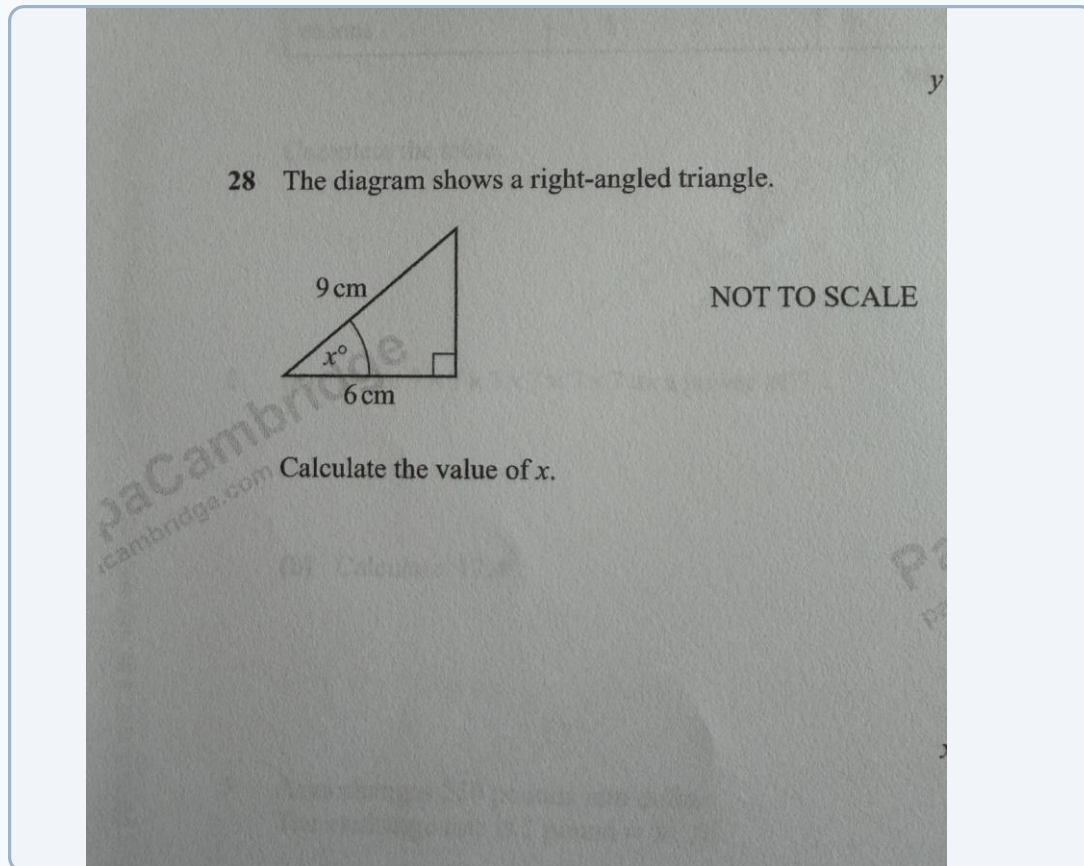


Image Quality

GOOD – All texts / symbols clearly readable

MODERATE – minor blur / skew but all essential information recoverable

POOR – visible degradation affecting text / symbol yet question remains testable

Question Interpretation Output – Edit Fields Node

OUTPUT  

1 item

1

content : 28 The diagram shows a right-angled triangle.
[Diagram: A right-angled triangle with sides labelled 9 cm, 6 cm, and an angle x° opposite the 6 cm side.]
Calculate the value of x .

Interpretation

PASS

FAIL

REASON : Required diagram information was not extracted from the source image.

Mathematical Reasoning Output – Solver Node

OUTPUT  

1 item

output

0

```
id : msg_0c0e137c004c2739006aa7890f80ac87d0ad6
8baa37bcf633b
```

```
type : message
```

```
status : completed
```

```
content
```

0

```
type : output_text
```

```
annotations : [empty array]
```

```
logprobs : [empty array]
```

```
text : Step 1:\n In the right-angled triangle, the
hypotenuse is 9 cm and the side opposite angle x is
6 cm.\n sin x = opposite / hypotenuse = 6/9 =
2/3\n \n Step 2:\n x = sin^-1(2/3) =
41.8103...\n \n Final Answer:\n 41.8°
```

```
role : assistant
```

Official Mark Scheme Answer

28	48.2 or 48.18 to 48.19
2	M1 for $\cos [\dots] = \frac{6}{9}$ oe or better or $90 - \sin^{-1} \left(\frac{6}{9} \right)$ oe or better

LINE End-to-End

PASS

FAIL

Final Answer

PASS

FAIL

REASON : Required visual information was not extracted from the source image; therefore, the solver could only provide a wrong reasoning.

Evaluation basis: Official Cambridge mark scheme + predefined validation rubric.

Solution Completeness Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = fully aligned with Cambridge-style method, wording, and marks (total score weight performance across all subparts based on official mark scheme)

Assessment Alignment Score — [0 / 1 / 2 / 3 / 4]

0 = unsuitable; 1 = weak; 2 = acceptable; 3 = strong; 4 = complete / fully aligned (mark-weighted performance across all subparts based on official mark scheme)

Success Layout – LINE Reply Message



Topic: Trigonometry
Difficulty: Easy
Assessment Objective: AO1

Solution

Step 1:

In the right-angled triangle, the hypotenuse is 9 cm and the side opposite angle x is 6 cm.
 $\sin x = \text{opposite} / \text{hypotenuse} = 6/9 = 2/3$

Step 2:

$x = \sin^{-1}(2/3) = 41.8103\dots$

Final Answer:
41.8°

M1: Uses $\sin x = \text{opposite}/\text{hypotenuse}$ and substitutes $\sin x = 6/9$.
A1: Calculates $x = \sin^{-1}(2/3) \approx 41.8^\circ$.

Total: 2

12:41 PM

Case Recording Rule

Record Excel values only after comparing the generated output with the official Cambridge mark scheme and the predefined rubric. If no final LINE output is delivered, mark educational-quality fields as N/A, not zero.

Excel Field	Value to Record	Evidence Location / Methodology
Test Case ID	MI-TC-xxx	Case slides + Excel row
Interpretation Pass	Pass / Fail	Input & Interpretation slide
Quality Gate Outcome	Pass / Fail	Success or failure evidence slide
Failure Stage	None / Analyze Image / Question Analyzer / Solver / Examiner / Moderator / Quality Checker / Formatter / LINE Delivery	Root-cause stage where the end-to-end failure originated
Failure Detected by	None / Analyze Image / Question Analyzer / Solver / Examiner / Moderator / Quality Checker / Formatter / LINE Delivery	Node/gate that identified and rejected the failure
LINE End-to-End Success	Success / Fail	LINE output or failure evidence
Final Answer Pass	Pass / Fail / N/A	N/A when no delivered final answer
Sub-question Accuracy (%)	0–100% / N/A	N/A when no assessable final output
Solution Completeness	0 / 1 / 2 / 3 / 4 / N/A	Weighted aggregate; N/A excluded in Excel formulas
Assessment Alignment	0 / 1 / 2 / 3 / 4 / N/A	Weighted aggregate; N/A excluded in Excel formulas
Evidence Ref	PPT/PDF slide + GitHub folder path	All evidence files

Repository rule: export completed deck to PDF and keep PPTX as editable source. Use the same MI-TC-xxx identifier in Excel, PPT/PDF and GitHub evidence folders.